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AI-driven Emotion Recognition for Adaptive Learning

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This is to certify that:

- (i) the thesis comprises only my original work toward the Bachelor Degree of Science (B.Sc.) at the German University in Cairo (GUC),
- (ii) due acknowledgment has been made in the text to all other material used

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June 8, 2026

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Abstract

An intelligent tutor can accurately determine what a student has mastered yet still lose their engagement, since a major driver of disengagement and dropout is affective rather than cognitive, and few tutoring policies treat affect as a control signal. This thesis addresses this gap by developing a two-stage affect-aware tutoring architecture whose perception and decision components are validated independently. The perception module performs multi-label academic-affect recognition from ten-second webcam clips, predicting boredom, engagement, confusion, and frustration as four binary states that may co-occur within the same clip. It combines an ImageNet-pretrained ResNet-50 backbone, Squeeze-and-Excitation channel recalibration, a two-layer bidirectional LSTM, and temporal attention pooling over 24 frames into four sigmoid heads. The decision module formulates tutoring as a partially observable Markov decision process solved by a Deep Q-Network (DQN), with state incorporates affective channels and latent mastery, while a multi-objective reward function jointly optimises Hake’s normalised learning gain and affect regulation across eight pedagogical actions, while the environment dynamics are derived from established theoretical models. To address of incomparable evaluation in prior DAiSEE-based work, six competing architectures were retrained from scratch under a unified experimental pipeline, with class imbalance handled consistently at the data, loss, and decision-threshold levels. The policy was trained on 10,000 heterogeneous synthetic learners and evaluated over ten random seeds against both a rule-based expert tutor and a random controller. The proposed model achieves the highest Macro F1 among standardised configurations (0.453), with a Weighted F1 of 0.798 and a mean per-label binary accuracy of 0.782 across the four affect labels. In simulation, the DQN policy outperforms both baselines across all metrics in both populations, improving learning gain by 0.088 over the expert tutor on the mixed population ($d = 2.01$, $p_{\text{Holm}} < 10^{-2}$, effect sizes ranging from 1.81 to 3.30 overall). It also shows interpretable behavioural patterns and demonstrates that a policy trained on a mixed population generalises effectively to the canonical learner rather than overfitting. Among fourteen prior systems surveyed, this is the only approach that unifies all seven design dimensions of affect-aware reinforcement learning-based tutoring. It combines a benchmark-validated perception layer with a deep value-based policy through a defined interface, enabling a shift from cognition-only adaptation toward tutoring systems that respond to the learner as they are, rather than solely to recorded performance.

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List of Abbreviations

FER	Facial Expression Recognition
AEQ	Achievement Emotions Questionnaire
AEQ-M	Achievement Emotions Questionnaire–Mathematics
AET-SSBD	Academic Engaged Time Code of the Systematic Screening for Behavior Disorders
BOSS	Behavioral Observation of Students in Schools
DOF	Direct Observation Form
COC	Classroom Observation Code
SECOS	State-Event Classroom Observation System
ADHD-SOC	Attention-Deficit Hyperactivity Disorder School Observation Code
SOS	Student Observation System
ASSISTments	ASSISTments web-based tutoring platform
LIWC	Linguistic Inquiry and Word Count
Coh-Metrix	Cohesion Metrics
AutoTutor	AutoTutor intelligent tutoring system

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Chapter 1

Introduction

1.1 Motivation

In a small classroom, a good teacher rarely relies on test scores alone. A furrowed brow during a derivation, a distant gaze halfway through an explanation, a brief smile after a difficult problem, and similar cues quietly tell the teacher when to slow down, offer a hint, adjust the pace, switch explanatory strategy, or offer a word of encouragement. Decades of research in affective neuroscience and educational psychology confirm that this teacherly intuition has deep roots. Emotion is not a byproduct of thinking but an integral part of it, and learning, as it unfolds in schools and in the real world, is not a disembodied or purely rational process [4]. Emotional states modulate attention, encoding, and memory consolidation, so that what a student feels while studying directly shapes what they later remember and how they will reason about it [5]. In tutoring contexts specifically, a distinctive family of academic emotions, namely engagement, confusion, frustration, and boredom, emerges repeatedly, each carrying pedagogical meaning of its own [6]. Pekrun’s Control-Value Theory explains *why* these emotions matter: fluctuations in learners’ appraisals of control over a task and of its personal value arouse achievement emotions that alter how information is processed, stored, and applied, feeding back into engagement, motivation, and ultimately performance [7]. A learner’s face, therefore, is not decorative information; it is a window into the very processes that education is trying to shape.

However, the contextual cues that a skilled teacher reads almost unconsciously largely disappear when learning moves online or to large-scale digital platforms. Modern e-learning and intelligent tutoring environments have expanded rapidly, especially after the pandemic, and are now a widely used channel for student learning[8]. Yet these systems typically treat learners as receivers of information, with little concern for their moment-to-moment emotional or cognitive condition. Under Pekrun’s Control-Value Theory, this neglect carries direct cognitive costs: whilst positive emotions such as enjoyment sustain motivation and broaden attentional

resources, negative states such as anxiety and boredom produce cognitive interference that impairs how information is encoded and retained [7]. The academic emotions most characteristic of tutoring interactions, namely confusion, frustration, and boredom, identified by D’Mello and Graesser as the states that recur most reliably in learning dialogues [6], can therefore persist silently and significantly hinder learning performance [9]. Studies of online and STEM learning repeatedly show that emotions shape engagement, motivation, and satisfaction, and that ignoring them leaves platforms unable to respond when students disengage [10]. Effective tutoring responds to such shifts not merely by selecting a harder or easier item, but by adjusting pace, offering scaffolding or hints, switching explanatory strategy, or providing motivational support; this multi-dimensional responsiveness is precisely what digital platforms most lack. The effect is compounded by the fact that supervision is thin in online settings: when nobody notices that attention is drifting, sustained inattention tends to be one of the first things that breaks down, eroding the very benefits that the medium was meant to deliver [11, 12]. A system capable of reading the learner’s face in real time would offer gains on several of these fronts simultaneously: it could detect confusion or boredom well before a failing assessment reveals the problem; adjust content difficulty, pacing, and support strategies continuously for each individual rather than for the average learner; assist teachers in monitoring emotional engagement across large cohorts at once; and generate richer behavioural and affective data than click-logs or response times alone can provide. The result would more closely resemble the personalised quality of one-to-one tutoring, and it would align with what learning science recommends: timely corrective feedback, guided reflection, and conditions that support long-term retention. The picture that emerges is therefore a paradox: precisely in the environments where such personalisation across content, pacing, and learner support is most needed, the richest source of human feedback, the learner’s face, is the one that digital systems most systematically ignore.

1.2 Problem Statement

A widely proposed response to this paradox is adaptive learning: systems that tailor the sequence, difficulty, feedback, and pacing of material to each individual learner rather than delivering a fixed path [13, 8]. Recent intelligent tutoring systems (ITS) have taken up this promise with increasingly sophisticated tools. Reinforcement-learning approaches have been used to adjust task difficulty and instructional complexity in simulated learners [14, 15], and have been combined with knowledge tracing to optimise personalised learning paths across students’ evolving states [16]. Comparative studies show that such policies can meaningfully improve task selection over rule-based baselines [17], and recent work couples knowledge tracing with reinforcement-learning-based problem selection to produce measurable learning gains in short human studies [18]. Even outside the reinforcement-learning paradigm, biologically inspired memory architectures have been used to recommend the next exercise within each learner’s Zone of Proximal Development [19]. Despite this progress, two patterns emerge clearly from

recent surveys of the field. First, a surprising number of adaptive systems are described in the literature but remain experimental, with many not validated in authentic classroom settings and often disconnected from the concrete problems students face [20, 21]. Second, and more importantly for this thesis, most adaptive systems still operate only on cognitive-behavioural traces, such as answers, response times, clicks, and hint requests, and treat the learner as affect-free [14, 16, 17]. Broader comparative work echoes that behavioural traces predominate when signals are chosen for the policy loop [18, 22]. Urbaite warns that a persistent risk in such systems is collapsing the multidimensional construct of engagement into “easy-to-log proxies” such as clicks and time-on-page [23]. Meng and Yang similarly observe that existing RL-driven tutoring solutions often overlook engagement factors altogether and, when they do attempt to model them, rely primarily on textual interaction patterns rather than on perceptual signals of the student [24]. A related but distinct limitation is that the action spaces of most RL-based tutors remain narrow by design: policies typically select among difficulty levels or choose which item to present next, but do not cover the broader pedagogical repertoire that a skilled tutor employs, including scaffolding, hints, pacing changes, motivational messages, and explanatory reformulations. Authors of recent RL-based tutors explicitly flag the integration of affective computing as open future work, acknowledging that their policies are, for now, both emotionally blind and pedagogically narrow [14, 18].

Facial Expression Recognition (FER) offers a practical way to recover part of the signal that adaptive systems currently ignore. Over the last decade, deep learning has substantially advanced FER accuracy, with convolutional networks, vision transformers, and hybrid temporal models all contributing to stronger recognition in unconstrained settings [25, 26, 27]. Attention-heavy variants further tighten performance in challenging conditions [28]. In educational contexts, researchers have moved beyond Ekman’s six basic emotions, which rarely occur in actual study sessions, towards academic affect dimensions such as engagement, boredom, confusion, and frustration, the very states singled out by earlier discourse-analytic work on tutoring [29, 6]. The DAiSEE dataset has become a common anchor for this line of research, and a growing body of deep-learning models, ranging from facial-embedding and landmark approaches to more recent dynamic FER models with attention and temporal mechanisms, has been trained on it [29, 30, 31]. Related architectures and training regimes continue to expand in the latest DAiSEE-based studies [32, 26, 33]. Parallel efforts in online-learning contexts use modern CNN and transformer architectures to monitor students in live classes [27, 12]. Additional classroom-oriented deployments extend the same design space [28, 11]. Systematic reviews of academic FER confirm that deep learning now dominates this subfield and that real-time deployment is becoming realistic, while also noting ongoing challenges from pose, illumination, occlusion, and cross-cultural variability [25, 34]. Overall, the evidence that facial signals carry meaningful information about engagement and cognitive state in online learners has become difficult to dispute [31, 11, 35]; yet accurate FER is necessary but insufficient for adaptive tutoring unless those signals are coupled to concrete instructional decisions that extend

beyond difficulty adjustment or next-item selection alone.

Some recent work does connect affect to adaptation, including reinforcement-learning formulations that map inferred emotional state to instructional adjustments such as difficulty, pacing, scaffolding, hints, or metacognitive prompts. In practice, however, many of these systems lean on multimodal or physiological channels, such as heart rate, EEG, respiration, text, voice, or tone [9, 36]. Such setups are often heavy in practice, since signals must be acquired, synchronised, and processed alongside the teaching interaction, which typically implies specialised hardware and careful integration before the approach can be deployed at scale in ordinary online courses [37, 38, 39]. A second limitation, within this same emotion-aware adaptation literature, is that affect is not always carried through to the decision process that selects a pedagogical intervention; many systems stop at detection or dashboard-style reporting, so the intended loop from detecting emotion, to using it in the adaptation decision, to executing a concrete instructional action is rarely closed [34, 40, 41]. Using facial expression alone offers a lighter alternative that is less intrusive than body-worn sensors and realistic to collect with commodity webcams for large cohorts. Prior reinforcement-learning work that uses facial affect as feedback for adaptation shows that such a loop is feasible, but evaluation is often simulation-based, the affective signal may be coarse or averaged across whole tasks rather than supplied as a continuous input at decision time, and the policy action space typically remains narrow [42]. This thesis therefore adopts an FER-only, webcam-based pipeline in which facial affect drives real-time pedagogical action selection through a reinforcement-learning policy over a multi-action instructional space, moving beyond the difficulty-centric sequencing envisaged in earlier project iterations.

While facial expression recognition in education and adaptive tutoring have both matured, they have largely matured in parallel. Most FER-for-learning systems stop at recognition: they train a classifier on DAiSEE or a similar dataset, report per-class accuracy, and at best propose an analytics dashboard or teacher-facing feedback channel [29, 33, 31]. Studies that span DAiSEE-trained models to online prototypes typically remain at the recognition stage for design reasons [32, 26, 27]. Further deployments discussed in the same literature often mirror that scope [12]. Where these systems do discuss adaptation, it is typically aspirational, leaving the pedagogical response either to a human instructor or to unspecified future work [30, 28, 43]. In parallel, adaptive tutoring systems that do implement closed-loop decisions rarely use facial affect on its own; as noted above, the face, when included, tends to be one channel among several, with non-facial signals typically carrying most of the weight. Scoping reviews make the same observation at a field level. Affective intelligent tutoring systems, it has been noted, “recognize more than they regulate”, still leaning on single-modality detection and rarely closing the loop with real-time interventions [44]. Much of the underlying emotion-recognition work, moreover, still relies on static datasets and predefined viewing scenarios, which can underrepresent how academic affect fluctuates during live online study. There remains, in short, a lack of complete pipelines in which webcam-based academic-affect recognition drives

a closed-loop instructional policy: a reinforcement-learning tutor that selects among diverse pedagogical interventions rather than limiting adaptation to the difficulty of the next exercise alone, together with an evaluation design that isolates the contribution of the affective channel. This thesis is motivated by that gap.

1.3 Objectives

This thesis develops an adaptive intelligent tutoring system that couples real-time recognition of learner affect with a sequential decision-making policy that selects pedagogical interventions in response to the learner’s state. The objectives below state, for each component and for the integrated system, what the work set out to achieve and how its success is measured.

1. To build and evaluate a standalone multi-label academic facial expression recognition (FER) model on DAiSEE that infers the co-occurring affective states of boredom, engagement, confusion, and frustration from video of the learner [45, 31]. The model performs multi-label affective state inference with particular emphasis on the reliable detection of minority affective states under the severe class imbalance of the benchmark [28]. Recognition performance is evaluated under a single standardised evaluation protocol using per-label and aggregate F1, accuracy, precision, and recall, so that the reported results reflect model quality rather than differences in data preparation.
2. To formulate adaptive tutoring as a discrete-time Markov decision process (MDP) that represents the learner’s evolving cognitive and affective state and selects among a broad repertoire of pedagogical interventions rather than difficulty adjustment alone [14, 17]. The transition dynamics and the multi-objective reward are grounded in established cognitive and affective learning theory, with the reward jointly optimising cognitive learning gain and affective well-being. Success is assessed by whether the formulation yields a well-posed, theory-consistent decision problem capable of supporting policy learning.
3. To learn and evaluate a value-based reinforcement-learning tutoring policy that selects pedagogical interventions from the learner’s observed state so as to maximise the multi-objective reward [46]. The learned policy is evaluated in simulation against an expert rule-based tutor and a random baseline on normalised learning gain, episode success rate, cumulative reward, consistency with theory-derived emotion-conditioned interventions, and the diversity of its action selection [34, 44, 25].
4. To develop a unified closed-loop architecture integrating facial expression recognition and reinforcement learning, and to evaluate its affect-driven decision-making in relation to established educational and affective theories. [44].

1.4 Thesis Outline

The remainder of this thesis is organised as follows:

- **Background** reviews the theoretical foundations of emotion in learning, facial expression recognition, and adaptive learning methods.
- **Literature Review** surveys prior work on academic emotion detection, adaptive tutoring, and emotion-aware adaptation in intelligent tutoring systems.
- **Methodology** presents the design of the FER module, the reinforcement-learning adaptation module (including the Markov decision process formulation, reward function, and learned policies), and their integration into a closed-loop pipeline.
- **Results** reports the experimental setup and quantitative findings, first for the FER component in isolation and then for reinforcement-learning policy comparison against expert rule and random baselines.
- **Conclusion** interprets the findings in relation to the gap identified above and discusses limitations and ethical considerations.
- **Future Work** outlines directions for extending the FER module and the tutoring system beyond the scope of this thesis.

Chapter 2

Background

This chapter introduces the key concepts and theoretical foundations underpinning the thesis. It covers affect recognition in educational settings, deep-learning approaches for visual affect analysis, adaptive learning and reinforcement learning, and concludes with the stochastic modelling of temporal affective dynamics.

2.1 Emotion Recognition Formulation

Affective sensing formulation. Let $x \in \mathcal{X}$ denote an observable input signal (e.g. a facial image or short video clip) captured during a learning activity. Emotion recognition is formulated as a supervised mapping problem $f : \mathcal{X} \rightarrow \mathcal{Y}$ that predicts a latent affective state $y \in \mathcal{Y}$ from x [25, 47]. When the input is restricted to facial imagery or video, the task is specifically referred to as Facial Expression Recognition (FER) [25].

2.2 Deep Learning Foundations

2.2.1 Neural Networks and Supervised Learning

A neural network is a type of machine learning model that learns to map inputs to outputs by passing data through multiple layers of transformations. Each layer applies a linear operation followed by a nonlinear activation function, enabling the model to learn increasingly complex patterns from the data [48]. The basic structure of a neural network is illustrated in Figure 2.1.

In supervised learning, the model is trained using labeled examples (x_i, y_i) , where each input is paired with a corresponding target output. The objective is to learn model parameters that minimize a loss function over the training dataset. This optimization is commonly performed using stochastic gradient descent.

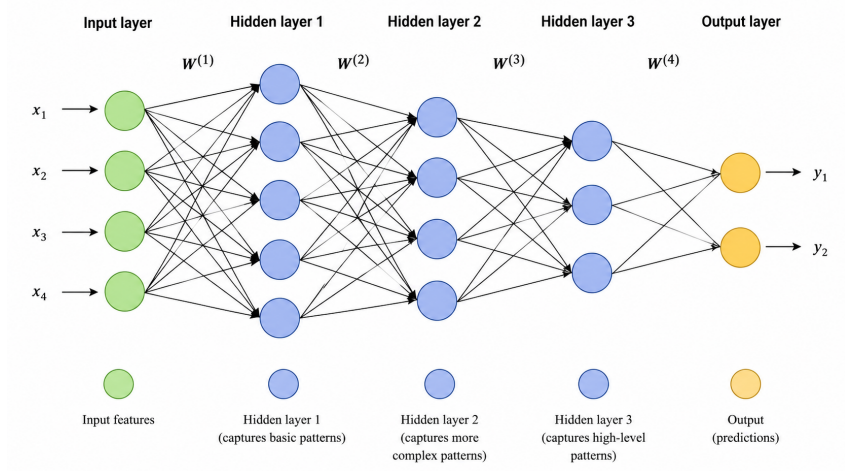


Figure 2.1: Illustration of a basic neural network architecture with input, hidden, and output layers.

Deep learning refers to neural networks with multiple hidden layers. The presence of depth enables hierarchical representation learning, where earlier layers capture low-level features and deeper layers encode more abstract and task-specific representations [48].

2.2.2 Activation Functions

Activation functions introduce nonlinearity into neural networks, allowing them to learn complex patterns beyond linear mappings. The most commonly used activation functions in this thesis are [49]:

$$\text{ReLU: } \sigma(z) = \max(0, z), \quad [1]$$

$$\text{Sigmoid: } \sigma(z) = \frac{1}{1 + e^{-z}}, \quad [2]$$

$$\text{Tanh: } \sigma(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}, \quad [3]$$

$$\text{Softmax: } \sigma(\mathbf{z})_i = \frac{e^{z_i}}{\sum_j e^{z_j}}. \quad [4]$$

ReLU is the standard activation for hidden layers due to its simplicity and efficiency [49]. Sigmoid is typically used for binary or multi-label outputs [49]. Tanh is commonly used in recurrent networks due to its zero-centered output [49]. Softmax is used in multi-class classification to produce a probability distribution over classes [49].

ReLU outputs zero for all negative inputs and grows linearly for positive ones, making it

fast to train and free of the vanishing gradient problem that affects bounded activations [49]. Sigmoid squashes any input to the range $(0, 1)$, which makes it suitable for independent binary predictions but susceptible to vanishing gradients at saturation [49]. Tanh maps inputs to $(-1, 1)$ and is zero-centred, which reduces the gradient update bias present in sigmoid and makes it the standard choice for recurrent hidden states [49]. Softmax normalises a vector of scores into a probability distribution summing to one, making it appropriate for mutually exclusive multi-class outputs rather than the independent binary predictions of sigmoid [49].

2.2.3 Convolutional Neural Networks

A Convolutional Neural Network (CNN) is a type of neural network designed for image data. Instead of using fully connected layers, CNNs apply convolution operations that slide small filters (kernels) over the input image to extract features such as edges, textures, and shapes.

Mathematically, a convolution layer computes feature maps by applying learned kernels across local regions of the input image, producing outputs that preserve spatial structure [27].

CNNs are widely used in computer vision because they rely on three key ideas: local connectivity (each neuron sees only a small region), weight sharing (the same filter is reused across the image), and translation equivariance, meaning that shifting the input leads to a corresponding shift in the output [27, 11].

A CNN is usually used as a feature extractor (backbone), where the network converts an image into a compact representation before classification. When pretrained on large datasets such as ImageNet, these models can be reused for new tasks through transfer learning, where the learned features are adapted to a smaller dataset [32, 27]. The convolutional neural network pipeline used for feature extraction is shown in Figure 2.2.

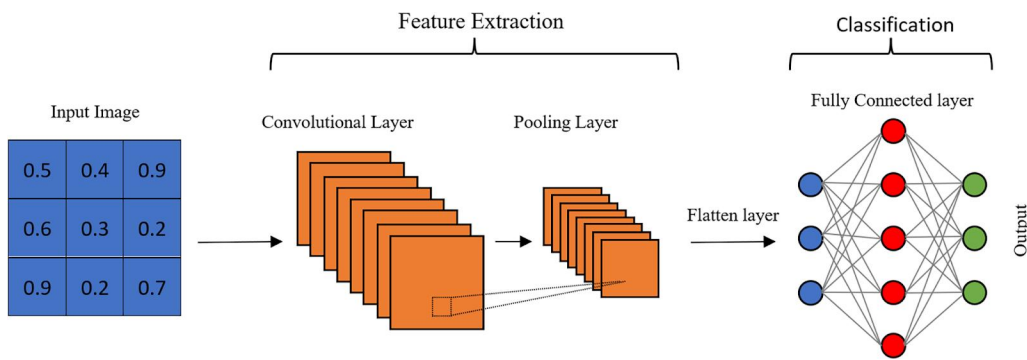


Figure 2.2: Basic architecture of a Convolutional Neural Network showing convolution, pooling, and classification stages. Adapted from [1].

2.2.4 Residual Learning and Major CNN Backbone Families

As convolutional neural networks become deeper, they can become more difficult to train. Adding additional layers does not always improve performance because gradients may become weaker as they propagate through the network [50]. This issue is known as the degradation problem.

To address this challenge, [50] introduced residual learning in the ResNet architecture. Instead of learning the desired mapping directly, each block learns a residual function that is added to the original input through a skip connection.

ResNet [50]. Residual Network (ResNet) is a widely used CNN architecture that introduces skip connections between layers. These connections allow information and gradients to flow more easily through the network, making it possible to train much deeper models than traditional CNNs.

Several versions of ResNet have been proposed, including ResNet-18, ResNet-34, ResNet-50, ResNet-101, and ResNet-152, which differ mainly in network depth. Among them, ResNet-50 is one of the most popular models because it provides a good balance between accuracy and computational cost. Due to its strong feature extraction capabilities and availability of pre-trained weights, ResNet-50 is widely used as a backbone network in facial expression recognition and other computer vision tasks [50].

EfficientNet [51, 52]. EfficientNet is a family of convolutional neural networks designed to achieve high accuracy while using fewer parameters and computational resources than many traditional CNNs. The architecture scales network depth, width, and input resolution in a balanced manner, allowing larger models to improve performance efficiently. EfficientNetV2 further improves training speed and computational efficiency while maintaining strong accuracy. Due to these advantages, EfficientNet models are widely used in computer vision tasks, including facial expression recognition [51, 52].

ShuffleNetV2 [53]. ShuffleNetV2 is a lightweight CNN architecture designed for fast and efficient inference on devices with limited computational resources. It reduces computational cost while maintaining competitive accuracy through an efficient network design. As a result, ShuffleNetV2 is commonly used in real-time and mobile vision applications where speed and efficiency are important [53].

2.2.5 Recurrent Neural Networks and LSTM

Recurrent Neural Networks (RNNs) are designed to process sequential data such as time series or text. Unlike feedforward networks, RNNs maintain a hidden state that is updated at each

time step, allowing the model to capture temporal dependencies in a sequence [54].

However, standard RNNs struggle to learn long-term dependencies due to vanishing and exploding gradients, which makes training unstable on long sequences [54].

To address this issue, Long Short-Term Memory (LSTM) networks were introduced. LSTMs improve RNNs by adding a memory cell that can store information over long time periods. This memory is controlled by gates that decide what information to keep, forget, or output, allowing the model to selectively retain important information while discarding irrelevant details [54].

A Bidirectional LSTM (BiLSTM) further improves this idea by processing the sequence in both forward and backward directions. This allows the model to use both past and future context when representing each time step [29, 30]. The BiLSTM architecture, illustrating the forward and backward state processing across time steps, is shown in Figure 2.3.

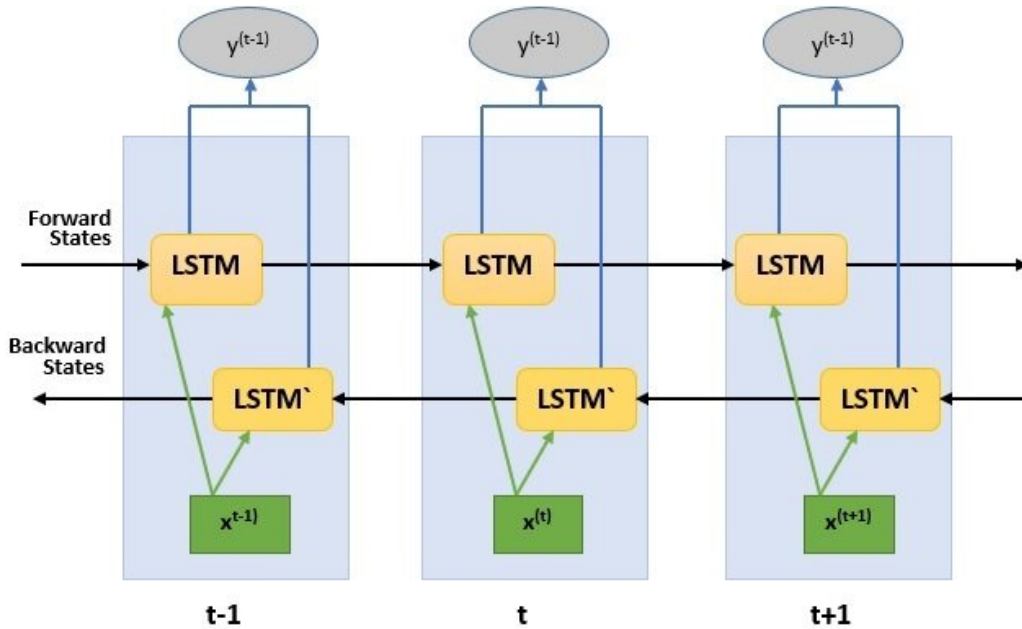


Figure 2.3: Bidirectional LSTM architecture showing forward and backward hidden state propagation across time steps (adapted from [2]).

2.2.6 Training Regularisation and Stability

Dropout [55]. Dropout is a technique used to reduce overfitting in neural networks. During training, it randomly disables some neurons so the model does not rely too much on specific features. During inference, all neurons are used with scaled outputs to maintain consistency [55].

Batch Normalisation [56]. Batch Normalisation improves training stability by normalising layer inputs. This helps the network train faster, reduces sensitivity to initialization, and allows

higher learning rates [56].

Gradient Clipping. Gradient clipping prevents unstable training by limiting very large gradients during backpropagation. This is especially useful in recurrent models, where gradients can explode over long sequences.

2.2.7 Adaptive Gradient Optimisers: Adam and AdamW

Adam is an adaptive optimisation algorithm that adjusts the learning rate for each parameter using running averages of past gradients [57]. This makes training faster and more stable compared to standard SGD.

AdamW improves Adam by separating weight decay from the gradient update. This leads to more effective regularisation and better generalisation performance [58].

2.2.8 Supporting Architectures and Classical Methods

DenseNet [59]. DenseNet connects each layer to all later layers, improving feature reuse and gradient flow while using fewer parameters.

3D DenseNet [31]. 3D DenseNet extends DenseNet to video data using 3D convolutions for spatial-temporal learning.

Xception [60]. Xception uses depthwise separable convolutions to reduce computation while keeping strong feature extraction ability.

VGG-19 [61]. VGG-19 is a simple deep CNN made of stacked 3×3 convolutions. It is mainly used as an early transfer learning backbone.

GRU and Bi-GRU [62]. GRU is a simpler version of LSTM with fewer gates. Bi-GRU processes sequences in both directions to capture past and future context.

C3D [63]. C3D uses 3D convolutions to learn spatial and temporal features directly from video clips.

LRCN [64]. LRCN combines CNNs for frame feature extraction and LSTMs for modeling temporal sequences.

TCN [65]. TCN replaces recurrent networks with dilated convolutions for efficient sequence modeling.

CBAM [66]. CBAM is an attention module that improves CNN features by focusing on important channels and spatial regions.

HOG [67]. HOG is a hand-crafted feature descriptor based on image gradients, used in classical vision tasks.

LBP [68]. LBP describes local texture patterns by comparing pixel intensities with neighbors.

SVM [69]. SVM is a classical classifier that finds a decision boundary with maximum margin, often used with HOG or LBP features.

2.3 Video Representation Learning

A video is a sequence of frames represented as a tensor. Unlike images, video data contains both spatial information (what appears in each frame) and temporal information (how content changes over time) [30, 31, 70].

Frame sampling. Processing all frames in a video is usually unnecessary and computationally expensive. Therefore, a fixed number of frames is selected from each video. This reduces redundancy and makes training more efficient, but may also lose some temporal details [70, 30].

Spatio-temporal model approaches. There are three common ways to model video data:

1. **CNN + RNN:** A CNN extracts features from each frame, and an LSTM or BiLSTM processes the sequence over time.
2. **3D CNN:** A single model learns both spatial and temporal features directly using 3D convolutions.
3. **Transformer-based models:** Frames are treated as tokens and related using self-attention to capture temporal dependencies.

These approaches are different ways of combining spatial and temporal information, and the choice depends on the application and computational constraints.

2.4 Attention Mechanisms

Attention mechanisms allow a model to focus on the most relevant parts of an input instead of treating all features equally. They assign different importance weights to different elements and combine them into a weighted representation [71].

2.4.1 Self-Attention and Multi-Head Attention

Self-attention is a mechanism where each element in a sequence compares itself with all other elements to determine which parts are most relevant. This allows the model to capture long-range dependencies more effectively than traditional sequential models [71].

In self-attention, each input element is transformed into queries, keys, and values. The similarity between queries and keys is used to compute attention weights, which are then applied to the values to produce the final output [71].

Multi-head attention [71]. Multi-head attention improves self-attention by using multiple attention “heads” in parallel. Each head learns different relationships in the data, and the results are combined to form a richer representation.

A Transformer layer is built by stacking multi-head attention with a feed-forward network and residual connections. This architecture allows efficient modeling of relationships across entire sequences [71]. Vision Transformers extend this idea to images by treating image patches as sequence tokens [72].

2.4.2 Channel and Spatial Attention

For a CNN feature map $F \in \mathbb{R}^{H \times W \times C}$, attention can be applied in different ways to improve feature representation.

Channel attention. Channel attention learns the importance of each feature channel and reweights them accordingly. The Squeeze-and-Excitation (SE) block is a common approach, where global information from each channel is first aggregated and then passed through a small neural network to generate channel-wise importance scores. These scores are used to rescale the feature channels, allowing the model to focus on the most useful features [73].

Spatial attention. Spatial attention focuses on *where* important information appears in the feature map. It learns a spatial mask that highlights informative regions while suppressing less relevant areas, improving the model’s ability to localise important visual cues [27].

2.4.3 Temporal Attention

Temporal attention is used to model how important different frames are in a video. Instead of treating all frames equally, it assigns higher weights to more informative frames.

The final video representation is computed as a weighted combination of frame features, where the weights reflect the importance of each frame. This allows the model to focus on key moments where emotional or semantic information is more evident [31, 72].

Temporal attention is especially useful when relevant information appears only in a few frames rather than uniformly across the entire video.

2.5 Intelligent Tutoring Systems and Knowledge Tracing

2.5.1 Intelligent Tutoring Systems

An Intelligent Tutoring System (ITS) is an AI-based learning system that adapts instruction to each learner. It typically considers three main parts: the learner model (student state), the domain model (learning content), and the pedagogical model (how the system teaches) [21, 20].

By continuously observing the learner and updating its internal state, an ITS can adjust feedback, difficulty, and learning content in a dynamic way.

2.5.2 Knowledge Tracing

Knowledge Tracing (KT) refers to the task of estimating a learner's knowledge over time based on their past interactions [23, 19]. It uses sequences of student responses to predict future performance and track learning progress.

Traditional approaches model knowledge as a hidden state that evolves over time, while modern deep learning approaches use recurrent neural networks to learn this progression directly from data (Deep Knowledge Tracing).

KT focuses on predicting student performance, but does not decide what action to take next. That role is handled by a separate decision-making module in intelligent tutoring systems.

2.6 Reinforcement Learning

Reinforcement Learning (RL) is a framework for sequential decision-making, where an agent learns to choose actions based on feedback from the environment [74]. The goal is to learn a policy that maximises long-term reward through trial and error interaction.

2.6.1 Markov Decision Processes

A Markov Decision Process (MDP) is a framework used to model sequential decision-making problems. It consists of states $\mathcal{S}$, actions $\mathcal{A}$, a transition function P , a reward function R , and a discount factor γ [74].

The key assumption is the *Markov property*, which states that the next state depends only on the current state and action, not on past history.

A policy π defines how an agent selects actions based on the current state. The goal of reinforcement learning is to learn a policy that maximises the expected long-term reward.

In simple terms, an MDP describes how an agent interacts with an environment step by step in order to maximise cumulative reward.

2.6.2 Partially Observable Markov Decision Processes

In many real-world problems, the agent cannot directly observe the full environment state. Instead, it receives partial or noisy observations. This setting is modeled using a Partially Observable Markov Decision Process (POMDP) [75].

A POMDP extends an MDP by introducing observations, meaning the agent must make decisions based on incomplete information.

Since the true state is not fully visible, the agent relies on past observations or an internal memory to estimate the current situation.

In practice, deep reinforcement learning models handle this limitation using recurrent networks or by summarising recent observations.

2.6.3 Model-Based vs. Model-Free Reinforcement Learning

Reinforcement learning methods are generally divided into two categories: model-based and model-free approaches.

Model-based methods use an explicit model of the environment to predict how states change and to plan future actions. In contrast, model-free methods learn directly from experience without modeling the environment dynamics [74].

Model-free approaches are more commonly used in complex real-world problems where building an accurate environment model is difficult or impractical.

2.6.4 Exploration and Exploitation

A key challenge in reinforcement learning is balancing exploration and exploitation.

Exploration means trying new actions to discover better strategies, while *exploitation* means choosing the best-known action based on current knowledge [74].

A common strategy is the ε -greedy policy, where the agent mostly chooses the best-known action but occasionally selects a random action to explore the environment.

2.6.5 Q-Learning

Q-learning is a reinforcement learning method that learns how good an action is in a given state without requiring a model of the environment [74].

It learns a function $Q(s, a)$ that estimates the long-term reward of taking action a in state s . The update rule improves this estimate using the reward received and the best estimated future action.

Q-learning is called *off-policy* because it learns from the best possible action rather than the action actually taken. It works well in small, discrete environments but does not scale to high-dimensional problems. The mapping from state representations to action-value outputs in Q-learning is illustrated in Figure 2.4.

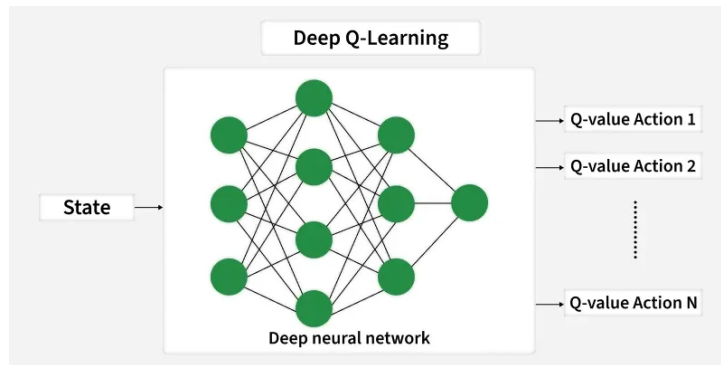


Figure 2.4: Mapping from input state to action-value outputs $Q(s, a)$, where each action receives a separate Q-score used for decision making.

2.6.6 Deep Q-Networks

Deep Q-Networks (DQN) extend Q-learning by using a neural network to approximate the Q-function instead of a table [46]. This allows the method to handle high-dimensional inputs such as images.

Two key ideas make DQN stable:

- **Experience replay:** past experiences are stored and randomly sampled during training to reduce correlation between updates.
- **Target network:** a separate, slowly updated network is used to compute learning targets, which stabilises training.

DQN is designed for discrete action spaces, where the best action is selected by comparing all possible actions.

2.7 Simulation-Based Evaluation in Reinforcement Learning

The reinforcement learning interaction loop between agent and environment is shown in Figure 2.5.

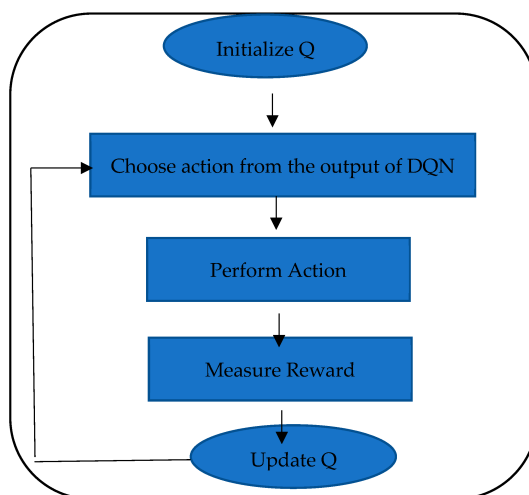


Figure 2.5: Reinforcement learning interaction loop showing the sequence of state observation, action selection, reward feedback, and environment transition.

Reinforcement learning methods are typically trained and evaluated in simulated environments due to the large number of interactions required for learning [74].

Simulation allows safe and scalable training before deploying models in real-world or human-centered systems such as intelligent tutoring systems.

However, results obtained in simulation may not fully transfer to real environments due to differences between the simulated and real dynamics, known as the sim-to-real gap [17].

Therefore, simulation results are best interpreted as controlled approximations of real-world behaviour.

The complete Deep Q-Network training pipeline, including experience replay and target network updates, is shown in Figure 2.6.

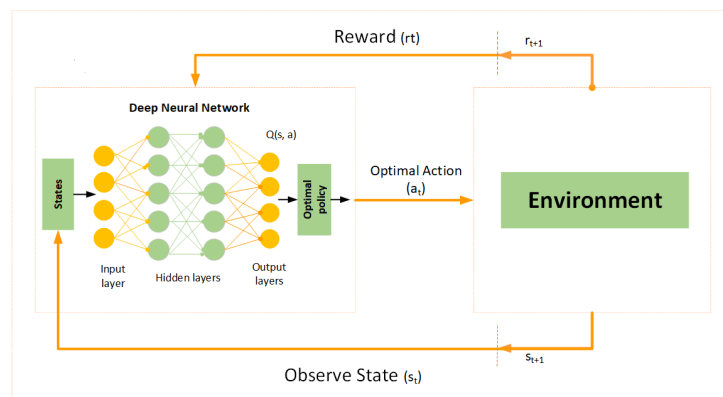


Figure 2.6: Deep Q-Network (DQN) training cycle including experience replay, target network updates, and Q-value optimization.

Chapter 3

Literature Review

3.1 Emotion in Learning

Early work in educational psychology treated emotion in achievement settings mainly through the lens of test anxiety. Foundational studies linked anxiety to interference with learning and performance in evaluative situations [76], and later reviews documented substantial correlational and experimental evidence on how anxiety relates to cognition and academic performance [77, 78]. Even within that limited focus, researchers were already trying to understand achievement-related emotions through the way students explain success and failure [79]. What remained comparatively open, however, was whether anxiety was the only emotion that mattered for how students actually learn in classrooms and during study.

By the early 2000s, that gap was addressed directly. Qualitative and questionnaire-based work showed that students report a wide range of emotions tied to instruction, studying, and testing, not only fear around exams [80]. The same line of work introduced the term *academic emotions* for this wider group and classified them using two simple dimensions: whether the emotion is positive or negative, and whether it increases or reduces a student's level of activity. Emotions were understood to influence learning and performance indirectly through motivation, learning strategies, mental resources, and self-regulation, while students' results and feedback could in turn influence how they feel in future learning situations [80]. In other words, researchers increasingly came to understand that learning is affected not only by one negative emotion, but by a broader set of emotional states that may appear during the same lesson and shape the way students pay attention, stay involved, participate, and make progress.

A parallel line of research developed around the concept of engagement rather than focusing only on specific emotion labels, although it still included a clear emotional dimension. In this view, engagement was not treated as a single construct, but as a combination of three related aspects: behavioural participation in learning activities, emotional reactions to school and

instruction, and cognitive effort directed towards understanding difficult material [81]. Emotional engagement in that sense covers interest, boredom, anxiety, and related feelings towards school and academic work, and it is treated as part of why students are willing to keep trying when tasks are hard [81]. Fredricks and colleagues connected these dimensions of engagement to long-standing motivational theories which suggest that students are more likely to remain involved in learning when they feel competent, have a sense of control over their learning, and feel socially connected to others [82]. They also linked them to expectancy-value perspectives, which suggest that students' decisions to engage are shaped by how interesting they find a task and how important or useful they believe it to be [83]. Engagement therefore became an additional way of understanding affect in education. It did not replace the study of specific emotions, but worked alongside it by offering a broader view of students' emotional involvement in learning.

From the mid-2000s onward, these educational perspectives were strengthened by more explicit theories explaining how emotions arise in learning situations, as well as by neuroscientific work on why affect matters for learning in the first place. One important example is Pekrun's control-value theory of achievement emotions, which argues that students' emotions depend largely on how much control they believe they have over academic tasks and how much value they attach to those tasks. According to this view, different combinations of control and value can give rise to emotions such as enjoyment, hope, anxiety, boredom, and shame, and these emotions can then influence students' interest, motivation, engagement, and academic achievement [7]. Independently, Immordino-Yang and Damasio argued that higher-level learning in school is not separate from emotion. They described emotional thought as the connection between feelings and the attention, memory, and reasoning processes that education tries to strengthen, and explained that emotion helps make experiences meaningful, which allows knowledge to be remembered and later used in real decisions [4]. This argument builds on earlier work which suggested that emotion is an essential part of rational behaviour and decision-making, rather than something that simply distracts from clear thinking [84]. Together, the appraisal perspective and the neuroscience perspective support the same general idea: emotion is not just an extra layer added to learning, but part of the process itself. In this way, these lines of work helped move educational research beyond a purely cognitive view of learning towards one that also recognises the role of motivation and affect.

Research on dialogue-based tutoring then helped explain more clearly how emotions appear during learning, especially when students interact with a tutor while solving problems, asking questions, and receiving feedback. Studies of long tutoring conversations showed that deep learning is often accompanied by a range of emotions, especially boredom, engagement, confusion, and frustration, which appear more often than the traditional basic emotion categories would suggest [6]. These four states later became the *de facto* target set in webcam-based educational facial expression recognition benchmarks such as DAiSEE, which the following sections review. Experimental studies showed that a moderate level of confusion can some-

times improve learning outcomes, partly because it increases attention and encourages deeper processing of the material [85]. A later review by Tyng and colleagues connected these findings from classrooms with brain imaging studies that showed how emotion affects thinking and learning [5]. Information with emotional meaning usually attracts attention more quickly and is given more focus than neutral information [86, 87]. At the brain level, areas such as the amygdala, medial temporal lobe, and prefrontal cortex help explain how emotional arousal influences the formation and strengthening of memories, which is why emotional events are often remembered more clearly than neutral ones [5, 88]. The same review stresses that stress and arousal can help or hurt learning depending on intensity and timing, and that motivational states such as curiosity support exploration and readiness to learn.

More recent review studies still show that emotion is a central part of how students experience learning, especially in difficult subjects and in online settings where students often receive less informal support. A systematic review of emotions in STEM online learning found that emotions strongly influence engagement, motivation, and satisfaction, and that teaching needs a better understanding of students' emotional experiences in order to respond to what they are actually facing [10]. This conclusion builds on a long tradition of educational psychology research that had already treated emotion as an important topic to understand and measure in schools [89]. Once emotion is treated as part of how students attend, remember, and stay engaged, the practical question becomes how to tell what a learner is feeling while they are actually working. The next section therefore turns from the conceptual case for emotion in learning to the ways researchers have tried to observe or measure it in educational settings.

3.2 Emotion Detection in Educational Systems

Once emotions were established as central to learning, motivation, and achievement, the field faced a more operational question: how can those emotions be detected while learning is actually taking place. Early work in affective computing had already described emotion detection as a measurement problem that could use several different signals rather than only one method [90], and educational researchers gradually translated that idea into classroom and tutoring contexts [91]. What came next was not one simple shift from one method to another, but several overlapping approaches, each with different strengths and limitations in terms of timing, scalability, and practical use in real educational settings.

At first, most educational studies relied on self-report. The achievement-emotions line of work around the Achievement Emotions Questionnaire (AEQ) gave researchers a structured way to measure recurring states such as enjoyment, pride, anxiety, anger, shame, and boredom in relation to classroom activity and assessment [7, 80]. The same type of questionnaires is still used in recent classroom AI studies. For example, Omar and colleagues used Achievement Emotions Questionnaire–Mathematics (AEQ-M) surveys before and after the in-

tervention, along with interviews, to track emotional changes in grade 8 algebra classes using an AI-supported learning environment [92]. Studies in distance learning also refined self-report methods instead of abandoning them. For example, some studies compared daily questionnaires completed after learning activities with emoticon tools used during the course, where students could mark moments when they felt certain emotions. These methods had clear advantages, including low instrumentation demands, direct access to students' own experiences, and easy deployment across different settings. However, they also had persistent limitations, such as dependence on students' self-awareness, delayed reporting, sensitivity to wording and social context, and difficulty capturing rapid emotional changes in real time [93].

Alongside these questionnaire-based approaches, researchers also used direct human observation to understand students' affect and engagement from their visible behaviour in real classrooms. In school psychology, structured coding systems such as Academic Engaged Time Code of the Systematic Screening for Behavior Disorders (AET-SSBD), Behavioral Observation of Students in Schools (BOSS), Direct Observation Form (DOF), Classroom Observation Code (COC), State-Event Classroom Observation System (SECOS), Attention-Deficit Hyperactivity Disorder School Observation Code (ADHD-SOC), and Student Observation System (SOS) were used to observe students at set time intervals and record whether they were focused, distracted, or showing signs of difficulty [94, 95]. Related work in learning technologies also used a similar observation-based approach in educational software and classroom interactions [96, 97]. Compared with questionnaires completed after the event, observation methods gave more detailed information about when emotions and behaviours appeared and linked interpretations more closely to what students were actually doing in the learning setting. However, these methods were demanding because they required observer training, checks for consistency between observers, repeated observations over time, and careful control of problems such as observer drift and reactivity, while still showing only small parts of students' experience during the full course [94].

As digital tutoring systems collected detailed data about what students did, researchers began to move from manual coding to large-scale behavioural inference. Instead of asking students about their feelings or observing them all the time, they used records such as answer correctness, hint use, timing, and help-seeking patterns to infer affect and disengagement [98, 99]. Pardos and colleagues are a good example of this shift. They built detectors in ASSISTments web-based tutoring platform (ASSISTments) that did not need sensors and were used to identify states such as boredom, engaged concentration, confusion, frustration, off-task behaviour, and gaming the system. They then followed these affective patterns over a full year and examined how they were related to students' end-of-year mathematics test results [100]. Studies in technology-rich classrooms also used coded behaviour data, sequence analysis, and Hidden Markov Models to study how students' emotions and behaviours changed over time [97]. These approaches made it easier to study many students and to follow them over time, but they still estimated emotion indirectly from behaviour. This is a limitation because the same behaviour

can reflect different emotions, the models depend on labelled training data, and their performance may vary across platforms, tasks, and teaching contexts [100]. The same behavioural traces later became the default input to many adaptive and reinforcement-learning tutors, even when facial or other perceptual signals were available [21, 24].

Dialogue-based tutoring introduced another approach by using language and conversation as signals of emotion. Research on tutoring dialogue showed that emotional states could, in principle, be inferred from what learners and tutors said, even when explicit emotion words were not used [101]. In typed AutoTutor intelligent tutoring system (AutoTutor) dialogues, D’Mello and Graesser demonstrated that discourse-sensitive features derived from Linguistic Inquiry and Word Count (LIWC) and Cohesion Metrics (Coh-Metrix) could predict distributions of boredom, confusion, flow/engagement, and frustration over whole sessions [6, 102]. Those results showed that language carries affective information, but only under conditions that are difficult to reproduce in ordinary online study. The models depend on extended tutoring conversation, domain-specific vocabulary, and session-level transcript summaries rather than on the short, ambiguous messages that dominate most digital exercises. A student who types “I don’t get it”, “ok”, or nothing at all may be confused, bored, or simply unwilling to write; the same string of words can therefore support more than one emotional reading, which makes text an inherently ambiguous channel for real-time adaptation [6, 93]. Recent chat-based and large-language-model tutors inherit the same problem: they can process what the learner chooses to write, but they cannot see the frustration or disengagement that appears on the face when the learner stops typing [37, 24]. Surveys of sentiment analysis in STEM online learning report the same weakness at scale: text-based methods struggle with sparse emotional language, context dependence, and the gap between what students write and what they actually feel during a task [10, 47].

Facial expression is a stronger and more direct alternative for educational affect monitoring. Comparative work during classroom use of an educational game found that face-only affect detectors were more accurate than detectors built from interaction logs alone, even though facial models could not be applied in every student case [103]. That pattern is consistent with a wider lesson from the field: visible affect on the face is closer to the emotional event than inferred labels derived from clicks, timings, or typed responses. Teachers already rely on this signal in classrooms; once webcams became standard in online learning, the same cue could be captured continuously without asking students to describe their feelings in words. Even early affective tutoring systems that experimented with text semantics treated facial expression as a necessary companion rather than an optional extra, because text alone did not provide a stable enough reading of the user’s state [104]. For adaptive systems that must respond while a student is still working, facial behaviour therefore offers a clearer, more continuous, and less ambiguous basis for detecting academic emotions such as confusion, boredom, frustration, and engagement than dialogue or log data alone [25, 41, 105]. The next section reviews how educational facial expression recognition has developed around that advantage.

3.3 Facial Expression Recognition in Education

Among the signals reviewed in the previous section, one stood out as visible and easy to capture during everyday learning. A learner's face is what teachers already read in the classroom, and an ordinary webcam is enough to record it during online sessions. This made Facial Expression Recognition (FER) an attractive option once deep learning became stable enough to be used outside the laboratory. Over the last decade, a dedicated line of educational FER work has grown around engagement, attention, boredom, confusion, frustration, and related academic emotions, and a systematic review by Lek and Teo confirms that this line is now the dominant computer-vision approach to emotion monitoring in digital learning environments [25]. More recent syntheses also describe facial behaviour as one of the most practical signals for studying affect in classrooms and online courses [41, 47].

A central piece of this development was the introduction of a public benchmark for in-the-wild educational affect. Gupta and colleagues released DAiSEE, a dataset of 9,068 short webcam clips collected from 112 students, where each clip is labelled with four affective states, namely engagement, boredom, confusion and frustration, and each state is rated on four ordinal levels [45]. Their benchmark also reported the first deep-learning baselines on the dataset, using InceptionNet at frame and video level, a 3D convolutional network (C3D), and a long-term recurrent convolutional network (LRCN). Top-1 accuracy was clearly higher for confusion and frustration than for engagement and boredom, with LRCN reaching 53.7% on boredom, 57.9% on engagement, 72.3% on confusion and 73.5% on frustration, while frame- and video-level InceptionNet stayed below 50% on boredom and engagement [45]. Two consequences shaped almost everything that came after. First, DAiSEE became the reference benchmark for academic FER [25]. Second, the dataset is genuinely difficult, especially for engagement and boredom, and is heavily class-imbalanced, which made temporal modelling, attention and imbalance-aware training the natural next steps. It is also worth noting that within the searched sources, no pure traditional machine-learning paper using DAiSEE as the main method was found; classical components such as facial landmarks, action units or handcrafted features only appear inside hybrid pipelines combined with deep networks.

Early educational FER work in the same period still kept one foot in classical computer vision. Rao and colleagues used a hybrid pipeline that merged Dlib landmark vectors with CNN features and a support vector machine to classify the four DAiSEE states, reporting 53.42% overall accuracy on DAiSEE alongside additional evaluation on JAFFE and CK+ [33]. Around the same time, Leong compared FaceNet embeddings and landmark-distance streams, each followed by an LSTM, to detect boredom and frustration on DAiSEE [29]. These two studies showed that educational FER had already moved away from generic basic-emotion labels toward states that matter for learning, but they also showed that single-frame or shallow-temporal architectures struggled with the hard DAiSEE classes.

A clearer step forward came in 2021 with the work of Abedi and Khan, who combined a

2D ResNet-18 with a dilated temporal convolutional network (TCN) and trained it end-to-end on raw video frames. They followed the DAiSEE-defined train, validation and test counts but merged training and validation for fitting, and reported 63.9% four-class engagement accuracy on the test set, improving over a ResNet+LSTM baseline at 61.15% [70]. Their study also made class imbalance an explicit design issue by using weighted cross-entropy and a balanced-batch sampling strategy. The same temporal direction was followed in educational FER more broadly: Asaju and Vadapalli stacked a ResNet-152 with a BiLSTM and mapped the four DAiSEE states onto a simpler positive–neutral–negative pedagogical scheme, including a small live validation with real students [30]. Together, these works confirmed that short video clips, rather than isolated frames, were a better unit of analysis for academic affect.

In 2022 the field moved toward attention and transformer-style processing of video. Mehta and colleagues introduced a 3D DenseNet built with spatial, temporal and hybrid non-local self-attention blocks and trained it with class-balanced focal and class-balanced MSE losses [31]. They reported 63.59% four-class engagement accuracy and 94.78% binary-engagement accuracy on DAiSEE, where the binary task is obtained by merging the two lowest engagement levels into “low” and the two highest into “high”, and is therefore not directly comparable to the four-level results. They also reported a regression MSE of 0.0347 on DAiSEE engagement and 0.0877 on EmotiW-EP. In the same year, Mandia et al. proposed a Vision Transformer with multi-head self-attention for automatic student engagement estimation on DAiSEE, using imbalance-aware losses and reporting an overall accuracy of about 55.18%, which the authors describe as outperforming Gupta et al.’s frame-level and video-level baselines by roughly 8–8.78 percentage points [106]. The full PDF for this paper was not accessible, so its method details are limited to the available metadata and abstract.

By 2023, two related directions became visible. The first was the hybrid deep-classical pipeline, well represented by Santoni et al., who combined OpenFace-derived facial landmarks, action units, head-pose and gaze features, dimensionality reduction with PCA or SVD, over-sampling with SMOTE, and a CNN classifier [107]. They reported a best maximum accuracy of 77.97% with the SVD+CNN configuration on a heavily resampled subset of DAiSEE rather than on the original benchmark split, which means the number is not directly comparable with results that follow the standard DAiSEE protocol. This work is important precisely because it shows that DAiSEE research is not only about replacing classical methods with deep networks: it is also about combining deep features with classical preprocessing, feature engineering and imbalance handling. A complementary 2023 study, although not on DAiSEE, illustrates the same hybrid trend in general FER: Mukhiddinov et al. used MediaPipe landmarks together with HOG features around those landmarks as input to a custom CNN with batch normalisation and Mish activation, reporting 69.3% accuracy across seven emotion classes on AffectNet under masked-face conditions [108]. Both studies confirm that handcrafted facial cues remain useful when they are integrated as front-ends to deep models rather than used alone.

The 2024 work continued in three directions. Shiri et al. replaced earlier CNN backbones with EfficientNetV2-L and combined it with GRU, Bi-GRU, LSTM or Bi-LSTM heads, reporting their best four-class engagement accuracy of 62.11% on DAiSEE with the LSTM variant [109]; the paper is sometimes cited under a co-author's name but the first author is Shiri. Tian et al. proposed a sequential ensemble model evaluated on both DAiSEE and EmotiW-EP, reporting an MSE of 0.0386 on DAiSEE and 0.0610 on EmotiW-EP, with the available abstract claiming stronger performance than prior approaches [110]; the full PDF was not accessible, so this entry should be treated as limited verification. Su and colleagues then pushed the field further into the transformer era with a hybrid architecture, called DTransformer, that combines a part-and-sensitive attention network on ResNet-18 patches with a 3D-CNN tube tokenizer feeding a multi-head self-attention transformer, fusing both branches for prediction [72]. They reported 64.0% four-class engagement accuracy on DAiSEE and a regression MSE of 0.0729 on EmotiW-EP, which makes their work one of the strongest current transformer-based engagement detectors.

Two recent 2025 studies show that hybrid deep-classical thinking is still alive. Rianto et al. took a graph-based view: they extracted 68 facial landmarks per frame, built a Delaunay triangulation graph, fed it to a graph convolutional network, and combined it with a 1D CNN over 52 action-unit values, training the bimodal model with the LDAM imbalance-aware loss [111]. They followed the DAiSEE engagement-only split and reported a best accuracy of 52.24% with macro-F1 of 0.49, while honestly noting that minority engagement levels remained largely mispredicted. In general FER, Türkeş and Aydın combined a pre-trained ResNet-50 feature extractor with a Support Vector Machine classifier [112]. The DAiSEE branch is therefore now bracketed by hybrid graph-and-CNN methods on one side, and by classical hybrid CNN+SVM pipelines on standard FER datasets on the other.

Educational FER has also matured in its target applications. Komaravalli and Janet fine-tuned an Xception-based transfer-learning model on a rebalanced DAiSEE subset and cross-tested it on OL-SFED [32]. Gupta and colleagues proposed EDFA, an ensemble that fuses VGG19 and ResNet50 outputs into a rule-based attention index for real-time classroom monitoring, deployed as a web application [11]. Aly combined a ResNet-50 backbone with the Convolutional Block Attention Module and a Temporal Convolutional Network for online student monitoring, reporting throughput of about 28 frames per second on a GPU, 12 on a CPU, and 15 on a Jetson Nano [27], and a follow-up study added a 3D CNN branch with an optimised fusion layer [28]. Tulsani and colleagues used a hybrid of MTCNN face detection, a Vision Transformer and a BiLSTM inside a learning-management system, with a near-real-time latency of about 0.8 seconds per frame sequence [113]. Liu and colleagues proposed an adaptive global attention block together with a differential temporal transformer, evaluated on DAiSEE, DFEW and FERV39k, and reported about 11.7 milliseconds per GPU inference [26]. Woolf and colleagues described the Face Readers line of work inside the MathSpring intelligent tutoring system, where attention, head pose and facial cues are used to predict problem-solving

outcomes and to trigger real-time interventions [35]. Together, these examples show that educational FER is no longer only about classification accuracy; it is slowly being connected to the feedback loops of adaptive learning systems.

Taken together, the DAiSEE evidence base evolved through deep-learning baselines, CNN-plus-temporal models, 3D and self-attention architectures, hybrid deep-classical pipelines, EfficientNet plus RNN systems, transformer-based detectors and ensemble models, and was complemented by hybrid CNN-and-classical FER work on standard expression datasets. Despite this progress, several limits remain. Datasets for academic emotions are still small and imbalanced; results on the four-level engagement task remain in the low-to-mid sixties even for strong transformer or attention models, and many studies do not use the official DAiSEE split or do not report per-class metrics, which makes head-to-head comparison difficult [70, 107, 111, 109, 72]. Generalisation across cultures and recording conditions remains weak, and visually similar states such as confusion and frustration are still confused [25, 47, 41]. More importantly for this thesis, almost all of the studies above stop at engagement classification, regression or visualisation; very few connect the FER output to a concrete next-task selection or adaptation policy, and live deployment in real classrooms is reported only in a handful of cases. Despite these concerns, facial behaviour remains one of the few signals that can be captured continuously, without attaching sensors to students, without interrupting the lesson, and with hardware that most classrooms and online learners already have. This practical profile is one of the reasons why FER, on its own, continues to be central to recent educational-affect research, and why a webcam-only FER pipeline that is explicitly connected to an adaptive next-task selector still defines an open and useful design point. Table 3.1 summarises representative DAiSEE-based FER studies for benchmark comparison. The table spans two landscape pages: the first lists each study, dataset, and task; the second gives the full architecture description and evaluation metrics for the same studies.

Table 3.1: Comparison of representative DAiSEE-based FER studies

Study	Dataset	Task
Gupta et al. (2016) [45]	DAiSEE	4 academic states, 4 ordinal levels
Abedi and Khan (2021) [70]	DAiSEE	4-class engagement classification
Mehta et al. (2022) [31]	DAiSEE, EmotiW-EP	4 academic states + engagement regression
Su, He, Luo (2024) [72]	DAiSEE, EmotiW-EP	4-class engagement (DAiSEE); regression (EmotiW-EP)
Liu et al. (2026) [26]	DAiSEE, DFEW, FERV39k	4 academic states (in-the-wild video FER)
Rianto et al. (2025) [111]	DAiSEE	4-class engagement; imbalance-focused

Table 3.1 (continued): architectural details and evaluation metrics.

Study	Architecture	Evaluation Metric(s)
Gupta et al. (2016)	Backbone: InceptionNet (frame/video), C3D (3D CNN), LRCN (CNN encoder). Temporal: 3D convolution (C3D) or CNN+LSTM (LRCN). Attention: none. Classification: per-state ordinal top-1 (4 labels).	Per-class top-1 acc. (LRCN): B 53.7%, E 57.9%, C 72.3%, F 73.5%
Abedi and Khan (2021)	Backbone: ResNet-18 (ImageNet-pretrained). Temporal: dilated TCN over frame features. Attention: none. Classification: 4-class softmax; weighted cross-entropy and balanced-batch sampling.	63.9% 4-class acc.

Note. Each architecture entry summarises the full pipeline as reported by the original study. Metrics are reported as published; four-class engagement accuracy, per-class accuracy, and regression MSE are not directly comparable across rows. Abbreviations: B, boredom; E, engagement; C, confusion; F, frustration.

Table 3.1 (continued).

Study	Architecture	Evaluation Metric(s)
Mehta et al. (2022)	Backbone: 3D DenseNet-121. Temporal: 3D convolution (tube modelling). Attention: spatial, temporal, and hybrid non-local self-attention blocks. Classification: 4-class + regression heads; class-balanced focal and CB-MSE losses.	63.59% 4-class acc.; MSE 0.0347 (regression)
Su, He, Luo (2024)	Backbone: ResNet-18 patches (PANet) + 3D CNN tube tokenizer (STformer). Temporal: 3D CNN tubes + transformer encoder. Attention: part-sensitive spatial attention + multi-head self-attention; late fusion. Classification: 4-class engagement / regression; dual-branch fusion head.	64.0% 4-class acc. (DAiSEE); MSE 0.0729

Note. Each architecture entry summarises the full pipeline as reported by the original study. Metrics are reported as published; four-class engagement accuracy, per-class accuracy, and regression MSE are not directly comparable across rows. Abbreviations: B, boredom; E, engagement; C, confusion; F, frustration.

Table 3.1 (continued).

Study	Architecture	Evaluation Metric(s)
Liu et al. (2026)	Backbone: ResNet-18. Temporal: Differential Temporal Transformer (DTFormer). Attention: Adaptive Global Attention block. Classification: 4-class softmax over fused spatiotemporal features.	62.56% (DAiSEE, 4-class); evaluated also on DFEW and FERV39k (cross-dataset FER)
Rianto et al. (2025)	Backbone: Delaunay GCN on 68 landmarks + 1D-CNN on 52 action units (bimodal). Temporal: none (per-frame streams). Attention: none. Classification: 4-class engagement; LDAM imbalance-aware loss.	52.24% acc.; macro-F1 0.49

Note. Each architecture entry summarises the full pipeline as reported by the original study. Metrics are reported as published; four-class engagement accuracy, per-class accuracy, and regression MSE are not directly comparable across rows. Abbreviations: B, boredom; E, engagement; C, confusion; F, frustration.

3.4 Adaptive Learning Systems

While the previous section focused on recognising learner states through facial behaviour, adaptive learning systems focus on deciding how instruction should respond to learner needs. These systems are digital tools that change how they teach depending on the individual learner. Instead of presenting the same material in the same order to every student, they adjust things like the content, the pace, the difficulty of the tasks, the feedback given after each answer, hints, and presentation format [8]. A common way to describe this behaviour is as a closed loop: the system collects data from the learner, uses that data to judge how the learner is progressing, and then decides what to show next [13]. Reviews of intelligent tutoring systems stress that adaptation is not limited to picking the next exercise: the tutor model may also schedule scaffolding, remediation, motivational messages, interface changes, and assessment pacing. The goal is to give each learner tasks that match their current level, so that they can keep improving without feeling bored or overwhelmed [21].

Early adaptive systems focused on explicit rules written by experts. These were called intelligent tutoring systems (ITS), and examples such as GUIDON and the LISP Tutor were built in the 1980s around hand-crafted if-then rules that decided when to give a hint, when to move on, and when to mark a concept as learned. The strength of these systems was that their decisions were transparent and easy to explain, but the rules had to be prepared for every topic, which made them hard to extend beyond a single subject. Rule-based logic has not disappeared, though. Sayed et al., for example, used a small set of threshold rules based on a student's penalised grade to move an exercise up or down in difficulty inside an online platform for third-grade mathematics [114].

Later work introduced student models that tried to estimate what each learner already knew. Instead of relying only on teacher-written rules, these systems built a picture of the learner from their interaction history and updated it after every answer [21]. This line of research eventually led to knowledge tracing methods, in which the system predicts how likely the learner is to answer the next question correctly based on their earlier answers. Pardos and colleagues showed that year-long patterns of boredom, engaged concentration, confusion, and gaming-the-system behaviour in ASSISTments logs correlate with end-of-year mathematics test scores [100]. Kulik and Zitnik built a recent exercise recommender that tries to pick questions just above the learner's current level [19]. Educational psychologists call this range the Zone of Proximal Development: the set of tasks that are challenging but still reachable. Their system ranks candidate questions by predicted difficulty and then chooses items around the 66th percentile, so that the next exercise is hard but not impossible [19].

Over time, researchers began to treat task selection as a decision problem that could be learned from experience. Reinforcement learning (RL) then became a common tool for this. In RL, the system tries different actions, observes the results, and slowly learns which actions lead to better outcomes. Azoulay et al. compared several adaptation methods within the same simu-

lated environment. The study tested different reinforcement learning algorithms, a bandit-style method, and a Bayesian approach that keeps a probability estimate of each learner's ability [17]. The Bayesian method reached about 89 to 93 percent of the performance of an ideal algorithm that already knew each student's real ability, while the reinforcement learning methods reached lower but still useful levels [17]. The study has become an important reference because it placed very different adaptation families on equal ground, and later adaptive tutoring work extended these baselines [115, 16].

After Azoulay et al. showed in 2021 that tutoring adaptation could be studied as a sequential decision problem inside a shared simulator, the next step was to ask whether a policy could be trained against an explicit learner model rather than only compared as a fixed algorithm [17, 115]. Kubotani et al. (2021) addressed spaced review scheduling, where the tutor must decide which previously studied item to present before forgetting occurs. They replaced live learners with a DAS3H-based mathematical student model and trained a Proximal Policy Optimisation agent to select review items over 30 items across 15 days and 30 sessions. The state came from estimated memory rates produced by the inner student model, and the action space selected the next item to review. Evaluation used only the mathematical simulator and compared RLTutor against random, Leitner, and threshold teachers; RLTutor maintained higher average retention in early sessions, although the gap narrowed later. The study matters because it shows why synthetic learners appear before classroom deployment, but its student model was still narrow and its action space was limited to review selection rather than full tutoring control.

The same year, Jezuina Korovesi and Ktona (2021) moved closer to the architecture used in the present thesis by training a pedagogical module with a Deep Q-Network in a Python intelligent tutoring system [116]. Their educational problem was lesson sequencing: the tutor had to decide whether a learner should remain in the current lesson or advance after concept mastery. The state was a binary vector encoding the active lesson, the concepts taught in that lesson, and the concepts already learned by the simulated student. The student model was a simple simulated learner with a fixed learning probability of 0.7, where each taught concept was learned or missed according to that probability. The action space contained two choices: stay in the current lesson or move to the next lesson. The reward function gave +10 when the action matched the mastery condition and -10 when it did not. Training used DQN with experience replay, a target network, batch size 64, discount factor $\gamma = 0.85$, and epsilon decay over up to 4,000 episodes in two scenarios, one with empty prior knowledge and one with random prior knowledge. Evaluation reported episode reward and episode length curves over 100 test episodes, but did not compare against a rule-based tutor baseline. This paper is the earliest clear DQN tutoring design in the set reviewed here, although its state did not include knowledge tracing or engagement sensing.

By 2023, the question shifted from whether DQN could train in simulation to whether any part of the tutoring policy could survive contact with a real classroom. Sayed et al. (2023) built

the APPEAL platform for grade-three mathematics and split adaptation into two decision layers [114]. Exercise difficulty and navigation were controlled by an online rule-based algorithm using penalised grades, attempts, and hints across Bloom taxonomy levels. Content presentation was handled separately by a Deep Q-Network formulated as a Markov decision process over student interaction traces. The DQN state included time spent per lesson, exercise scores, hint use, and repeated attempts within taxonomy levels and difficulty bands. The action space encoded VARK presentation modes as four binary switches that showed or hid visual, aural, read/write, and kinesthetic material. The reward combined academic improvement with savings in time, hints, and attempts. Training used DQN with experience replay; live evaluation ran for two months with 26 learners and reported post-test gains relative to pre-test, with the largest improvements for students who started with lower performance. The study shows that DQN can operate inside a deployed school platform, but only for presentation choice, not for full tutoring control, and the sample remained small.

Once DQN tutors existed both in simulation and in partial live deployment, the next development in 2025 was to strengthen the student model before policy learning. Fu (2025) proposed RL-DKT, which uses a recurrent dynamic knowledge tracing network to estimate skill mastery and a REINFORCE agent to select the next task from offline logs [16]. The educational problem was learning-path sequencing rather than live hint or feedback control. The state was the DKT hidden vector updated after each response; the action space was the set of candidate tasks available at each step; and the reward combined correctness, task difficulty, and progress terms rather than a single binary outcome. Evaluation used ASSISTments, KDD Cup 2010, and Cognitive Tutor logs and reported AUC of 0.874, F1 of 0.812, average completion time of 34.5 minutes, dropout of 3.2%, and a reported 12.5% reduction in completion time relative to baselines. RL-DKT is not a DQN tutor, but it clarified an important design pattern for this thesis: separate student modelling from policy optimisation. Its limitation is offline replay rather than an online tutor loop with synthetic or live students.

Guzman and Cruz-Mercado (2025) extended that pattern with a larger action space and a two-stage evaluation design [18]. Their RL-Tutor system targeted introductory Python skill development. A Dynamic Key-Value Memory Network maintained the student model from correctness, response time, and attempts, producing a knowledge-tracing vector concatenated with recent learning-object context and session meta-features such as consecutive errors and hint requests. A Proximal Policy Optimisation agent then selected hierarchical actions: problem versus instructional page versus review quiz, difficulty level, target skill, hint type, and pacing decision. The reward combined performance, efficiency, learning gain, engagement, and retention terms. Because online RL sample collection is slow, the agent was first trained for about 400,000 steps against 10,000 synthetic students and then tested in a between-subjects study with 90 human learners against rule-based and static tutors. Simulated mean cumulative reward reached 315 ± 22 versus 205 ± 15 for the rule-based tutor ($p < 0.001$). Live normalised learning gain was 0.72 for RL-Tutor, 0.58 for the rule-based tutor, and 0.45 for the static tutor

($p < 0.01$), with similar retention advantages one week later. This study improved both the student model and the action space, but it used PPO rather than DQN and still inferred engagement only from behaviour, not from facial signals.

The most recent DQN work in this line returns to large-scale synthetic evaluation. Zou and Xiong (2026) built a Double DQN path optimiser for a knowledge-graph curriculum with 537 nodes and 1,243 relations [117]. The state represented learner mastery and progress over the graph; the action space selected the next course item; and the reward combined learning effectiveness, time cost, and knowledge coverage. Training used experience replay and epsilon-greedy exploration in a simulation with 5,000 virtual learners. The system reached an 87.3% completion rate, mastery of 0.823, and outperformed random, collaborative-filtering, and standard Q-learning baselines. For beginner learners, average mastery rose from 0.512 to 0.789 in ablation analysis. This paper is the strongest recent example of DQN plus synthetic cohort evaluation, but it optimises path order rather than hints, feedback, or pacing, and it was not followed by a classroom study.

Viewed across these dates, the DQN tutoring line changed in a clear order. Jezuina Koroveshi and Ktona (2021) first mapped lesson progression to DQN actions in a simulated Python tutor with a simple mastery vector and a fixed reward table [116]. Sayed et al. (2023) kept DQN for presentation only and left difficulty to rules, which reduced the learned portion of the policy but provided rare live-school evidence [114]. Zou and Xiong (2026) then moved to Double DQN with a structured knowledge-graph state and 5,000 virtual learners, improving scale but narrowing the problem to path selection [117]. What improved over time was the realism of evaluation settings and the richness of student models in hybrid systems such as RL-DKT and RL-Tutor [16, 18]. What remained unresolved was a DQN tutor that combines an explicit student model, a multi-action pedagogical policy, and reusable synthetic evaluation without splitting control across rules and learned components.

3.4.1 Simulation, Synthetic Students, and the Ethics of Live Policy Learning

Deep reinforcement-learning algorithms require interaction volumes that real classrooms structurally cannot supply during policy development. Sanz Ausin and colleagues confronted this directly: their offline ITS corpus contained fewer than 800 complete student trajectories, a fraction of the interaction budget that DQN typically consumes on even simple benchmarks [118]. Dulac-Arnold and colleagues formalise the same constraint for production systems more broadly, arguing that online policy search must be both sample-efficient and rapidly performant because exploratory actions on real users carry immediate cost [119]. Kubotani et al. reach the same conclusion for adaptive review scheduling: the interaction count required for PPO optimisation cannot be extracted from live learners without prohibitive delay, which is precisely why RL-Tutor trains against a virtual student model [115]. Li's survey of RL in education con-

firms that sample intensity and exploration risk remain persistent barriers across algorithmic families [120]. The gap between what policy search demands and what a classroom can supply is structural, not incidental.

The practical constraints extend beyond data volume. During policy search, a tutor must try actions whose value is still uncertain, and delayed, noisy educational outcomes make harmful policies difficult to detect after deployment [119]. Reward misspecification compounds this risk: a proxy that over-weights engagement can produce policies that reach high measured reward while delivering negligible mastery gain. Olukola and Rahimi demonstrate this concretely in a simulated Python tutor, where an engagement-heavy policy drove the Reward Hacking Severity Index to 0.317; architectural safety constraints reduced it to 0.102 [121]. Exploratory policy learning on real learners is therefore constrained by safety, cost, and reward uncertainty, not by a categorical prohibition on classroom research. The important distinction is between the training phase, where a policy is still searching and reward signals are uncertain, and the evaluation phase, where a fixed, already-validated policy is tested under ethical approval. Sanz Ausin et al. follow exactly this separation: offline deep RL on historical logs, followed by randomised classroom studies once a policy is selected [118].

Because of these constraints, synthetic student environments have become the standard first-stage training platform in RL tutoring research, and the models used to populate them have grown progressively richer. Early DQN work by Jezuina Korovesi and Ktona trained a lesson-progression policy against a simulated learner with a fixed concept-learning probability across thousands of simulated episodes [116]. Kubotani et al. replaced that with a DAS3H memory model pre-trained on EdNet, so the RL agent optimised review scheduling by interacting with a calibrated inner student model rather than with students directly [115]. Guzman and Cruz-Mercado extended this design with a DKVMN synthetic cohort of 10,000 virtual learners and approximately 400,000 PPO training steps before a between-subjects study with 90 human participants [18]. At larger scale, Azoulay et al. used a shared virtual-student environment to benchmark competing adaptation families under identical conditions [17], and Zou and Xiong sampled 5,000 virtual learners to evaluate a Double DQN path optimiser on a 537-node knowledge graph [117]. Across all of these systems, simulation provides the controllable, reproducible, and scalable stage that precedes any contact with real classrooms and from which policies graduate to live evaluation.

The present thesis follows this simulation-first pipeline. Existing synthetic environments represent learners through knowledge-tracing estimates [17, 16, 115, 116]; the next natural step is to extend the synthetic student state to include a multidimensional affective state representation derived from facial expression recognition. Rather than treating emotion as a single engagement proxy, the thesis integrates four discrete academic emotions identified in prior educational FER work: boredom, confusion, engagement, and frustration. This affective state representation is integrated into the DQN state space and trained inside a synthetic student

environment before any classroom use. Table 3.2 summarises the studies that most directly inform this design.

Table 3.2: Simulation-based RL tutoring studies most relevant to the present thesis design

Paper	Student model	RL method	Learning task	Evaluation setup	Main result
Jezuina Koroveshi & Ktona (2021) [116]	A simple simulated learner that learns with a fixed probability	DQN	Deciding whether to stay in the current lesson or advance (Python)	Simulation only	Learning improved steadily in simulation training; no live student test
Zou & Xiong (2026) [117]	5,000 simulated learners across different ability levels	DQN	Selecting the next course item from a knowledge graph	Large-scale simulation only	87.3% course completion; strong learning performance (mastery 0.823)
Sanz Ausin et al. (2019) [118]	Trained on historical student data (no live students used)	DQN (offline)	Choosing worked examples vs. problem-solving in logic	Trained offline, then tested with real students	Significant learning gain for stronger learners; real-time training with students is not practical
Guzman & Cruz-Mercado (2025) [18]	A knowledge-tracing model trained on 10,000 simulated learners	Reinforcement learning	Selecting problem type, difficulty, hints, and pacing (Python)	Trained in simulation, then tested with 90 real students	Better learning quality (0.72 vs. 0.58) than a rule-based tutor
Sayed et al. (2023) [114]	Real classroom learners (no simulated cohort)	DQN	Choosing how content is presented (visual, audio, read/write, hands-on)	Two-month classroom pilot with 26 students	Post-test gains, largest for lower-performing students

Note. All studies train RL tutoring policies using simulated students or historical data, since exploratory RL training is typically performed in simulation or offline due to safety and cost constraints. Real-student studies are used only for controlled evaluation after a policy has been validated. None of these systems uses facial expression recognition in the learner state.

3.5 Emotion-Aware Adaptation

The previous section reviewed reinforcement-learning tutors that adapt from knowledge and performance traces alone. This section looks at systems that also use learner emotion. The work in this area has shifted over time. Early systems read emotion from costly multi-sensor rigs or from weak text signals. Later systems moved toward webcam facial expression recognition (FER) joined with learned adaptation policies. That shift is where the present thesis sits: a single webcam reads the face, and a learned policy decides what to do next. The studies below trace that shift and report what each system actually measured.

The first practical systems read emotion from dialogue with the tutor. AutoTutor held short natural-language dialogues and asked whether dialogue signals could show how the learner felt [102]. In 2005, D’Mello and colleagues combined affect sensors with AutoTutor to identify boredom, confusion, frustration and engagement [122]. Later work showed that ordinary features of tutoring conversation carried information about boredom, confusion, flow and frustration [101, 6], and that confusion, when handled well, can help learning [85]. This was the historical start of the field: emotion can be detected during instruction. The response, at that stage, was small hint and encouragement changes. A more sensor-heavy line followed. Woolf and colleagues built the Wayang Outpost tutor with a four-sensor rig: a webcam, a pressure-sensitive chair, a pressure mouse and a skin-conductance bracelet [123]. It estimated states such as frustration and engagement and changed problems, encouragement, or an animated learning companion. The reported direction was less frustration and more interest when companions were used, and the effect was stronger for girls. This shows that multi-sensor emotion-aware tutoring is old, but it was costly: four sensors per student, custom hardware, a sensor server and a research team for one school. A cheaper line then tried a webcam with text. Lin and colleagues combined FER with text analysis in a tutor for digital art, using an Active Shape Model on webcam images and a semantic voting method on typed input [104]; the evaluation tested usability only, not learning gains. Fwa proposed a rule-based affective tutor for novice programmers that mapped detected emotions to predefined tutoring strategies [124]. A classroom trial with 39 learners (21 control, 18 test) reported faster task completion (median 470.5 s versus 831 s, $p < 0.05$) and more exercises attempted (median 7.5 versus 5, $p < 0.05$). Leong then studied efficient deep models that detect boredom and frustration from webcam video in real time, which made real-time webcam reading realistic without extra hardware [29].

Petrovica and Ekenel reviewed fourteen affective tutoring systems and named the trade-off that still shapes the field [105]. They described three paths. Sensor-free log methods scale but lose accuracy. Sensor-heavy physiological or camera methods are accurate but costly and uncomfortable. They proposed a sensor-lite middle path: use only the webcam or microphone that classrooms already have. They also reported that most systems of that period lacked rich real-time adaptation.

Two systems around 2020 sat at the two ends of this trade-off. Megahed and Mohammed

combined FER with fuzzy-logic rules to change difficulty and presentation [125]; no learning result was reported. Standen and colleagues evaluated the MaTHiSiS system for learners with intellectual disabilities, using five sensing channels (face, gaze, body pose, voice and a gesture accelerometer) [126]. The study ran with 67 learners across six schools in the United Kingdom, Italy and Spain. Sessions driven by affect and achievement gave significantly more engagement and significantly less boredom than achievement-only sessions, but there was no significant difference in learning gains.

From 2024 onward, several systems connected emotion recognition to reinforcement learning [42, 36, 9, 38]. Pérez, Dapena and Aguilar used two Q-learning tables over a circumplex model and chose when to exploit or explore based on the learner's emotional region [42]. The affect data came from a static expression corpus, not from a live webcam module. The method was tested in simulation only, with one thousand runs over two hundred tasks and no human participants, and reached a learning gain of 0.49 at task 200, above all baselines (next best 0.32). Govea and colleagues used multimodal sensing (cameras, microphones and biometric sensors) with deep reinforcement learning, trained on data from 500 students; the reported emotion detection accuracy rose to 89.3%, with reported gains in real-time adaptability and personalisation [36]. Kong and colleagues used multimodal sensing (webcam, voice, physiological signals and logs) with a reinforcement-learning controller built as a DQN-backed contextual bandit [38]. A lab and real-classroom randomised study with 271 learners reported affect recognition accuracy of 82 to 88% and about 15 percentage points higher learning gain for mid-ability learners over a static baseline.

Gong and colleagues built a DQN multimodal system (webcam, microphone and text) that detects basic emotions, namely happy, sad, angry, neutral and surprised, rather than academic emotions [9]. It reported a fusion accuracy of 0.91 and an AUC of 0.94. These numbers come with clear limits: the system was evaluated on public datasets with no real students, and it targets basic emotions rather than the academic emotions that matter for learning. A scoping review of twenty-seven systems found that facial expression was the dominant sensing channel and scaffolding the most common pedagogical response [34].

Table 3.3 compares these systems and points to the gap addressed in the next section: emotion is rarely joined to a learned, webcam-only tutoring policy.

Table 3.3: Emotion-aware tutoring systems: sensing, adaptation, and reported results.

Study	Sensing modality	Emotions detected	Adaptation method	Deployment setting	Main result
Fwa (2018) [124]	webcam plus key-board/mouse logs plus eye tracker	frustration, (dis)engagement	rule-based (Rete engine)	real classroom trial, n=39	faster task completion (median 470.5 s vs 831 s, $p<0.05$); more exercises attempted
Standen (2020) [126]	five sensors: face, gaze, body pose, voice, gesture	engagement, boredom	affect plus achievement rules	six schools, 67 learners	more engagement and less boredom vs achievement-only; no learning gain difference
Pérez (2024) [42]	static expression corpus (not a live webcam)	valence, arousal (anxiety, boredom)	Q-learning, two tables	simulation only (no students)	learning gain 0.49 at task 200, above all baselines (next best 0.32)
Govea (2024) [36]	multimodal: cameras, microphones, biometric sensors	happiness, sadness, anger, surprise, fear, disgust (six basic)	deep RL	data from 500 students	emotion detection accuracy rose from 72.4 to 89.3% (k-fold, k=10)
Kong (2025) [38]	multimodal: webcam, voice, physiological, logs	engagement, confusion, frustration, boredom, delight, anxiety, neutral, surprise	DQN-backed contextual bandit	lab plus real classroom, n=271	affect accuracy 82 to 88%; about +15 points learning gain for mid-ability learners
Gong (2025) [9]	multimodal: webcam, mic, text	basic emotions (happy, sad, angry, neutral, surprised)	DQN (MDP)	public datasets only; no real students	fusion accuracy 0.91, AUC 0.94 (dataset-only, basic emotions)

Note. Facial expression is the most common sensing channel across these systems. Several systems add extra hardware cost. A webcam-only FER signal joined to a learned policy is the configuration this thesis targets.

3.6 Research Gap

Facial expression recognition for education has matured from handcrafted features and shallow classifiers into deep temporal and transformer-based models that run on an ordinary webcam in real time [25, 72, 28, 26]. Public benchmarks such as DAiSEE made the academic emotions of boredom, confusion, engagement and frustration a shared target [45]. Two limits still hold. Recognition accuracy on the four-level engagement task stays in the low-to-mid sixties even for strong models, and visually similar states such as confusion and frustration are still confused [70, 111, 25]. More important for this thesis, almost all of this work stops at classification, an engagement index, or a dashboard, and does not feed its output into a tutoring decision [33, 32, 29, 28]. Facial expression recognition has therefore solved a perception problem without closing the loop to action.

In parallel, reinforcement learning has given tutoring systems a principled way to choose what to do next. Adaptation has been framed as a sequential decision problem, and policies have been learned for exercise selection, review scheduling, hints, presentation and learning-path sequencing [17, 115, 114, 16, 117]. Because online exploration on real learners is costly and slow, and because reward functions are easy to misspecify, this work has settled on simulation with synthetic students as the standard initial training stage [119, 118, 120, 121]. These tutors are strong on the cognitive side, but they represent the learner almost only through correctness, response time, hint use and knowledge-tracing estimates [17, 16, 18, 117]. The affective state of the learner does not enter the decision policy.

Emotion-aware tutoring has tried to bring affect back into the loop. The idea is more than fifteen years old, from multi-sensor classroom rigs [123] to recent systems that join emotion recognition with reinforcement learning [42, 36, 38, 9]. A field-level scoping review still reports that these systems recognise affect more reliably than they act on it, with scaffolding the most common and often hand-coded response [34]. Three problems recur. The strongest results depend on multimodal or physiological sensing, such as cameras combined with voice, posture, skin conductance or other biosignals, which is hard to reproduce in ordinary classrooms or at scale [123, 126, 36, 38]. Several systems detect only the basic emotions rather than the academic emotions that matter for learning [36, 9]. And the affect-to-action mapping is often fixed by hand or organised around a single control such as difficulty or tone [125, 42, 37, 114].

Read together, these three lines leave a specific problem open. Affect can be sensed from an ordinary webcam, and tutoring actions can be learned in simulation, but the two have not been joined in a way that is at once deployable, learnable and testable. The affective signal is rarely part of the learned state and reward of the policy, since it is added as a rule or used only for detection [34, 125]. The affective controllers that do perform well rely on sensing that does not scale, which leaves the trade-off between sensing richness and deployment feasibility unresolved [105, 38, 36]. The contribution of affect is also seldom isolated: the closest system to this thesis combines facial valence and arousal with Q-learning, but it draws affect from

a static corpus rather than a live webcam [42]. Reproducibility makes the problem harder, because the state, reward, transition assumptions and hyperparameters of these controllers are often left unspecified [36, 38].

This open problem matters for more than engineering. The evidence that emotion-aware adaptation improves learning is still weak. The most carefully evaluated multi-sensor study found more engagement and less boredom under affect-aware adaptation, but no significant difference in learning gains [126]. At the same time, placing affect inside the reward is risky, because a policy that optimises an engagement proxy can reach a high reward while producing little mastery [121]. Deciding whether a webcam-derived affective signal genuinely changes a tutoring policy, and whether that change helps learning or only inflates the reward, therefore needs a controlled comparison under matched conditions. Without an explicit and reproducible protocol that isolates the affective channel, the question of whether sensed emotion should drive instruction cannot be answered.

The present thesis addresses this problem directly. It uses facial expression recognition from a single webcam, trained for the academic emotions of boredom, confusion, engagement and frustration, following the sensor-lite path argued for ordinary classrooms [105, 25]. It places the resulting affective state inside the state of a reinforcement-learning tutor that selects from a discrete set of pedagogical actions, rather than a single difficulty rung [17, 16]. It then evaluates the policy in a synthetic student environment with a stated state, reward and transition model, comparing an emotion-aware policy against a matched emotion-free baseline under the same student population and hyperparameters [118, 121, 22]. This design keeps sensing within reach of real classrooms while integrating affective information directly into the reinforcement-learning state representation rather than relying on fixed pedagogical rules. Building on the research gaps identified in this chapter, the proposed framework combines a facial expression recognition module with an adaptive tutoring system to create a complete emotion-aware learning environment. The thesis also proposes a lightweight facial expression recognition architecture for academic emotion detection that achieves strong classification performance and serves as the affective input to the adaptive tutoring system. The next chapter describes the FER architecture, the Deep Q-Network (DQN) reinforcement-learning tutor, the full Markov decision process, the synthetic student environment, and the experimental framework used to evaluate the proposed emotion-aware tutoring framework under controlled simulation conditions.

Chapter 4

Methodology

4.1 Chapter Overview

This chapter specifies the methodology of an affect-informed adaptive tutoring framework that operates as a two-stage pipeline. The first stage is a perception module that performs Facial Expression Recognition (FER) on short webcam clips of the learner. The second stage is an adaptive tutoring system that consumes the FER output to select pedagogical actions for the learner in real time.

The FER module maps each input clip to a four-dimensional affect vector whose components correspond to Boredom, Engagement, Confusion, and Frustration. These four states are the academic affective states annotated in the dataset used in this thesis. Each component of the vector is a continuous score in the interval $[0, 1]$ produced by an independent sigmoid output, so the four scores may be simultaneously high and support co-occurring affective states.

The output of the FER module is passed directly into the downstream adaptive decision-making system. That system is formulated as a Markov Decision Process (MDP) and is solved with a Deep Q-Network (DQN) agent. The four-dimensional affect vector produced by the FER module forms part of the state representation observed by the DQN agent at each decision step, alongside the learner’s knowledge and engagement signals. The formal definition of the MDP, the action space, the reward function, and the training procedure of the DQN agent are presented in the adaptive tutoring methodology that follows in Section 4.6; the present chapter is concerned with the perception stage and with the data-flow interface that connects it to the decision stage.

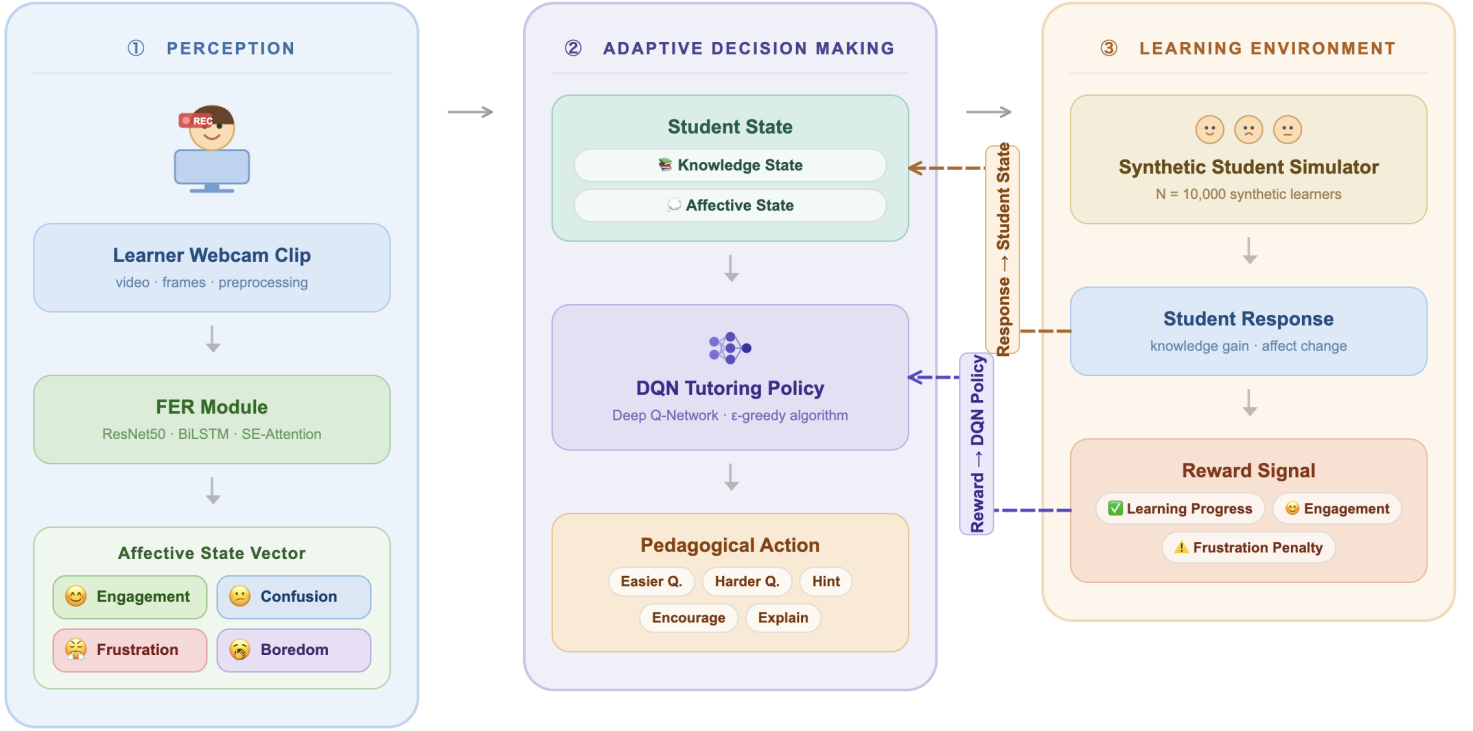


Figure 4.1: Affect-informed adaptive tutoring framework. The Perception zone maps a learner webcam clip to an affective state vector over Boredom, Engagement, Confusion, and Frustration. The Adaptive Decision Making zone combines this vector with the learner’s knowledge state to form the student state consumed by the DQN tutoring policy, which selects a pedagogical action. The Learning Environment zone houses the synthetic student simulator and the reward signal that closes the reinforcement-learning loop.

The rest of this chapter is organised as follows. Section 4.2 describes the dataset and label structure. Section 4.3 describes the data preparation pipeline applied before training, including the ordinal-to-binary label mapping. Section 4.4 presents the FER architecture component by component and summarises the design elements of the proposed model. Section 4.5 describes the training and evaluation protocol applied to the FER module, including the loss function, the imbalance-handling strategy, the optimisation procedure, the per-label threshold tuning, and the evaluation metrics. Section 4.6 then specifies the adaptive tutoring methodology that consumes the FER output, including the Markov Decision Process formulation, the DQN agent, and the synthetic student simulator used for training and evaluation.

4.2 Dataset Description

The FER module is trained and evaluated on the DAiSEE dataset [45]. DAiSEE is a publicly available collection of approximately 9,068 short webcam clips of learners recorded in

study-like conditions. Each clip is annotated for four academic affective states: Boredom, Engagement, Confusion, and Frustration [45].

The annotation scheme assigns each of the four states an ordinal intensity level on the integer scale $\{0, 1, 2, 3\}$, corresponding respectively to absent, slight, moderate, and very strong manifestations of the state. In this thesis these ordinal annotations are mapped to binary present/absent labels following the adaptation policy of Malekshahi et al. [127]; the conversion rule is formalised in Section 4.3.

The task is formulated as a multi-label binary classification problem. A single clip can be marked positive for several states at the same time, so the four binary labels are predicted independently and may co-occur on the same clip [45].

The dataset is partitioned into the official DAiSEE training, validation, and test splits as released by Gupta et al. [45]. The training set contains 5,358 clips, the validation set contains 1,429 clips, and the test set contains 1,784 clips. The training set is used to fit the model parameters, the validation set is used to select the best epoch and to tune per-label decision thresholds, and the test set is held out and is used only at the end to report final metrics. The frequencies of the four affective categories in the training set are unequal, which is a natural property of the dataset and is reflected without alteration in the splits used in this work.

4.3 Data Preparation

4.3.1 Ordinal-to-Binary Label Mapping

The original DAiSEE annotations use the ordinal intensity scale $\{0, 1, 2, 3\}$ for each of the four affective states. For the multi-label binary formulation used in this thesis, each ordinal label is converted into a binary label according to the rule [127]

$$y_\ell = \begin{cases} 0, & \text{if } \text{intensity}_\ell \in \{0, 1\}, \\ 1, & \text{if } \text{intensity}_\ell \in \{2, 3\}, \end{cases} \quad \ell \in \{\text{Boredom, Engagement, Confusion, Frustration}\}. \quad [1]$$

Rule 1 treats absent and slight manifestations as the negative class and moderate and very strong manifestations as the positive class. The same rule is applied identically to every affective state and to every clip in the training, validation, and test splits.

This binary adaptation policy is inspired by the engagement-detection methodology of Malekshahi et al. [127], in which the engagement-level rule $\text{Engagement} \leq 1 \rightarrow \text{Disengaged}$ and $\text{Engagement} \geq 2 \rightarrow \text{Engaged}$ is used to convert the original DAiSEE ordinal labels into a two-level classification problem. Their work was developed primarily for the engagement label; in the present thesis the same intensity threshold is extended uniformly to all four academic affective states (Boredom, Engagement, Confusion, and Frustration), so that each

clip receives an independent binary label per state and the multi-label co-occurrence structure of academic affect is preserved.

4.3.2 Frame Extraction and Sampling

Each DAiSEE clip is decoded into image frames in an offline preprocessing step that samples each ten-second clip at a fixed low frame rate, producing a sequence of sixteen frames per clip stored as JPEG images on disk. The FER model expects a temporal axis of length $T=24$; a per-batch sampling step therefore produces twenty-four indices using nearest-neighbour mapping over the interval $[0,15]$, resulting in repeated frames when multiple indices map to the same source frame. The temporal length $T=24$ was selected empirically as it improved validation Macro F1 compared to $T=16$, despite the presence of repeated frames.

4.3.3 Spatial Preprocessing

Each frame is decoded with OpenCV, converted from BGR to RGB, resized to 224×224 , scaled to the unit interval by division by 255, and standardised using the ImageNet channel mean $(0.485, 0.456, 0.406)$ and standard deviation $(0.229, 0.224, 0.225)$ [50]. The same preprocessing is applied at training, validation, and test time.

4.3.4 Training-Time Augmentation

Frame-level augmentations are applied only at training time and never at validation or test time. Each frame can be horizontally flipped with probability 0.5. Brightness and contrast are jittered with probability 0.5, by sampling a contrast factor uniformly between 0.85 and 1.15 and a brightness shift uniformly between -15 and $+15$ pixel values. A random rotation between -7° and $+7^\circ$ is applied with probability 0.25 using reflective borders. A Gaussian blur with a 3×3 kernel is applied with probability 0.15. A random crop is applied with probability 0.25 by sampling a crop ratio uniformly between 0.90 and 1.0 of the frame size and resizing back to 224×224 .

4.3.5 Dominant-Group Resampling

A dominant-group resampling step is applied to the training set before training. Each training clip is assigned to a single dominant affective group according to the priority order Frustration, Confusion, Boredom, Engagement; clips with no positive label are assigned to a separate *None* group. Clips in the Engagement group are retained once, clips in the Boredom group are duplicated twice, clips in the Confusion group three times, and clips in the Frustration group four times. The resulting duplicated training set is shuffled with a fixed random seed for reproducibility. The resampling is applied at the data level only; the validation and test sets remain unchanged.

4.4 FER Architecture Overview

The FER module is a single end-to-end neural network that takes a short video clip of the learner as input and produces a four-dimensional affect vector as output. The network input is a tensor of shape $[B, T, C, H, W]$, where B is the batch size, $T = 24$ is the temporal axis, $C = 3$ is the number of colour channels, and $H = W = 224$ is the spatial resolution. The output is a vector of four sigmoid scores per clip, one for each of Boredom, Engagement, Confusion, and Frustration.

The architecture comprises five blocks arranged in sequence: a spatial backbone, a channel-attention block, a temporal recurrent encoder, a temporal attention pooling layer, and a multi-label classification head. The complete pipeline is illustrated in Figure 4.2. A summary of the five blocks together with the additional components that operate on the model output is also given in Table 4.1; each block is described in detail in the subsections that follow.

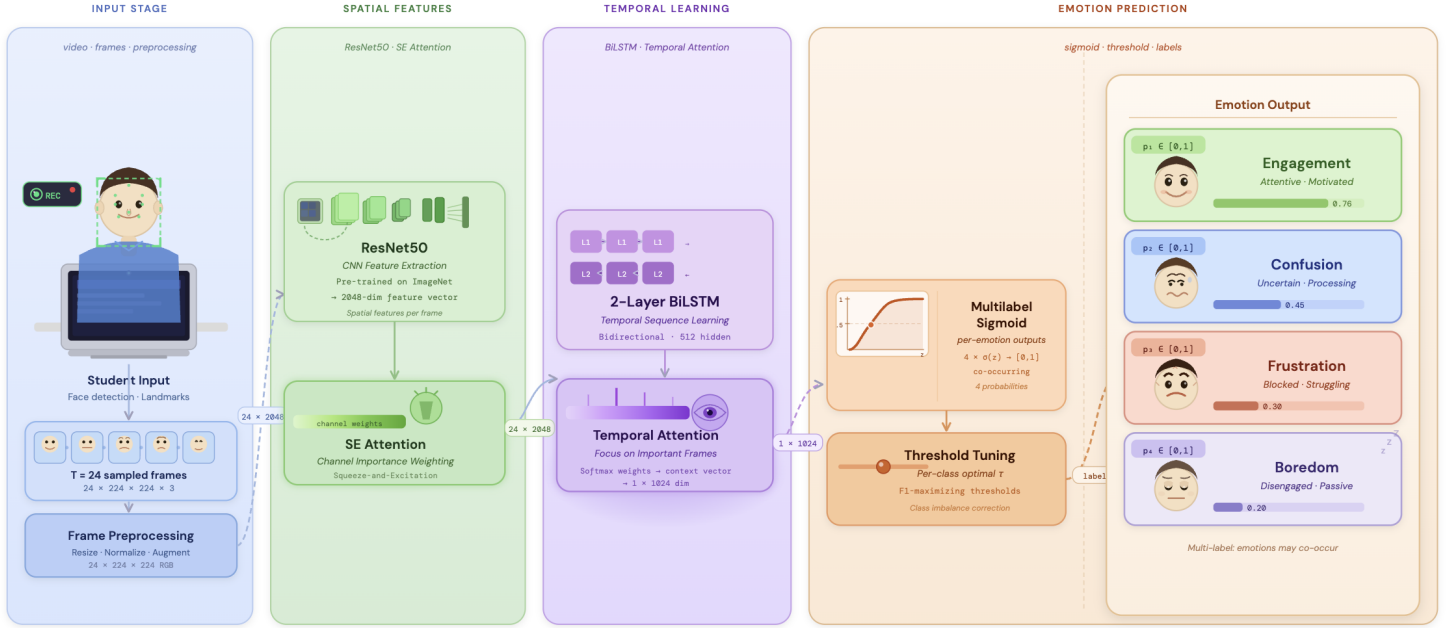


Figure 4.2: End-to-end FER architecture for multi-label academic affect recognition. A learner video clip is sampled to a fixed length of $T = 24$ frames and preprocessed, per-frame spatial features are extracted with a ResNet-50 backbone and recalibrated by a Squeeze-and-Excitation channel-attention block, their temporal pattern is encoded by a two-layer Bidirectional LSTM, the time steps are aggregated by a temporal attention pooling layer, and a multi-label classification head produces four independent sigmoid scores for Boredom, Engagement, Confusion, and Frustration. Per-label thresholds tuned on the validation set turn these scores into the final binary affect labels.

4.4.1 Spatial Backbone: ResNet-50

The spatial backbone is ResNet-50 [50], a fifty-layer convolutional neural network with residual connections that add the input of each block back to its output. The backbone is loaded with weights pre-trained on ImageNet and is fine-tuned end-to-end. The ImageNet classification head is removed, leaving the convolutional body and the global average pooling layer. For each input frame the body produces a 2048-dimensional feature vector. The $T = 24$ frames of a clip are passed through the backbone in a single batched call by reshaping the input to $[B \cdot T, C, H, W]$, and the resulting features are reshaped back to $[B, T, 2048]$ before the next block.

4.4.2 Squeeze-and-Excitation Channel Attention

The per-frame feature vectors are passed through a Squeeze-and-Excitation (SE) block [73]. The SE block re-weights the 2048 feature channels of each frame using two fully connected layers. The first layer projects the channel vector to a hidden size of $\max(2048/16, 64) = 128$ with a ReLU activation, and the second projects it back to 2048 with a sigmoid activation, producing one importance weight per channel. The weights are multiplied element-wise onto the original feature vector. The output has the same shape $[B, T, 2048]$ as the input.

4.4.3 Temporal Encoder: Two-Layer BiLSTM

The recalibrated per-frame features are passed to a two-layer Bidirectional Long Short-Term Memory (BiLSTM) network [54]. The BiLSTM has two stacked layers with hidden size 256 in each direction and a dropout of 0.5 between layers. It reads the $[B, T, 2048]$ sequence produced by the SE block and outputs a sequence of shape $[B, T, 512]$, where $512 = 2 \cdot 256$ comes from concatenating the forward and backward hidden states at each time step.

4.4.4 Temporal Attention Pooling

The temporal attention pooling layer aggregates the BiLSTM outputs into a single clip-level vector [71]. For each of the $T = 24$ time-step vectors, the layer computes an unnormalised attention score with a fully connected layer of hidden size 128 and a tanh activation, followed by a second fully connected layer that produces a single scalar. The T scalars are normalised by a softmax across time, and the clip representation is formed as the weighted sum of the time-step vectors with these weights. The output has shape $[B, 512]$.

4.4.5 Multi-label Classification Head

The clip representation is mapped to four label logits by a classification head consisting of a dropout layer, a linear projection to 256 units, a ReLU activation, a second dropout layer,

and a final linear projection to four logits. Both dropout layers use a dropout rate of 0.5. A sigmoid activation is applied independently to each of the four logits, producing four scores in the interval $[0, 1]$. The four scores correspond, in order, to Boredom, Engagement, Confusion, and Frustration [45]. They are converted into binary predictions by the per-label thresholds described in Section 4.5.3.

Table 4.1: Summary of the components that constitute the proposed FER pipeline and the supporting studies in which each component, or a close relative of it, has been reported. The first five rows correspond to the five architectural blocks described in Section 4.4; the last two rows correspond to the training-time loss reweighting and the decision-time threshold calibration described in Section 4.5.

Component	Supporting Studies	Role in the Pipeline	Function Reported in Prior Work
ResNet-50	[50, 27]	Per-frame spatial feature extractor producing a 2048-dimensional vector	Deep convolutional network with residual connections, pre-trained on ImageNet and adopted as a stable backbone for transfer learning in vision tasks
Squeeze-and-Excitation block	[73, 27, 26]	Channel attention applied to the per-frame feature vectors	Lightweight module that recalibrates feature channels to emphasise informative responses
Two-layer BiLSTM	[54, 113]	Temporal sequence encoder operating on the recalibrated per-frame features	Recurrent architecture that captures bidirectional temporal dependencies across video frames
Temporal Attention	[71, 31, 72]	Time-step weighting mechanism that aggregates the BiLSTM outputs into a clip-level representation	Attention layer that assigns importance scores to temporal features prior to aggregation
Multi-label classification head	[45, 111]	Four independent sigmoid outputs producing the affect vector	Standard formulation for multi-label classification tasks in which categories may co-occur
Weighted binary cross-entropy	[70, 31, 111]	Loss reweighting at the optimisation level	Loss-level reweighting reported in the literature as a means of addressing class-frequency differences in multi-label settings
Per-label threshold tuning	[111, 31]	Decision-boundary calibration on the validation set	Validation-driven calibration of the per-label decision threshold used to convert sigmoid scores into binary predictions

4.5 Training and Evaluation Protocol

The FER module is trained end-to-end: both the ResNet-50 backbone and the upper temporal blocks are updated by gradient descent.

4.5.1 Loss Function and Imbalance Handling

The loss function is the per-label binary cross-entropy with logits (`BCEWithLogitsLoss`), with a positive-class weight per label. The four labels are treated as independent binary decisions and the per-label losses are summed across the batch.

The class-frequency differences described in Section 4.2 are handled at three complementary levels of the training pipeline. First, at the **data level**, the dominant-group resampling step defined in Section 4.3.5 duplicates training clips so that minority dominant categories are seen more often by the model during each epoch. Second, at the **loss level**, a per-label positive-class weight is incorporated into the binary cross-entropy with logits. The weight for label ℓ is computed from the training set as

$$w_\ell = \text{clip}\left(\frac{N_\ell^-}{\max(N_\ell^+, 1)}, 0.5, 6.0\right), \quad [4.2]$$

where N_ℓ^+ and N_ℓ^- are the numbers of positive and negative training examples for label ℓ and $\text{clip}(\cdot, a, b)$ restricts the weight to the interval $[a, b]$. The bounded weight ensures that a missed positive on a minority label is more costly than a missed negative without allowing a single rare label to dominate the gradient. Third, at the **decision level**, a per-label decision threshold is tuned on the validation set as described in Section 4.5.3, so that the conversion from sigmoid score to binary prediction is calibrated separately for each of the four affective labels.

4.5.2 Optimisation

The model is optimised with AdamW. The learning rate is set to 5×10^{-5} , the weight-decay coefficient to 10^{-5} , and the batch size to 4. Gradients are clipped at a maximum global norm of 5.0. A learning-rate scheduler of type `ReduceLROnPlateau` monitors the validation Macro F1 score; if the metric does not improve for two consecutive validation rounds, the learning rate is multiplied by 0.5. The model is trained for up to twelve epochs, with an early-stopping rule that interrupts training if the validation Macro F1 does not improve for four epochs in a row. The checkpoint with the highest validation Macro F1 is retained.

4.5.3 Per-Label Threshold Tuning

The FER model produces a continuous sigmoid score in $[0, 1]$ for each of the four affective labels. Converting these scores into binary predictions requires a decision threshold per label.

A single global threshold of 0.5 is not optimal for this task because the DAiSEE dataset is class-imbalanced and because each affective state exhibits a different score distribution; a uniform threshold therefore yields uneven performance across labels. Label-specific thresholds are consequently calibrated independently on the validation set.

After every training epoch, the current model is applied to the full validation set, producing a probability matrix of size $N_{\text{val}} \times 4$. For each label ℓ , a grid of candidate thresholds is evaluated to identify the value that maximises the per-label F1 score:

$$\left[\tau_{\ell}^* = \arg \max_{\tau \in \{0.10, 0.15, \dots, 0.70\}} \text{F1}_{\ell}(\tau) \right]. \quad [4.3]$$

The four label-specific thresholds τ_{ℓ}^* selected at the best-performing validation checkpoint are retained and applied at test time. This calibration step constitutes the final decision stage of the FER pipeline, converting sigmoid probability estimates into the binary affect predictions consumed by the adaptive tutoring module.

4.5.4 Evaluation Metrics

The FER module is evaluated using a set of multi-label classification metrics, all computed at the per-label decision thresholds calibrated in Section 4.5.3. The following notation is used throughout this section. For a given affective label, a *True Positive* (TP) is a clip correctly predicted as positive; a *True Negative* (TN) is a clip correctly predicted as negative; a *False Positive* (FP) is a clip incorrectly predicted as positive when the ground-truth label is negative; and a *False Negative* (FN) is a clip incorrectly predicted as negative when the ground-truth label is positive. Let $L = 4$ denote the number of affective labels and N the number of clips in the evaluation set. Every metric defined below is computed during both validation and final test evaluation, and every metric is reported in the experimental results of Chapter 5.

Per-label Accuracy. Per-label accuracy measures the proportion of all predictions for a given label, both positive and negative, that are classified correctly [128]:

$$\left[\text{Accuracy}_{\ell} = \frac{\text{TP}_{\ell} + \text{TN}_{\ell}}{\text{TP}_{\ell} + \text{TN}_{\ell} + \text{FP}_{\ell} + \text{FN}_{\ell}} \right]. \quad [4.4]$$

Per-label accuracy is reported individually for each of the four affective states. Because the dataset is heavily imbalanced, a high per-label accuracy can be achieved by a model that predicts predominantly the negative class; per-label accuracy is therefore interpreted alongside F1 metrics rather than in isolation.

Precision. Precision measures the proportion of positive predictions that are genuinely positive. It quantifies how reliably the model identifies the presence of an affective state without

raising false alarms [128]

$$\left[\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \right]. \quad [4.6]$$

Recall. Recall measures the proportion of truly positive instances that the model successfully detects. It reflects the model’s sensitivity to the presence of each affective state [128]:

$$\left[\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \right]. \quad [4.7]$$

A high recall indicates that the model misses few genuine cases, which is particularly important for minority affective states such as Confusion and Frustration in the DAiSEE dataset.

F1 Score. The F1 score is the harmonic mean of Precision and Recall. It provides a single balanced measure that penalises both false positives and false negatives, making it well-suited for evaluation under imbalanced label distributions [128]:

$$\left[\text{F1} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \right]. \quad [4.8]$$

Per-label F1 scores are reported individually for each affective state. They are also used during per-label threshold tuning on the validation set: for each label, the threshold that maximises the per-label F1 is selected, as defined in Equation [4.3].

Macro F1. Macro F1 is computed as the unweighted mean of the per-class F1 scores, treating all classes equally regardless of their frequency. This makes it sensitive to performance on minority classes. [129]

Micro F1. Micro F1 aggregates true positives, false positives, and false negatives across all classes before computing a single F1 score. It therefore reflects overall performance weighted by the total number of samples. [129]

Weighted F1. Weighted F1 computes the average of per-class F1 scores weighted by the number of true instances in each class, providing a balance between Macro F1 and class-frequency bias. [129]

Samples F1. Samples F1 is commonly used in multi-label classification and computes the F1 score per sample by comparing the set of true labels and predicted labels, then averaging across all samples. [130, 129]

4.5.5 Implementation Details

The FER module is implemented in Python with PyTorch and torchvision. Frame input/output uses OpenCV, and metrics are computed with scikit-learn. The dataset, model, training, and evaluation scripts are organised as separate self-contained modules. The frame extraction step and the construction of the resampled training CSV are executed once offline. Training and evaluation were run on a single NVIDIA Tesla T4 GPU on the Kaggle Notebooks platform. The full set of training hyperparameters is summarised in Table 4.2.

Table 4.2: FER training hyperparameters.

Hyperparameter	Value
Image size	224×224
Frames per clip T	24
Backbone	ResNet-50, ImageNet pre-trained, fine-tuned
SE reduction ratio	16
BiLSTM layers	2
BiLSTM hidden size	256 per direction
Temporal attention hidden size	128
Classifier hidden size	256
Dropout	0.5
Loss	BCE with logits, per-label <code>pos_weight</code> clipped to $[0.5, 6.0]$
Optimiser	AdamW ($lr = 5 \times 10^{-5}$, $weight_decay = 10^{-5}$)
Gradient clipping	global norm ≤ 5.0
LR scheduler	ReduceLROnPlateau on val. Macro F1 (factor 0.5, patience 2)
Batch size	4
Epochs	up to 12, early stop patience 4 on val. Macro F1
Threshold grid	0.10, 0.15, $\dots$, 0.70 per label, per-label F1 max
Train-time augmentation	flip 0.5, brightness/contrast 0.5, rotation 0.25, blur 0.15, crop 0.25
Oversampling multipliers	engagement $\times 1$, boredom $\times 2$, confusion $\times 3$, frustration $\times 4$

4.6 Adaptive Tutoring via Reinforcement Learning (DQN-Based)

4.6.1 Theoretical Foundations and Citation Mapping

This section documents the theoretical foundations of the proposed tutoring environment. All cognitive, affective, and pedagogical mechanisms are inherited from published sources; the DQN agent learns only to act within this theory-grounded environment. No new cognitive or

affective model is proposed.

The theoretical framework is presented in two layers. Table 4.3 provides a conceptual overview of the modelling decisions and their motivating sources. Tables 4.4 and 4.5 form a traceability matrix in which each row states one inherited mechanism, presents its core formal rule, and cites the originating source. Part A covers the cognitive model, affective dynamics, and emotion taxonomy. Part B covers the pedagogical interventions, reward function, partial observability, and the DQN learning algorithm. Every equation and transition rule introduced in the following subsections is traceable to a row in one of these two tables.

Table 4.3: Layer 1 – Conceptual Foundations. Each row identifies one theoretical idea that motivates a modelling decision. No equations or implementation details appear in this layer.

Modelling Aspect	Theoretical Contribution	Primary Source
Cognitive Learning Dynamics	Mastery evolves through saturating learning with diminishing gains near proficiency.	Guzmán & Cruz-Mercado [18]
Affective State Modelling	Academic emotions persist over time rather than resetting after each interaction.	Henke et al. [131]
Cognitive-Affective Coupling	Cognitive outcomes and affective states are mutually dependent and should be modelled jointly.	Wang [132]
Academic Emotion Taxonomy	Four discrete academic emotions govern learning behaviour and motivational states.	Pekrun [7]
Challenge-Skill Balance	Optimal learning occurs within a moderate difficulty band; mismatches drive negative affect.	Csikszentmihalyi [133]
Multi-Objective Reward Design	Tutoring rewards should reflect both learning progress and affective well-being simultaneously.	Kong et al. [38]
Utility Aggregation	Cognitive and affective objectives should be combined into a single unified optimisation target.	Hernández et al. [134]
Partial Observability	Tutoring agents operate without direct access to latent task parameters.	Ramachandran et al. [135]

Table 4.4: Layer 2 – Theory-to-Methodology Traceability (Part A: Cognitive Model and Affective State Structure). Each row states one inherited mechanism, its core formal rule, and the originating source. Specific coefficient values and hyperparameter settings appear in the body text. Part B (Table 4.5) and Part C (Table 4.6) follow.

Component	Methodological Rule	Formal Rule	Source
Cognitive Model			
Item correctness	Computes response correctness from the difference between learner mastery and task difficulty.	$\Pr(\text{correct}) = \sigma(\beta(k - \alpha d))$	Guzmán & Cruz-Mercado [18]
Mastery update	Mastery grows toward saturation, with stronger updates after incorrect responses than correct ones.	$k_{t+1} = \text{clip}(k_t + g_t(1 - k_t) - \delta_t)$	Guzmán & Cruz-Mercado [18]
Affective State Structure			
Factored learner state	Represents cognition and affect as separate components within a single learner state.	$\mathbf{s}_t = (k_t, e_t, f_t, c_t, b_t, \iota_t)$	Wang [132]
Cognitive–affective coupling	Affect evolves based on emotion, action, and mastery state..	$P(e' \mid e, a, d')$	Wang [132]

Table 4.5: Layer 2 – Theory-to-Methodology Traceability (Part B: Affective Dynamics, Challenge–Skill Mismatch, Emotion Taxonomy, Pedagogical Interventions, and Reward Function). Continuation of Table 4.4 (Part A); Part C (Table 4.6) follows.

Component	Methodological Rule	Formal Rule	Source
Affective Dynamics			
Affective AR(1) dynamics	Affect changes gradually toward targets influenced by action and task–skill mismatch.	$x_{t+1} = \lambda_x x_t + (1 - \lambda_x) \tau_x(a_t, m_t) + \varepsilon_t$	Henke et al. [131]
Challenge–Skill Mismatch			
Mismatch regimes	Task–mastery gap defines whether the task is too hard, too easy, or in the flow zone, shaping affective responses.	$m_t = d_t - k_t$; $m_t \gg 0 \Rightarrow \text{hard}$; $m_t \ll 0 \Rightarrow \text{easy}$; $m_t \approx 0 \Rightarrow \text{flow}$	Csikszentmihalyi [133]
Emotion Taxonomy and Discretisation			
Discrete emotion rule	Maps affect to four discrete emotion classes through thresholding.	$\iota_t \leftarrow \text{priority}(f_t, c_t, b_t)$ $\rightarrow \{\text{frustrated, confused, bored, engaged}\}$	Pekrun [7]
Pedagogical Interventions			
Action set and affective targets	Each intervention specifies how it modifies affective targets based on pedagogical theory.	$\mathcal{A} = \{\text{hint, scaffold, encouragement, simplify, harder, break, explanation, no-action}\}$; each a defines $\tau_x^{\text{act}}(a)$	Sudharsan et al. [136]
Reward Function			
Normalised learning gain	Measures progress as normalized gain in mastery.	$\Delta k_{\text{norm}} = \frac{k_{t+1} - k_t}{1 - k_t}$	Hake [137]
Multi-objective reward	Uses a single reward combining learning progress with affective regulation signals.	$r_t = w_k \Delta k_{\text{norm}} + w_e e_t - w_f f_t - w_b b_t - w_c c_t$	Kong et al. [38]

Table 4.6: Layer 2 – Theory-to-Methodology Traceability (Part C: Partial Observability, Episode Termination, and DQN Learning Algorithm). Continuation of Table 4.4 (Part A) and Table 4.5 (Part B), completing the traceability matrix.

Component	Methodological Rule	Formal Rule	Source
Partial Observability and Episode Termination			
Hidden task difficulty	Task difficulty is hidden from the agent, consistent with a POMDP tutoring setting.	$\mathbf{o}_t = (k_t, e_t, f_t, c_t, b_t, \iota_t); d_t \notin \mathbf{o}_t$	Ramachandran et al. [135]
Dropout termination	Episode ends when sustained frustration exceeds a dropout threshold, representing learner disengagement..	$\phi_t = 1$ sustained for $\geq T_{\text{drop}}$ steps $\Rightarrow$ episode terminated	Ramachandran et al. [135]
DQN Learning Algorithm			
DQN optimisation	The Q-network is trained by minimizing the difference between its predictions and stable targets generated by a frozen copy of the network, using sampled past experiences for stable learning.	$y_t = r_t +$ $\gamma \max_{a'} Q^-(\mathbf{s}_{t+1}, a'; \theta^-);$ $L(\theta) =$ $\mathbb{E}_{\mathcal{B}}[\text{smoothL}_1(y_t -$ $Q(\mathbf{s}_t, a_t; \theta))]$	Mnih et al. [46]; Raffin et al. [3]

The block diagram of Figure 4.3 renders the learner state as $\mathbf{s}_t = [k_t, e_t]$ for visual compactness. In the formal model, the symbol e_t expands into the four affective scalars and the discrete academic-emotion identifier defined in Equation [4.13]. The resulting formulation is consistent with established affective tutoring architectures in the literature [132, 18, 38].

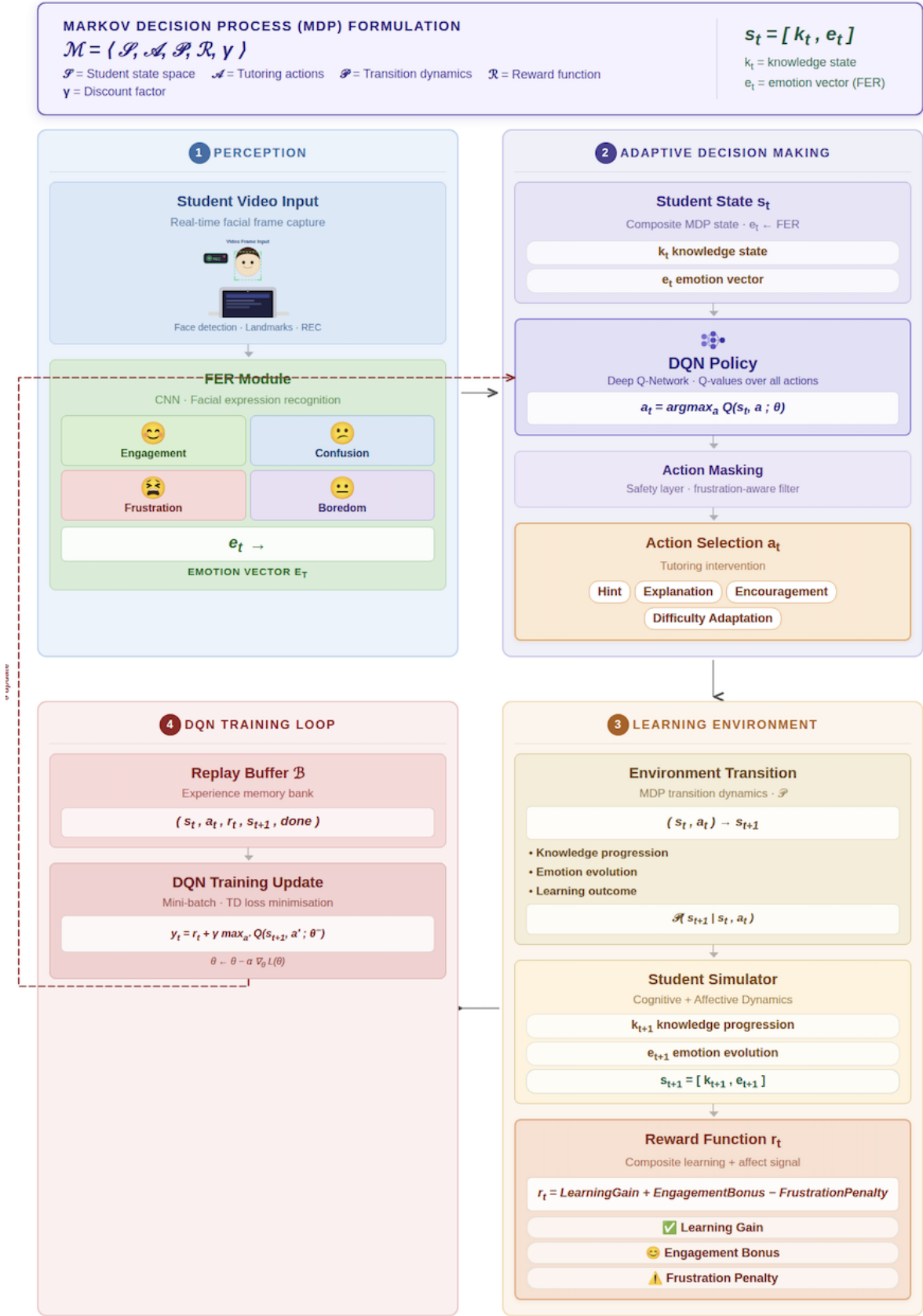
4.6.2 System Architecture and Reinforcement Learning Formulation

The learning environment is formalised as a discrete-time Markov decision process. Student and tutor interactions are modelled through learner observations, pedagogical actions, and stochastic transitions over a cognitive and affective state.

At each decision step t , the environment emits an observation $\mathbf{o}_t$ and an admissibility mask $\mathbf{m}_t \in \{0, 1\}^{|\mathcal{A}|}$ that encodes pedagogical safety constraints. The DQN policy then selects an admissible action a_t . The learner state evolves from $\mathbf{s}_t$ to $\mathbf{s}_{t+1}$ under the transition kernel of Sec-

tions 4.6.3 and 4.6.3. Finally, a scalar reward r_t is returned by the multi-objective formulation of Section 4.6.3.

Episodes terminate under three conditions: task completion when mastery exceeds a fixed threshold, persistent negative affect indicating learner disengagement, or reaching the maximum interaction horizon.



The system is structured as a reinforcement learning architecture consisting of interacting cognitive, affective, and policy components. A synthetic student simulator supplies the cognitive and affective dynamics. A Markov-decision-process environment orchestrates the transition kernel. An affective interface produces the discrete academic-emotion identifier. A reward formulation combines cognitive gain with affective regulation. A DQN agent learns the policy. An evaluation pipeline aggregates outcomes across episodes and seeds.

Training is performed offline against a synthetic population, consistent with prior reinforcement-learning-based tutoring simulations [18, 38]. The real-time facial-expression-recognition pipeline of Section 4.4 is treated as a modular extension that can replace the synthetic affective channel at deployment. It is not activated in the reported experimental setting.

4.6.3 Markov Decision Process and Environment Model

This section formally defines the Markov decision process underlying the adaptive tutoring system and specifies the models that govern cognitive state evolution, affective dynamics, learner heterogeneity, task structure, observation, pedagogical actions, and reward. Together these components constitute the theory-inherited environment within which the DQN policy is trained.

State Space and Cognitive Model

The tutoring task is formalised as a Markov decision process $\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, R, \gamma)$ with discount factor $\gamma = 0.99$ following standard practice in deep reinforcement learning and serves as a design hyperparameter controlling the emphasis on long-term rewards [46]. Following the factored intelligent-tutoring POMDP of Wang [132] and the partial-observability framing of Ramachandran et al. [135], the latent learner state combines one cognitive component, four continuous affective dimensions, and one categorical academic-emotion identifier:

$$\mathbf{s}_t = (k_t, e_t^{\text{eng}}, f_t, c_t, b_t, \iota_t) \in [0, 1]^5 \times \{0, 1, 2, 3\}, \quad [4.13]$$

where k_t is the scalar mastery, e_t^{eng} engagement, f_t frustration, c_t confusion, b_t boredom, and ι_t the academic-emotion identifier defined over the four-class taxonomy of Pekrun [7].

A single scalar mastery variable $k_t \in [0, 1]$ governs the cognitive state evolution. No auxiliary skill vector or item-level knowledge component is modelled. An internal latent task difficulty $d_t \in [0, 1]$ is maintained by the environment but is not exposed to the policy, in line with the partial-observability assumption of Ramachandran et al. [135]. The observation accessible to the agent is therefore a projection of $\mathbf{s}_t$ together with an optional affective-persistence channel introduced in Section 4.6.3.

The cognitive component follows the saturating learning model of Guzmán & Cruz-Mercado [18]. The probability of a correct response on an item of difficulty d by a learner of mastery k is modelled by a logistic function of their difference:

$$\Pr(\text{correct} \mid k, d) = \sigma(\beta(k - \alpha d)), \quad [4.14]$$

with discrimination $\beta = 3.0$ and difficulty scale $\alpha = 1.0$.

Mastery itself evolves by a saturating update on the remaining headroom $(1 - k_t)$, scaled by a learning gain g_t that depends on the chosen pedagogical action and on the learner's personality, together with a forgetting correction and Gaussian noise. The core learning rule is

$$k_{t+1} = \text{clip}_{[0,1]}(k_t + g_t(1 - k_t) - \delta_t + \eta_t). \quad [4.15]$$

The learning gain is defined as $g_t = g_{\text{base}}(a_t) \gamma_s \kappa(\text{correctness}_t)$, where $g_{\text{base}}(a_t)$ is the base increment associated with action a_t , γ_s is the personality-level learning rate, and κ is a multiplicative factor that depends on whether the learner's response was correct. The forgetting term δ_t subtracts a small fraction of mastery when the response is incorrect or when engagement falls below 0.3, capturing the disengagement-driven decay reported in the academic-emotion literature [7]. The noise term $\eta_t \sim \mathcal{N}(0, 0.02^2)$ models residual stochasticity.

The per-learner parameters $(\gamma_s, \beta_s, \lambda_s, \rho_s)$ scale, respectively, the learning rate, the sensitivity to challenge and skill mismatch, the forgetting rate, and the patience. Their population-level sampling distributions are specified in Section 4.6.3.

Affective State Dynamics

The affective dynamics are constructed in five conceptual steps and operationalised by Equation [[4.16]] below.

Step 1: cognitive update. The mastery update of Equation [[4.15]] produces an updated mastery k_{t+1} , which subsequently shapes the challenge and skill mismatch entering the affective targets.

Step 2: emotional persistence. Each continuous affective dimension $x \in \{e^{\text{eng}}, f, c, b\}$ retains a fraction λ_x of its previous value, following the first-order autoregressive account of academic affect of Henke et al. [131]. The empirically derived persistence coefficients are $\lambda_f = 0.69$, $\lambda_e = 0.60$, $\lambda_c = 0.47$, and $\lambda_b = 0.36$.

Step 3: mismatch effect. The challenge and skill mismatch $m_t = d_t - k_{t+1}$ drives a baseline target τ_x^{mm} for each affective dimension. When $m_t > 0.3$, the task is too hard and the targets push frustration and confusion upward. When $m_t < -0.3$, the task is too easy and the boredom target rises. Within the band $|m_t| \leq 0.3$, the targets fall in a moderate-engagement regime consistent with the flow-zone construct of Csikszentmihalyi [133].

Step 4: intervention modulation. The chosen pedagogical action a_t may override selected target dimensions with action-specific values $\tau_x^{\text{act}}(a_t)$, following the intervention catalogue of Sudharsan et al. [136]. For example, hints lower the confusion target, encouragement lowers the frustration target, and breaks raise the engagement target.

Step 5: final state composition. The effective target combines the mismatch baseline and the intervention overrides into a single value $\tau_x(a_t, m_t)$. Each affective dimension is then updated by

$$x_{t+1} = \text{clip}_{[0,1]}(\lambda_x x_t + (1 - \lambda_x) \tau_x(a_t, m_t) + \varepsilon_t), \quad \varepsilon_t \sim \mathcal{N}(0, 0.03^2). \quad [4.16]$$

Combined with the mastery update of Equation [[4.15]], this composition realises the factored cognitive and affective transition $P(e' \mid e, a, d')$ proposed by Wang [132].

Emotion taxonomy and two separate thresholding rules. The categorical academic-emotion identifier $\iota \in \{\text{confused}, \text{bored}, \text{frustrated}, \text{engaged}\}$ adopts Pekrun’s four-class taxonomy [7]. Two thresholding rules govern ι . These two mechanisms operate in separate pipelines and must not be conflated.

Rule A: synthetic emotion discretisation. In the synthetic-training regime, the continuous affective scalars are mapped to a discrete class by a priority rule applied to their intensities. The label *frustrated* is assigned when $f > 0.6$, *confused* when $c > 0.5$, *bored* when $b > 0.5$, and *engaged* otherwise. These intensity thresholds were not chosen arbitrarily but calibrated empirically on validation rollouts to maximise cumulative reward and stabilise learning dynamics.

Rule B: facial-expression-recognition confidence filtering. When the real-time facial-expression-recognition pipeline of Section 4.4 replaces the synthetic affective channel, a predicted emotion label is accepted only when the network’s predicted probability exceeds a confidence threshold of 0.5. If the confidence falls below this threshold, the previously emitted emotion is retained. This confidence threshold was likewise selected through empirical tuning on validation experiments, balancing label responsiveness against noise-induced policy instability.

Although the two rules share the numerical value 0.5, they apply to different quantities. Rule A thresholds affective intensities. Rule B thresholds classifier confidences. The two rules never coexist within a single inference path.

Synthetic Student Population

Training is performed against a fixed population of $N = 10,000$ synthetic learners. This sample size is consistent with prior reinforcement-learning tutoring simulations [18, 38, 135] and is chosen to ensure statistical stability of aggregate metrics across seeds and learner heterogeneity.

Each synthetic learner is parameterised by nine heterogeneity coefficients drawn independently from uniform distributions, covering the personality parameters $(\gamma_s, \beta_s, \lambda_s, \rho_s)$ of Section 4.6.3 together with auxiliary traits of frustration tolerance, boredom sensitivity, engagement recovery, baseline confidence, and persistence. For reporting purposes only, each learner is post-hoc classified into one of four illustrative types (fast learner, slow learner, easily frustrated, patient); this classification has no effect on transition dynamics.

At the start of each tutoring episode, the initial learner state is sampled from learner-specific uniform ranges over k_0 , e_0^{eng} , f_0 , c_0 , and b_0 , the initial academic-emotion identifier is set to *engaged*, and the latent task difficulty d_0 is drawn from a uniform range. The difficulty is subsequently modified only through the two difficulty-adjusting interventions defined in Section 4.6.3 and remains hidden from the policy, in accordance with the partial-observability assumption of Ramachandran et al. [135].

Observation, Transition, and Termination

State definition. The latent learner state $\mathbf{s}_t$ defined in Equation [[4.13]] carries the scalar mastery, the four continuous affective scalars, and the categorical academic-emotion identifier. The latent task difficulty d_t is internal to the environment and is updated only through the difficulty-adjusting interventions of Section 4.6.3.

Observation model. The observation vector emitted to the policy is the projection of $\mathbf{s}_t$ onto the channels visible to the agent:

$$\mathbf{o}_t^{(6)} = (k_t, e_t^{\text{eng}}, f_t, c_t, b_t, l_t)^\top. \quad [4.17]$$

When the affective-persistence channel is enabled, the observation is extended with an additional feature that measures how long frustration has been continuously active. This value is computed from the number of consecutive steps with active frustration, normalised and clipped to the range $[0, 1]$. Channel-masking ablations then set individual observation components to zero to evaluate their contribution without retraining the model.

Transition function. The state transition $P(\mathbf{s}_{t+1} \mid \mathbf{s}_t, a_t)$ is built from four steps: response correctness, mastery update, affect update, and emotion classification. First, student responses are generated using the logistic correctness model in Equation [[4.14]]. Next, learner mastery is updated using the rule in Equation [[4.15]]. Affective states are then updated gradually using the autoregressive model in Equation [[4.16]]. Finally, continuous affect is mapped to discrete emotion classes as described in Section 4.6.3.

In addition, the environment tracks a persistent-frustration flag $\phi_t \in \{0, 1\}$, which becomes active when frustration remains above 0.7 for three consecutive steps and is reset when it drops below 0.5.

Episode termination. Episodes terminate under three conditions. The first condition is task completion, triggered when $k_{t+1} \geq 0.95$. The second condition is a learner-dropout event, triggered when ϕ_t remains active for at least five consecutive interaction steps and treated as a dropout signal in the sense of Ramachandran et al. [135]. The third condition is the maximum interaction horizon, reached when the episode length attains fifty interaction steps.

Reward Function

The reward function is operationalised as a multi-objective scalar that combines cognitive gain with affective regulation, following the formulation of Kong et al. [38]. The theoretical motivation for jointly optimising cognitive and affective objectives is discussed in Section 4.6.1. The reward is constructed in five conceptual layers, summed, and then clipped.

(1) Cognitive term. The cognitive component is the normalised learning gain of Hake [137], defined on the remaining mastery headroom:

$$\Delta k_{\text{norm}} = \text{clip}\left(\frac{k_{t+1} - k_t}{1 - k_t + \varepsilon}, -1, 1\right), \quad \varepsilon = 10^{-8}. \quad [4.18]$$

This term rewards genuine mastery gains, normalised relative to the headroom still available to learn.

(2) Engagement term. A positive contribution $w_e e_{t+1}^{\text{eng}}$ rewards states of high engagement, following Kong et al. [38].

(3) Affective penalty terms. Three subtractive contributions $w_f f_{t+1}$, $w_b b_{t+1}$, and $w_c c_{t+1}$ penalise frustration, boredom, and confusion, respectively.

(4) Auxiliary penalties. The reward includes two additional penalties to handle undesirable tutoring situations. A persistent-frustration penalty is applied when the learner remains in a sustained high-frustration state. A second penalty discourages taking no meaningful pedagogical action when the learner is already in a struggling condition, defined by elevated confusion or frustration levels.

(5) Final clipped reward. The five layers are summed and clipped to $[-1, 1]$, following the value-stability prescription of Mnih et al. [46]:

$$r_t = \text{clip}_{[-1,1]}\left(w_k \Delta k_{\text{norm}} + w_e e_{t+1}^{\text{eng}} - w_f f_{t+1} - w_b b_{t+1} - w_c c_{t+1} - w_f \Pi \mathbb{K}[\phi_{t+1}] - \rho \mathbb{K}[\text{struggle}_t]\right). \quad [4.19]$$

The active weight vector is obtained through tuning and set to

$$(w_k, w_e, w_f, w_b, w_c) = (0.35, 0.20, 0.15, 0.15, 0.15).$$

All experiments use the reward formulation defined in Equation [[4.19]].

Action Space and Pedagogical Interventions

The action space consists of eight discrete pedagogical interventions:

$$\mathcal{A} = \{\text{hint, scaffolding, encouragement, simplify, harder, break, explanation, no-action}\}.$$

instantiating the intervention catalogue of Sudharsan et al. [136]. Each intervention is applied within the environment by modifying the affective dynamics through the autoregressive update in Equation [[4.16]]. Formally, each action induces an affect modulation vector

$$\tau(a_t) = (\tau_f, \tau_c, \tau_b, \tau_e),$$

which determines the directional influence on frustration, confusion, boredom, and engagement. When an entry is context-dependent, its value is governed by the challenge–skill mismatch mechanism defined in Section 4.6.3.

The interventions follow canonical pedagogical effects reported in Sudharsan et al. [136], where supportive actions reduce negative affect (e.g., frustration and confusion), while restructuring actions adjust task difficulty to sustain engagement.

Admissibility of actions is enforced prior to policy execution using a binary mask $\mathbf{m}_t \in \{0, 1\}^{|\mathcal{A}|}$. This mask is computed using three safety constraints applied in priority order.

First, a recency constraint suppresses recently executed interventions over a short cooldown horizon to prevent repetitive policy behaviour.

Second, the difficulty-increasing intervention is disabled when frustration exceeds a predefined threshold or when the learner is already classified as highly frustrated.

Third, under persistent-frustration conditions, only supportive interventions remain admissible, restricting the action set to recovery-oriented pedagogical strategies.

A separate emotion-to-action compatibility mapping is defined solely for post-hoc evaluation of pedagogical alignment (Section 4.6.5). This mapping is not used during training or reward computation and does not influence policy learning.

4.6.4 DQN Architecture and Training

This section specifies the Q-network architecture, the off-policy optimization procedure, and the full training protocol used to develop the DQN tutoring agent.

Q-Network Architecture and Learning Algorithm

The DQN agent learns a state-action value function following Mnih et al. [46], implemented using Stable-Baselines3 [3]. The network is a two-layer multilayer perceptron with ReLU

activations, mapping the observation vector in Equation [[4.17]] to a value for each pedagogical action in $\mathcal{A}$.

Training follows the standard DQN procedure with a target network Q^- whose parameters are periodically copied from the online network every 2,000 steps. The temporal-difference target is defined as:

$$y_t = r_t + \gamma \max_{a'} Q^-(\mathbf{s}_{t+1}, a'). \quad [4.20]$$

The network is trained by minimising a smooth- L_1 loss over transitions sampled from a replay buffer:

$$L(\theta) = \mathbb{E}_{\mathcal{B}} [\text{smoothL}_1(y_t - Q(\mathbf{s}_t, a_t))]. \quad [4.21]$$

All training hyperparameters, including optimisation settings, replay buffer configuration, and exploration schedule, are defined in the Implementation Details section (Section 4.6.6).

Pedagogical constraints are enforced through an action mask. Inadmissible actions are excluded by setting their value to $-\infty$ before action selection, ensuring that the policy never selects unsafe interventions. During exploration, any invalid action is resampled before execution.

Training and Inference Procedure

The policy is trained across ten independent random seeds, each instantiating a synthetic population of 10,000 learners and a budget of 50,000 interaction steps. Learning proceeds through continual interaction with the composite transition kernel of Section 4.6.3, in which cognitive, affective, and admissibility dynamics jointly evolve and shape the reward signal of Equation [[4.19]]. Policy parameters are refined by the DQN updates of Equations [[4.20]]–[[4.21]], while periodic checkpointing every 10,000 steps preserves the snapshot of highest held-out performance.

At inference, the trained policy operates over the admissible action set under one of three decision regimes, greedy, residual ε -greedy with $\varepsilon = 0.05$, or temperature-scaled softmax, supporting a spectrum from deterministic to exploratory behaviour. The resulting tutoring agent is intrinsically adaptive: interventions emerge from the evolving learner state rather than from any predefined pedagogical script, allowing identical observations to elicit different responses as mastery, affect, admissibility, and persistent frustration evolve.

4.6.5 Evaluation Protocol

Each trained policy is evaluated over 1,000 episodes per seed across all ten seeds. Per-seed means are aggregated into a normal-approximation 95% confidence interval, $\bar{x} \pm 1.96 s/\sqrt{n}$ with $n = 10$, reported for every metric defined below.

Performance Metrics

For each evaluation episode of horizon T , the following theory-grounded quantities are computed.

Cumulative reward.

$$R = \sum_{t=0}^T r_t. \quad [4.22]$$

Raw learning gain.

$$\Delta k = k_T - k_0. \quad [4.23]$$

Normalised learning gain. Δk_{norm} from Equation [[4.18]] [137], evaluated end-to-end:

$$\Delta k_{\text{norm}}^{(0:T)} = \frac{k_T - k_0}{1 - k_0}. \quad [4.24]$$

Success indicator. Successful learning is defined as the joint satisfaction of mastery attainment, regulated negative affect, and task completion, formulated symbolically as

$$\mathcal{S} = \mathbb{K} \left[k_T \geq k^* \wedge \max\{f_T, c_T\} \leq \eta^* \wedge \iota_T \in \mathcal{E}^+ \right], \quad [4.25]$$

where k^* denotes the mastery-attainment target, η^* the tolerable upper bound on negative affective intensities, and $\mathcal{E}^+$ the subset of academic-emotion classes consistent with productive engagement.

Convergence indicator.

$$\mathcal{C} = \mathbb{K}[\sigma_W(R) < \varepsilon_c], \quad [4.26]$$

with $\sigma_W(R)$ the rolling standard deviation of the cumulative reward over the last W episodes and ε_c a convergence tolerance.

As a degeneracy check, the empirical frequency of each intervention $a \in \mathcal{A}$ is recorded and a dominance warning is raised whenever any single frequency exceeds a usage-share threshold ρ^* .

Baselines and Comparisons

Baseline design draws on rule-based adaptive tutoring [39] and RL benchmarking practice for educational environments [36]. All baselines share the environment, synthetic population, and seed schedule of the DQN policy:

1. **Fixed policy baseline:** a uniform random agent over the admissible action set $\mathcal{A}$.

2. **Rule-based adaptive baseline:** a hand-crafted priority waterfall over admissible interventions, conditioned on ι_t and the mismatch regime, following [39].

All comparators are evaluated under the metrics and aggregation defined above. Robustness is verified by re-running the protocol under perturbations of the reward weights $(w_k, w_e, w_f, w_b, w_c)$ and the simulator coefficients on identical seeds, reporting whether the agent ordering is preserved.

4.6.6 Implementation Details

The reinforcement-learning module is implemented in Python on top of the Gymnasium simulation interface standard [138] and the Stable-Baselines3 reference DQN learner [3]. Reproducibility is enforced by fixing the random number generators of every stochastic component at the start of each run and by seeding the synthetic population sampler and the per-episode resets with deterministic offsets. Table 4.7 collects every hyperparameter referenced in the preceding subsections, together with an explicit provenance for each value.

Table 4.7: Hyperparameters of the DQN tutor with explicit provenance. Each parameter is classified as literature-derived (with citation), inherited as a Stable-Baselines3 default [3], fixed as an experimental setting, or selected through empirical tuning on validation rollouts. The classification is intended to make the dependence of each value on prior work transparent and to avoid implying that all settings were directly inherited from theory.

Hyperparameter	Value	Source
Reinforcement-learning algorithm	DQN	Literature-derived [46, 3]
Q-network architecture	Two hidden layers of width 64, ReLU activations	Literature-derived [39]
Optimiser	Adam	SB3 default [3]
Loss function	Smooth- L_1 (Huber)	SB3 default [3]
Learning rate α	5×10^{-4}	SB3 default [3]
Mini-batch size	64	SB3 default [3]
Replay-buffer capacity	100,000	Literature-derived [46, 3]
Discount factor γ	0.99	Literature-derived [46]
Gradient-step frequency	One update per four interaction steps	Literature-derived [46, 3]
Target-network update interval	2,000 interaction steps (hard copy)	Literature-derived [46]
Exploration schedule	Linear ε -greedy, $1.0 \rightarrow 0.05$ over the first 30% of training	Literature-derived [46]
Final exploration ε	0.05	Literature-derived [46]
Reward weights (w_k, w_e, w_f, w_b, w_c)	(0.35, 0.20, 0.15, 0.15, 0.15)	Empirically tuned
Persistent-frustration penalty Π	1.0	Empirically tuned
No-action struggle penalty ρ	0.02	Empirically tuned
Training budget per seed	50,000 interaction steps	Experimental setting, inspired by [14]
Maximum interaction steps per episode	50	Experimental setting
Evaluation episodes per seed	1,000	Experimental setting (statistical reliability)

Parameters labelled as *Tuned* were selected through empirical validation, whereas the remaining values were adopted from prior literature, Stable-Baselines3 defaults, or experimental design choices.

4.6.7 Ethical Considerations

Policy learning is conducted entirely within a synthetic learner population, and no human-subject data are used for training or policy optimisation. The use of simulated learners is consistent with established reinforcement-learning tutoring research and provides a safe environment for policy development prior to any live educational deployment [139, 140].

4.7 Chapter Summary

This chapter specified the methodological framework of the proposed affect-aware intelligent tutoring system.

- System architecture: FER-based perception module coupled with a DQN-based pedagogical adaptation module.
- Data pipeline: DAiSEE-based acquisition with frame extraction, temporal resampling, preprocessing, and augmentation.
- Perception model: ResNet-50 transfer-learning classifier for confidence-weighted academic-emotion inference.
- Emotion representation: four-class control-value taxonomy spanning boredom, engagement, confusion, and frustration.
- Affective interface: streaming mapping from FER outputs to the agent observation space.
- Synthetic learner model: mastery dynamics, first-order affect persistence, latent task-difficulty process.
- State formulation: factored representation over mastery, affective variables, and discrete emotion under POMDP constraints.
- Action space: pedagogical interventions with admissibility masking, cooldown regulation, and safety constraints.
- Reward model: multi-objective optimisation of normalised learning gain and affect-sensitive terms.
- Learning algorithm: Deep Q-Network with experience replay, target-network stabilisation, and ε -greedy policy.
- Evaluation protocol: multi-seed experimentation, baseline comparison, confidence-interval estimation, episode-level metrics.
- Reproducibility framework: deterministic execution, seed control, and hyperparameter traceability.

Empirical results obtained under this methodological specification are reported in Chapter 5.

Chapter 5

Results and Discussion

5.1 Chapter Overview

This chapter is in two parts. Part I, in Section 5.2, presents the FER module on DAiSEE [45]. Part II, in Section 5.3, presents the adaptation module in a Gymnasium-compatible synthetic tutoring environment. The two parts use different protocols and are interpreted separately. The main RL experiments use simulator-derived affect states (`USE_REAL_FER=False`) and do not quantify end-to-end webcam-based tutoring.

5.2 FER Module Results on DAiSEE

5.2.1 Experimental Standardization

Prior FER work on DAiSEE evaluates models under inconsistent conditions. Papers vary in preprocessing pipeline, frame count, class-balancing strategy, loss function, and evaluation metric, and several of these choices are not reported explicitly in the source papers. A direct numerical comparison between two published results is therefore unreliable. Any observed gap may reflect data preparation rather than model design, and the information needed to reproduce a published result is not always present.

To remove this confound, one standardised preprocessing and evaluation pipeline was designed and applied to every architecture in this chapter. The 24-frame sampling rule, per-frame normalisation, the official DAiSEE splits, and the evaluation metrics are identical across runs. Every model was retrained from scratch under this pipeline. Architectural and training differences therefore explain any performance gap reported below, not preprocessing differences. The single-label formulation predicts one dominant class per clip with a softmax head and a cross-entropy or focal cross-entropy loss; the multi-label formulation predicts four independent

binary labels with a sigmoid head and a binary cross-entropy loss. The two are evaluated with different metrics and are not compared numerically as if they solved the same task.

5.2.2 Single-Label Formulation

Single-label classification was evaluated first to test whether one dominant label per clip could represent learner affect. Table 5.1 reports each architecture retrained under the standardised pipeline, with its source paper cited inline.

Table 5.1: Single-label results on the DAiSEE test split under the standardised pipeline.

Architecture	Loss / Sampling	Frames	Macro F1	Accuracy	Inspired From
ResNet50 + BiLSTM + Temp. Attn	Focal CE / bal. CSV	16	0.311	0.702	[30, 141]
ResNet50 + 2-BiLSTM + Temp. Attn	Focal CE / bal. CSV	24	0.271	0.581	[30]
ShuffleNetV2 + BiLSTM + Temp. Attn	Focal CE / OS	16	0.266	0.631	[27]
ShuffleNetV2 + SE + 2-BiLSTM + Temp. Attn	Focal CE / bal. CSV	24	0.269	0.651	[27]
HOG + LBP + SVM (handcrafted baseline)	RBF SVM / bal. weights	8	0.214	0.750	—
ResNet50 features + SVM (frozen baseline)	RBF SVM / bal. weights	16	0.229	0.759	[50]

The strongest single-label run reaches accuracy 0.702 and Macro F1 0.311. Its per-label F1 is 0.834 on engagement but 0.126, 0.161, and 0.122 on boredom, confusion, and frustration. Every single-label deep model reproduces this pattern, and the SVM baselines do so more extremely: they collapse to predicting engagement on almost every clip and score zero F1 on the three minority labels while still reaching 0.75 accuracy. High accuracy on DAiSEE indicates collapse to the dominant class rather than detection of all four states. The single-label head also cannot represent that a learner is simultaneously engaged and confused, which is required by the adaptation module. Figure 5.1 contrasts matched architectures under both formulations.

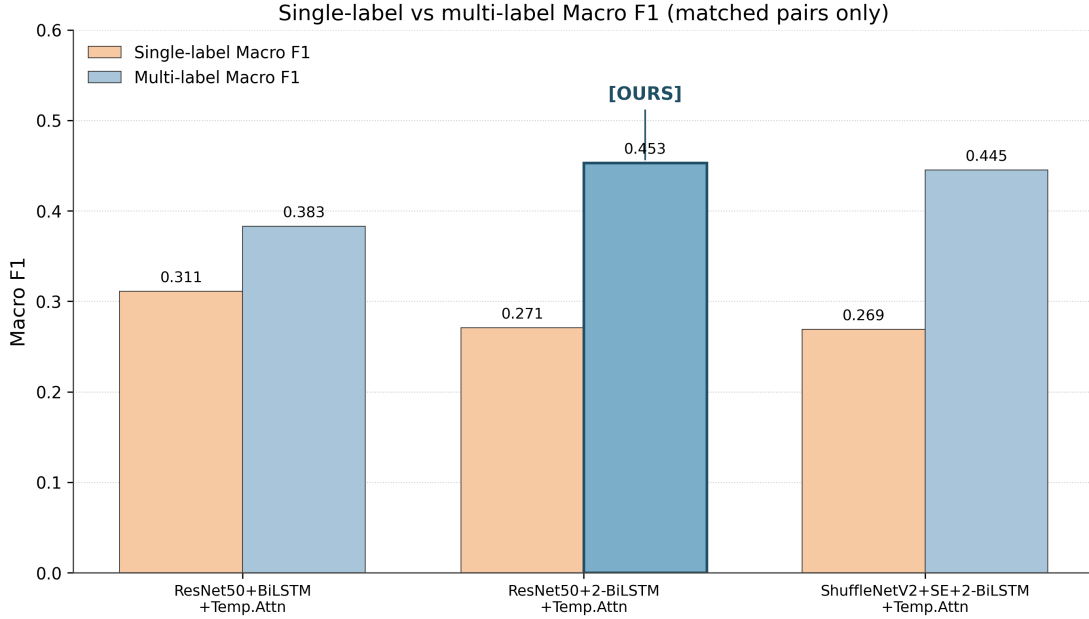


Figure 5.1: Macro F1 of architectures evaluated under both single-label and multi-label formulations. The proposed model is marked **[OURS]**.

5.2.3 Multi-Label Formulation

Table 5.2 reports the multi-label architectures retrained under the standardised pipeline. “WRS” denotes a WeightedRandomSampler; “OS-CSV” denotes fixed-multiplier oversampling (Eng. $\times 1$, Bor. $\times 2$, Con. $\times 3$, Fru. $\times 4$); “tuned” denotes per-label thresholds optimised on the validation split. The proposed model is marked **[OURS]** and its row is bold.

Table 5.2: Multi-label results on the DAiSEE test split under the standardised pipeline. The proposed system is **[OURS]**. Accuracy denotes mean per-label binary accuracy.

Architecture	Frames	Macro F1	Accuracy	Inspired From
ResNet50 + BiLSTM, weighted BCE	16	0.408	0.764	[30]
EfficientNetV2-S + BiLSTM	16	0.379	0.728	[109]
EfficientNetV2-S + Transformer	8	0.319	0.604	[109, 72]
ViT Transformer-based variant	16	0.280	0.751	[113]
ShuffleNetV2 + SE + 2-BiLSTM, tuned thr.	24	0.445	0.740	[27]
3D DenseNet + 3D Self-Attn	16	0.379	0.610	[31]
[OURS]: ResNet50 + SE + 2-BiLSTM + Temp. Attn	24	0.453	0.782	Proposed

Figure 5.2 plots Macro F1 across all multi-label runs. **[OURS]** is the highest entry.

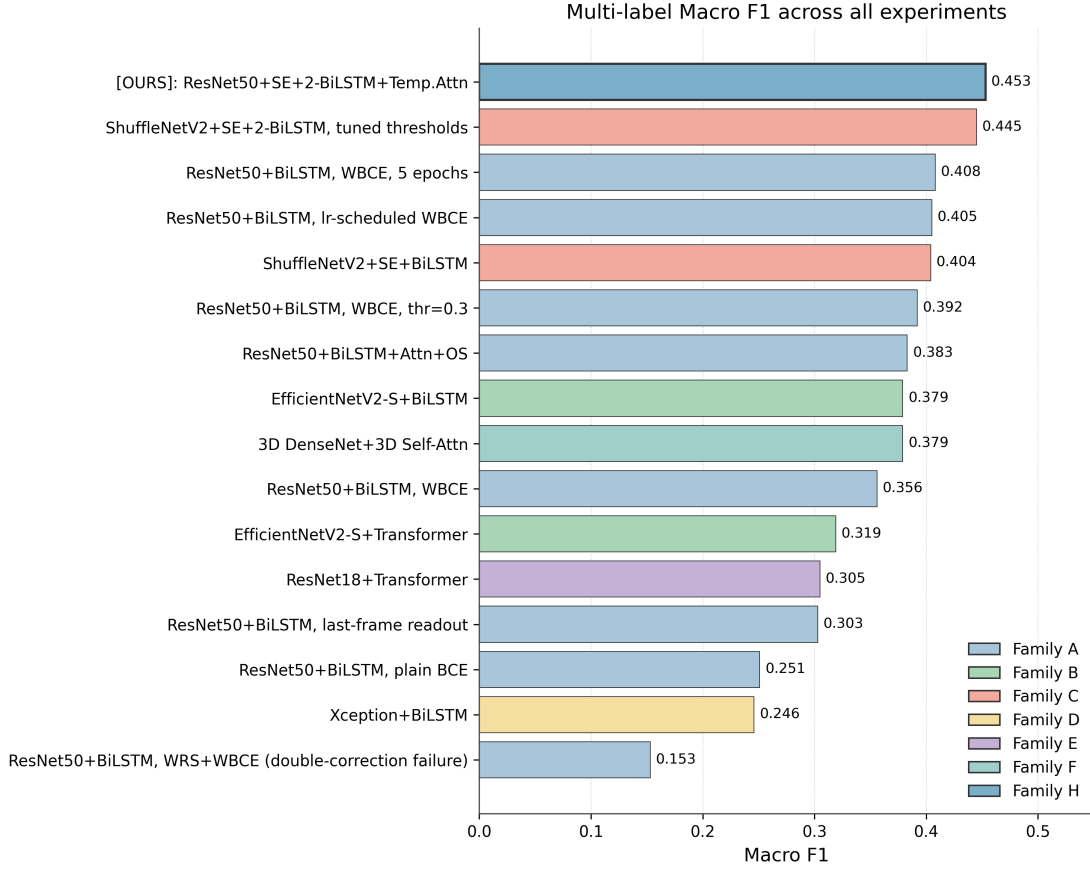


Figure 5.2: Multi-label Macro F1 for all architectures in Table 5.2, coloured by backbone family. [OURS] is the highest entry.

5.2.4 Test Set Performance of [OURS]

[OURS] was evaluated on the 1,784 DAiSEE test clips with per-label thresholds saved at the validation checkpoint (0.30 for boredom, 0.10 for engagement, 0.60 for confusion, 0.60 for frustration) and fixed at test time.

At the overall level, [OURS] reaches Macro F1 0.453, Micro F1 0.724, Weighted F1 0.798, Samples F1 0.746, and overall label accuracy 0.782. Weighted F1, Samples F1, and Micro F1 are dominated by engagement, the most frequent label. Macro F1 weights all four labels equally and falls below the others because boredom, confusion, and frustration remain harder than engagement under imbalance-aware training. Macro F1 is the primary metric in this thesis because it is the only metric that does not let engagement dominance hide poor minority-label performance.

At the per-label level, engagement reaches precision 0.951, recall 1.000, F1 0.975, and label accuracy 0.951. With 1,696 positive test clips and the lowest tuned threshold (0.10), recall is complete; the visual cues for engagement, alert eyes, stable head pose, and forward-leaning

posture, are also more reliable than the cues for the rare states [6, 30, 113]. Boredom reaches precision 0.239, recall 0.708, F1 0.358, and label accuracy 0.462, an intentional recall-over-precision trade-off appropriate for an adaptation system that must not miss disengagement. Confusion reaches precision 0.171, recall 0.325, F1 0.224, and label accuracy 0.802 on 157 positive clips, while frustration reaches precision 0.205, recall 0.338, F1 0.255, and label accuracy 0.911 on 80 positive clips. Both per-label accuracies above 0.80 are inflated by the dominant negative class (“not confused” prevalence 0.912, “not frustrated” prevalence 0.955). Both results clear the relevant benchmark of non-zero recall, which early baselines did not.

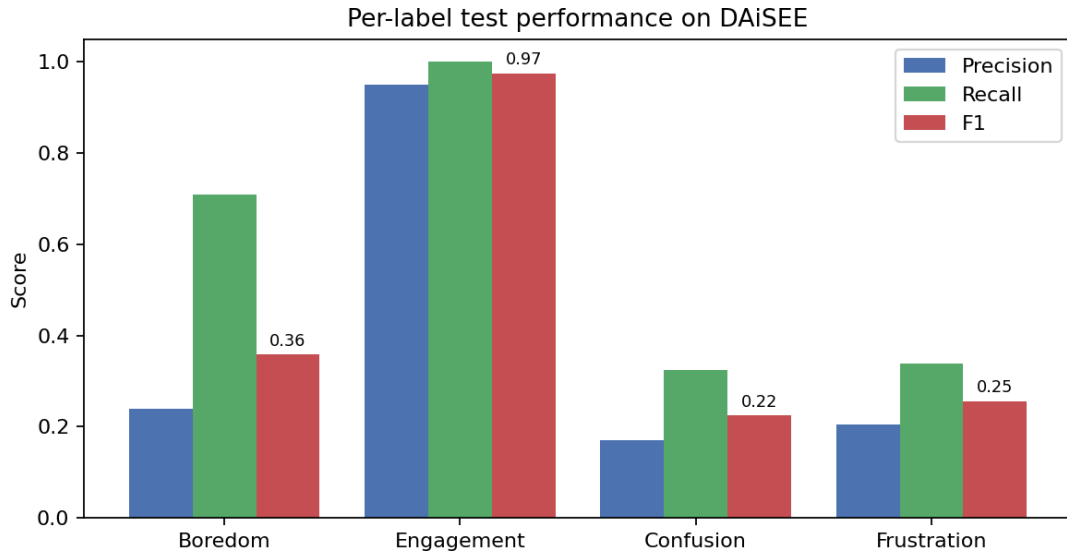


Figure 5.3: Per-label precision, recall, and F1 of **[OURS]** on the DAiSEE test split.

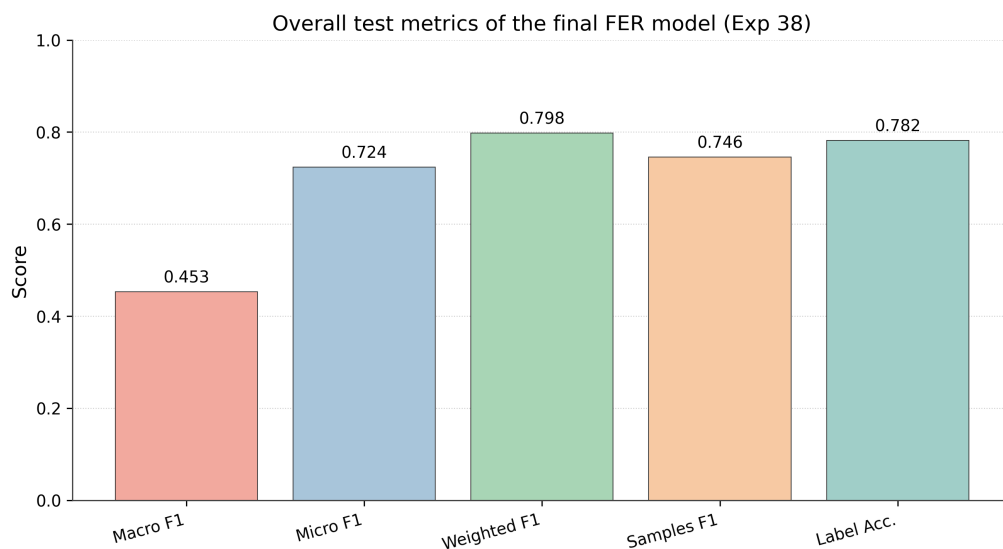


Figure 5.4: Overall test metrics of **[OURS]** on the DAiSEE test split: Macro F1, Micro F1, Weighted F1, Samples F1, and per-label accuracy.

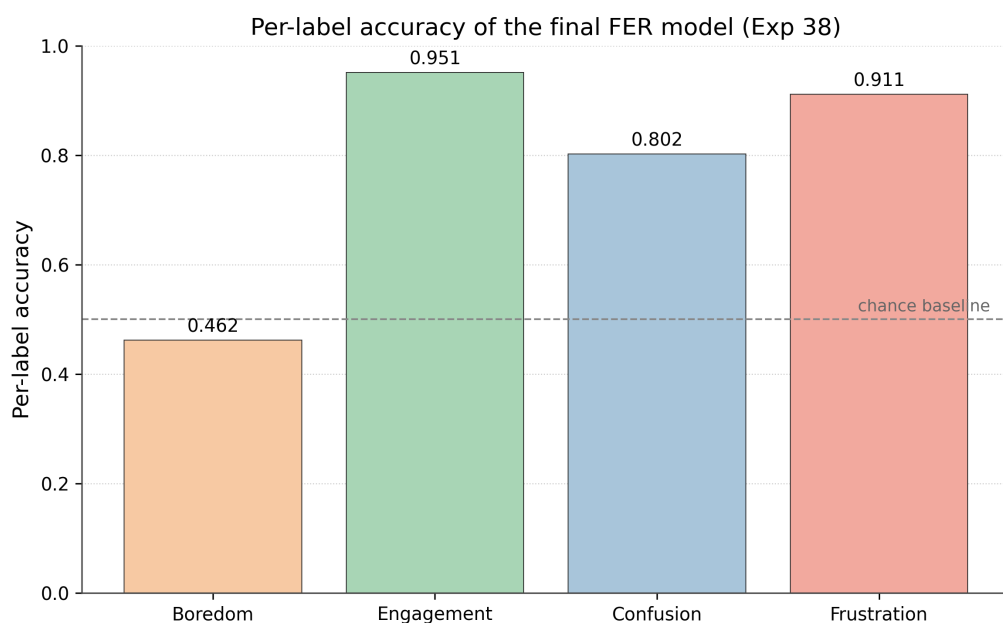


Figure 5.5: Per-label accuracy of **[OURS]**. The dashed line at 0.5 marks the balanced-binary chance baseline.

5.2.5 Training Behaviour

Training loss of **[OURS]** decreases monotonically from 0.7605 at epoch 1 to 0.4938 at epoch 5, while validation Macro F1 peaks at epoch 1 with 0.5117 and stays between 0.4795 and 0.5000

thereafter. Validation Micro F1 rises mildly from 0.6883 to 0.7348 and validation Weighted F1 is flat around 0.76 across the run. The training loss continues falling after epoch 1 while validation Macro F1 has already stopped improving, the standard signature of mild overfitting reported by other CNN-plus-recurrent DAiSEE pipelines at small batch sizes [70, 31, 109]. Early stopping selects the epoch-1 checkpoint because the ResNet50 backbone, initialised from ImageNet and fine-tuned at a small learning rate, has most of the visual representation needed for academic affect already in place at the start of training (Figure 5.6).

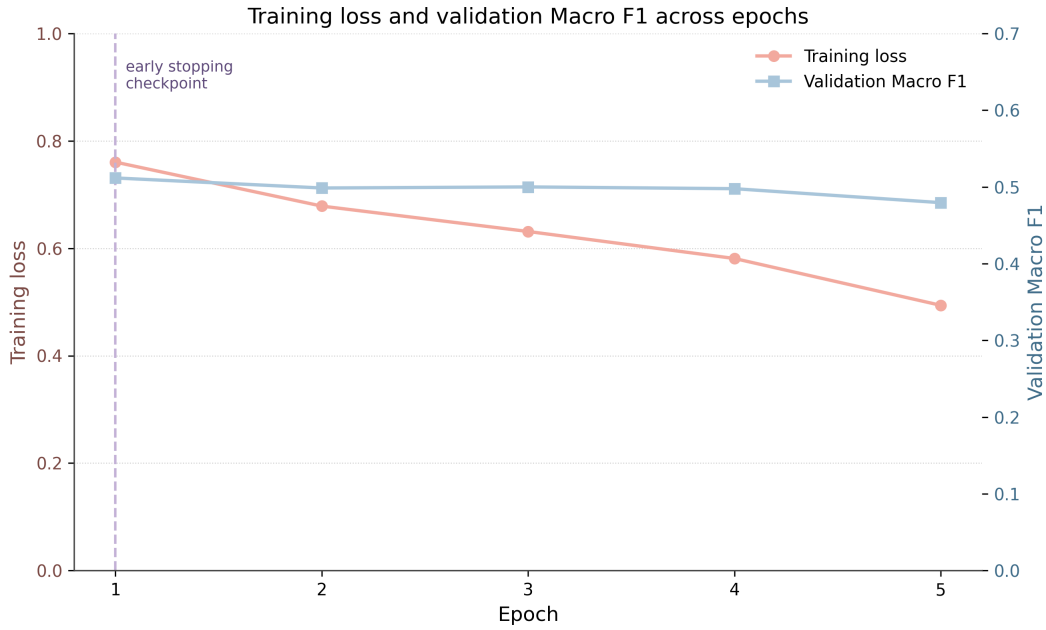


Figure 5.6: Training loss and validation Macro F1 across epochs for [OURS]. The dashed line at epoch 1 marks the early-stopping checkpoint.

5.2.6 Literature Context

Most prior DAiSEE work simplifies the task to a single-label four-class engagement problem and evaluates under undisclosed preprocessing. Reported metrics therefore reflect heterogeneous evaluation environments. Table 5.3 lists the prior work and the relationship each source has to this thesis. The Relationship column uses *Inspired by* when only a backbone or component was borrowed, *Reproduced under our pipeline* when the architecture was retrained from scratch under the standardised setup, and *Component borrowed* for a temporal or attention module re-used in modified form.

Table 5.3: Prior FER work on DAiSEE and its relationship to this thesis.

Work	Backbone	Temporal / At- tention	Labels	Accuracy (%)	Macro F1 [‡]	Relationship
Gupta et al. [45]	C3D / I3D	3D CNN	4-state ML	57.9 [†]	—	Dataset paper, used as baseline
Abedi and Khan [70]	ResNet	TCN	Eng. only	63.9	—	Inspired by, com- ponent borrowed (TCN head)
Mehta et al. [31]	3D DenseNet	3D self-attention	Eng. only	63.6	—	Reproduced under our pipeline
Shiri et al. [109]	EfficientNetV2- L	RNN	Eng. only	62.11	—	Backbone bor- rowed, downsized to V2-S
[OURS]	ResNet50 + SE	2-BiLSTM temp. attn.	+ 4-state ML	78.2	0.453	Proposed system

[†] The 57.9% for Gupta et al. is the average of per-class top-1 accuracy across four affective states (Boredom 53.7%, Engagement 57.9%, Confusion 72.3%, Frustration 73.5%) for their best model (LRCN); a single overall accuracy is not reported.

[‡] Macro F1 is not reported by any prior work in this table; these works use single-label multiclass accuracy on engagement-only or per-class top-1 accuracy, and Macro F1 across all four affective states cannot be inferred from their reported numbers.

Table 5.4: Extended DAiSEE accuracy comparisons from works using non-standard or engagement-only evaluation protocols (reproduced from cited sources).

Method	Accuracy (%)	Source
<i>Komaravalli & B [32] — customized DAiSEE subset[§]</i>		
Video-level InceptionNet	46.40	[32]
Frame-level InceptionNet	47.10	[32]
C3D feature extraction	48.60	[32]
I3D	52.35	[32]
ResNet + TCN with weighted loss	53.70	[32]
C3D fine tuning	56.10	[32]
C3D (focal loss)	56.20	[32]
LRCN	57.90	[32]
DFSTN	58.84	[32]
C3D + TCN	59.97	[32]
DERN	60.00	[32]
ResNet + LSTM	61.15	[32]
ResNet + TCN	63.90	[32]
3D DenseAttNet	63.59	[32]
Ours (Komaravalli & B)	64.00	[32]
<i>Rezaee et al. [142] — engagement-only, single-label DAiSEE[¶]</i>		
InceptionNet	46.40	[142]
C3D fine-tuning	56.10	[142]
LRCN	57.90	[142]
I3D	52.35	[142]
C3D (FL)	56.20	[142]
ResNet+TCN	53.60	[142]
DFSTN	58.84	[142]
DERN	60.00	[142]
Neural Turing Machine	61.30	[142]
ResNet+TCN with weighted loss	53.70	[142]
C3D+TCN	59.97	[142]
ResNet+LSTM	61.15	[142]
ResNet+TCN	63.90	[142]
DenseAttNet	63.59	[142]
EfficientNetV2-L+GRU	61.45	[142]
EfficientNetV2-L+Bi-GRU	61.56	[142]
EfficientNetV2-L+LSTM (proposed)	62.11	[142]
EfficientNetV2-L+Bi-LSTM	61.67	[142]

[§] Komaravalli & B trained and evaluated on a customized filtered subset of DAiSEE, not the standard train/val/test split. Results are not directly comparable to systems using the standard protocol.

[¶] Rezaee et al. evaluate on engagement-only (single label, 4-level ordinal) rather than all four affective states. Accuracy figures reflect engagement classification only and are not comparable to 4-state multilabel systems.

Reported values reflect each paper’s original evaluation environment. Direct numerical comparison with **[OURS]** is not appropriate.

The numbers in Table 5.3 cannot be ranked against each other or against **[OURS]**: each source uses its own preprocessing, splits, and label definitions, and Su et al.’s comparison table aggregates many methods across heterogeneous setups, so it cannot serve as a unified benchmark. Two characteristics nonetheless distinguish **[OURS]**: it is the only configuration evaluated in this thesis that performs four-state multi-label classification on DAiSEE under a single standardised pipeline, and its architecture is based on widely available deep-learning components that can be reproduced using commonly accessible GPU hardware.

5.2.7 Discussion

1. Accuracy is misleading on DAiSEE. The HOG+LBP+SVM baseline reaches 0.750 accuracy by predicting engagement on almost every clip, with zero F1 on the three minority labels. The strongest single-label deep model reaches 0.702 accuracy with minority F1 below 0.13. **[OURS]** reports per-label accuracy of 0.911 on frustration despite F1 of 0.255 because 95.5% of test clips are not frustrated. Macro F1 is the only metric that does not hide this collapse. Figure 5.7 makes the same point visually.

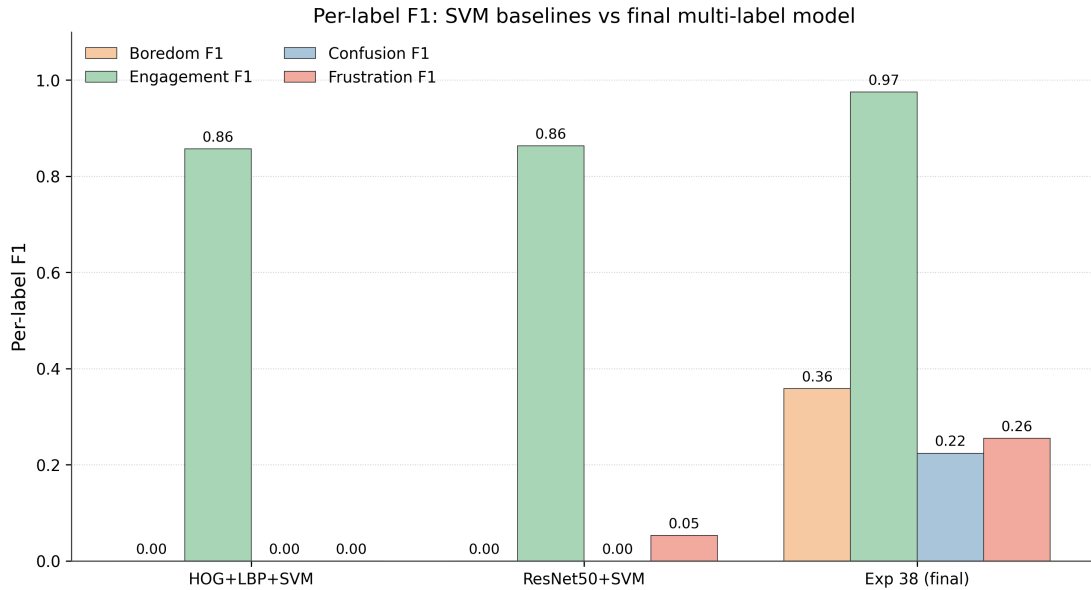


Figure 5.7: Per-label F1 of the SVM baselines and **[OURS]**. The SVM baselines have non-zero F1 only on engagement.

2. Imbalance must be addressed at every level. Plain BCE collapses to Macro F1 0.251. Loss-level correction alone, weighted BCE, reaches 0.408. Stacking a WeightedRandomSampler on top of weighted BCE applies two corrections to the same imbalance and over-corrects:

engagement F1 falls to zero and Macro F1 drops to 0.153. The combination used by **[OURS]**, mild fixed-multiplier oversampling with clipped positive-class weights and per-label tuned thresholds, reaches 0.453. Figure 5.8 traces this progression. No single intervention is sufficient; the three levels must be calibrated jointly.

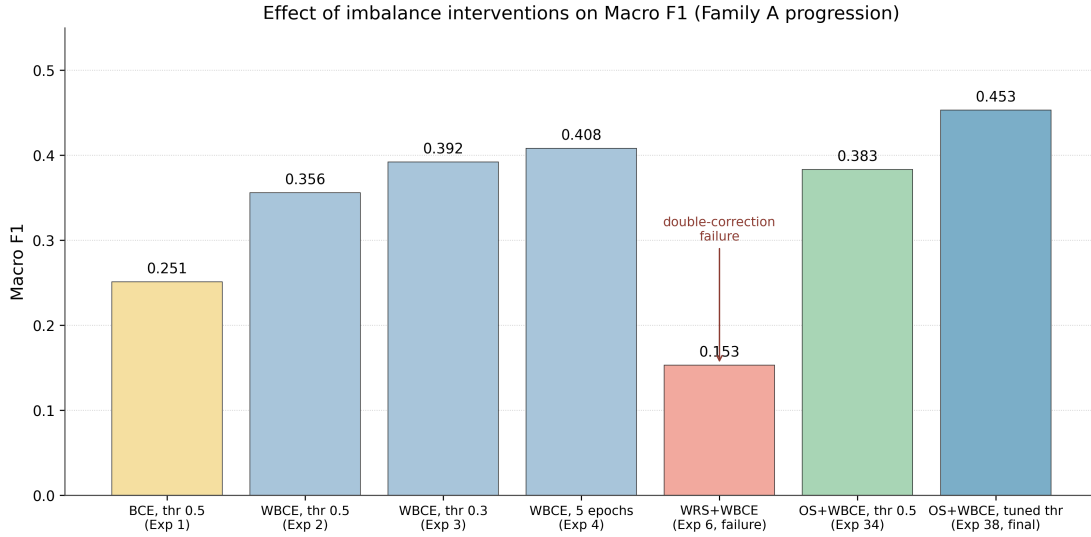


Figure 5.8: Macro F1 progression through the ResNet50+BiLSTM ablation. The WRS+WBCE double-correction failure is highlighted in coral; **[OURS]** is the highest entry.

3. Heavier or more recent backbones do not help at this scale. EfficientNetV2-S [109] reaches Macro F1 0.379, below the lightweight ShuffleNetV2+SE [27] at 0.404. The 3D DenseNet with self-attention [31], the heaviest model tested, also reaches 0.379. With only 5,358 DAiSEE training clips, backbone scaling does not behave as on ImageNet, a pattern consistent with recent educational FER reviews [25, 41, 47].

4. Per-label threshold tuning is the most cost-effective intervention. Moving a fixed threshold from 0.5 to 0.3 on the same network raises Macro F1 from 0.356 to 0.392. Per-label tuned thresholds reach 0.445 and 0.453, the two highest entries in Table 5.2, and require no retraining once the validation pass is complete.

5. Confusion and frustration remain hard. Confusion F1 0.224 on 157 positive clips and frustration F1 0.255 on 80 positive clips are the principal limitations of **[OURS]**. Both labels are rare, ambiguous, and often suppressed by learners facing a webcam. The per-label thresholds favour recall, which is the appropriate operating point for the adaptation module but caps precision. Pushing these two labels further would require additional modalities (audio, posture, log-stream signals) or a larger affective video corpus.

5.2.8 Summary

[OURS] achieves the following on the DAiSEE test set:

- **Architecture:** ResNet50 + SE block + two-layer BiLSTM + temporal attention, trained on 24-frame clips.
- **Training:** Weighted BCE loss with fixed-multiplier oversampling (OS-CSV) and per-label thresholds tuned on the validation split.
- **Macro F1:** 0.453. **Weighted F1:** 0.798. **Micro F1:** 0.724. **Samples F1:** 0.746.
- **Per-label F1:** Boredom 0.358 | Engagement 0.975 | Confusion 0.224 | Frustration 0.255.
- **Principal limitation:** Precision-recall trade-off on confusion and frustration, the two rarest and visually most ambiguous labels.
- **Distinction:** The only configuration combining four-state multi-label DAiSEE classification with calibrated, reproducible, deployment-ready thresholds under a single standardised pipeline.

The four-dimensional emotion vector produced by [OURS], one binary label per affective state per clip, serves as the affect input to the reinforcement-learning adaptation module described in the following section.

5.3 Adaptation Module Results and Discussion

5.3.1 Experimental Overview

Section 5.3 reports the evaluation of the adaptation module specified in Chapter 4. Two student populations are used: a single canonical learner (Configuration A) that exposes overfitting and a heterogeneous mixed population (Configuration B) that mirrors the deployment setting. The trained DQN policy is compared with a rule-based expert tutor and a uniform random policy under identical evaluation episodes and ten independent seeds; a twenty-seed confirmation run was performed for Configuration B. Section 5.3.2 reports the headline comparison, Section 5.3.3 reports robustness and generalisation, Section 5.3.4 positions the system against prior work, Section 5.3.5 discusses the educational implications and Section 5.3.6 closes the loop with the FER perception layer of Part I.

Baselines. The rule-based waterfall tutor encodes the affect-keyed expert heuristic that is the canonical hand-crafted reference for affect-aware tutoring [123, 114]; the uniform random policy samples actions over the legal action mask and serves as the distribution-free lower-bound reference [115, 118].

Metric legend. Three outcome metrics drive the headline comparison: normalised learning gain, mean episode reward and episode success rate. The first measures Hake’s session-level knowledge growth [137] and is the standard knowledge-acquisition measure in RL-based tutoring [115, 72]. The second is the multi-objective return that the policy optimises. The third counts episodes terminating with high knowledge, low frustration and low confusion. Two further measures are reported as *diagnostics* and are not interpreted as performance scores: policy-emotion alignment is the proportion of steps at which the chosen action matches the rule-based expert heuristic and therefore measures similarity to that heuristic rather than tutoring quality; Shannon entropy of the action distribution is a behavioural indicator of action diversity and policy-collapse risk. Formal definitions of every quantity are given in Chapter 4.

5.3.2 Comparative Performance Evaluation

Table 5.5 consolidates the headline outcome metrics. The DQN policy is the highest-scoring policy on every outcome metric and in both populations, and the rule and random baselines are clearly separated from each other. Pairwise Cohen’s d effect sizes lie between 1.81 and 3.30 and all Holm-corrected p -values are below 10^{-2} .

Table 5.5: Headline outcome metrics for the DQN policy and the rule and random baselines on both configurations. Effect sizes and Holm-corrected p -values are computed against the DQN policy. Mean values with 95% bootstrap confidence intervals across ten seeds.

Config	Policy	Learning gain	Reward	Success rate
A	DQN	0.602 [0.560, 0.644]	2.192 [2.029, 2.354]	0.248 [0.174, 0.321]
A	Rule	0.491 [0.488, 0.494]	1.882 [1.874, 1.890]	0.096 [0.089, 0.102]
A	Random	0.443 [0.440, 0.446]	1.423 [1.408, 1.439]	0.062 [0.055, 0.068]
A	DQN vs Rule	$d = 2.30, p_H = 3.5 \times 10^{-3}$		
A	DQN vs Random	$d = 3.30, p_H = 4.7 \times 10^{-4}$		
B	DQN	0.548 [0.510, 0.586]	2.031 [1.814, 2.248]	0.295 [0.234, 0.356]
B	Rule	0.460 [0.453, 0.467]	1.938 [1.921, 1.955]	0.156 [0.147, 0.165]
B	Random	0.414 [0.409, 0.420]	1.399 [1.370, 1.427]	0.101 [0.092, 0.110]
B	DQN vs Rule	$d = 2.01, p_H = 5.1 \times 10^{-3}$		
B	DQN vs Random	$d = 3.05, p_H = 6.2 \times 10^{-4}$		

Figures 5.10 and 5.11 display the same outcome values graphically. The rule-based expert improves on the random policy by virtue of its affect-keyed action choice, but it remains reward-suboptimal because the waterfall ordering forces it to commit early to a single action class. The DQN policy exceeds both baselines because it learns to integrate cognitive and affective penalties into a multi-step return, rather than committing to either a fixed action chain or to uniform sampling. Figure 5.9 shows that the training reward of the DQN policy crosses the

rule level within approximately 10,000 steps and stabilises near step 30,000, with a narrow confidence band that justifies the choice of 50,000 steps as the training budget.

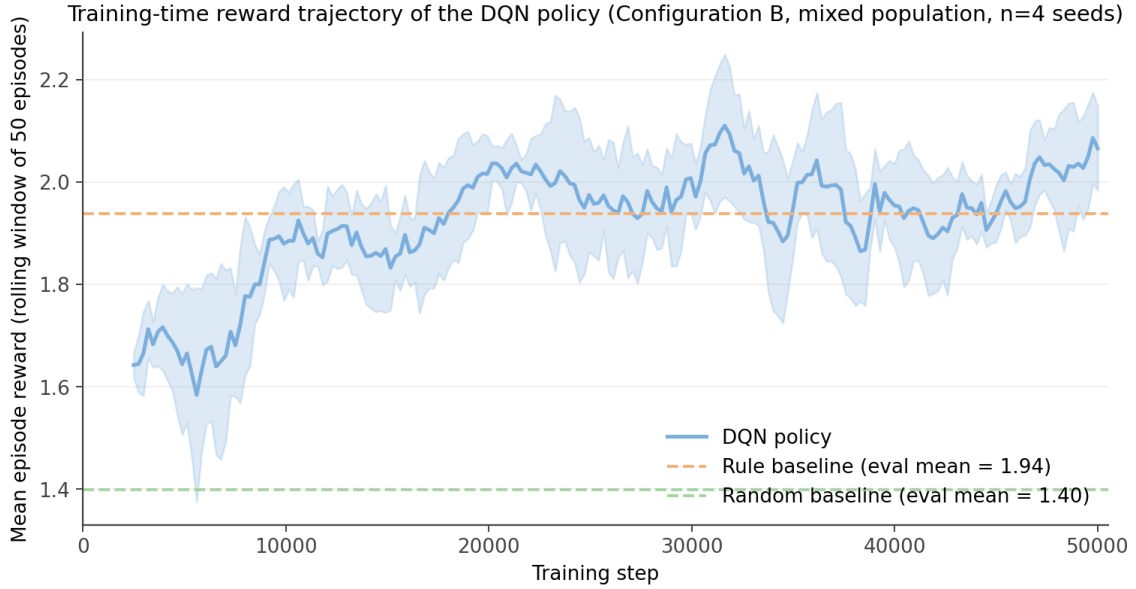


Figure 5.9: Training-time reward trajectory of the DQN policy on Configuration B with the eval-time reward of the rule and random baselines as horizontal references. The confidence band is ± 1.96 standard errors across four seeds. The rule and random baselines do not undergo training; their lines are the average evaluation reward of fixed policies and are shown as horizontal references so that the DQN trajectory can be read against a stable benchmark.

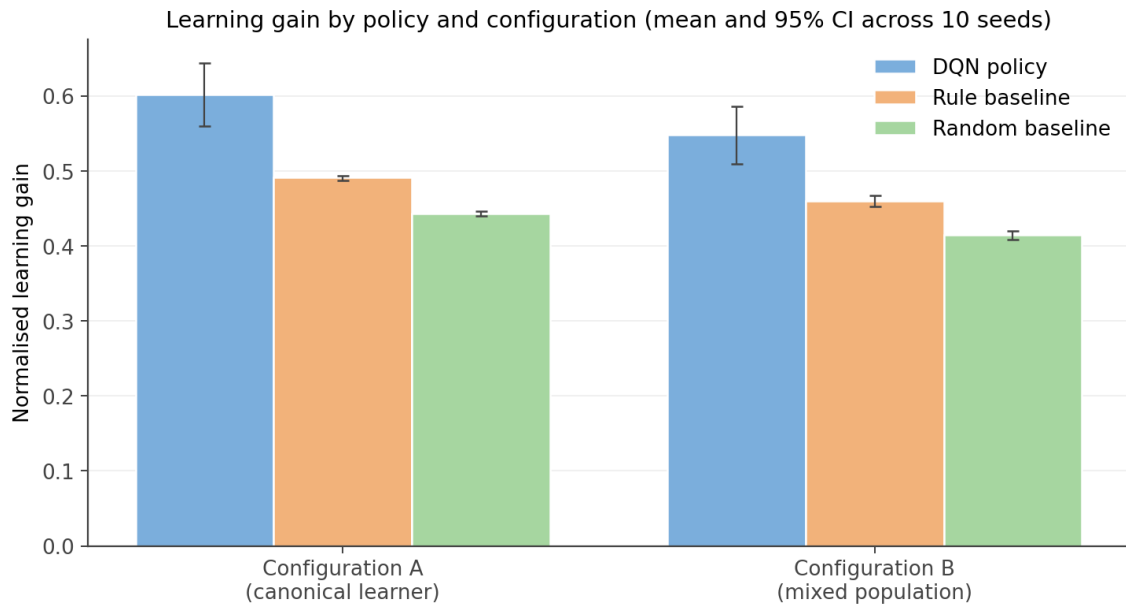


Figure 5.10: Normalised learning gain by policy and configuration with 95% bootstrap confidence intervals across ten seeds.

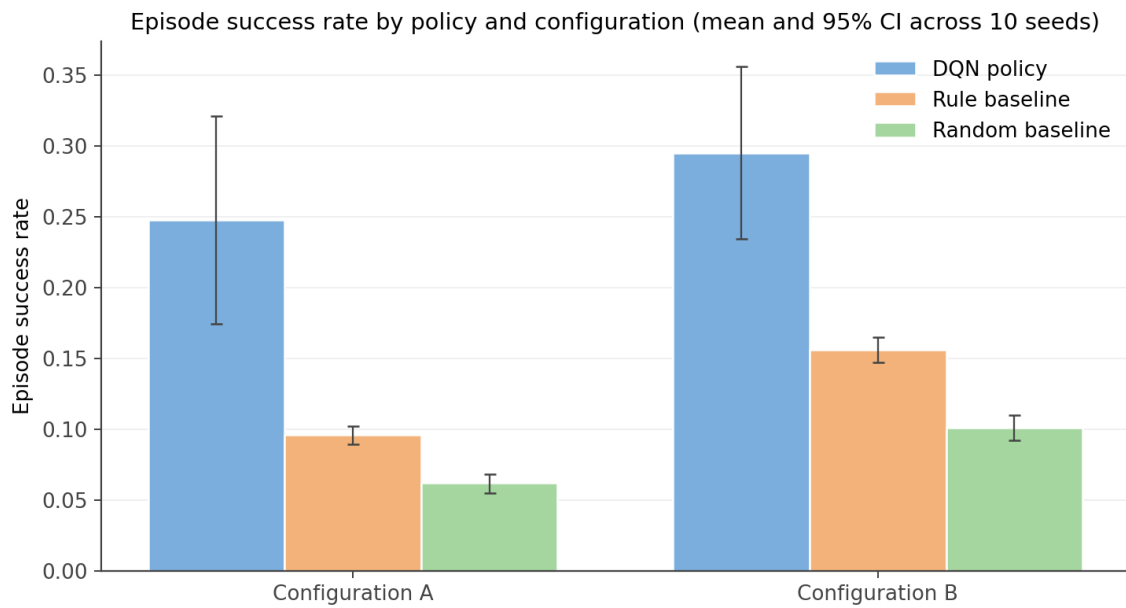


Figure 5.11: Episode success rate by policy and configuration with 95% bootstrap confidence intervals across ten seeds.

5.3.3 Robustness and Generalisation

Figure 5.12 shows the per-seed distribution of normalised learning gain. The standard deviations across the ten seeds are 0.0679 on Configuration A and 0.0614 on Configuration B, giving coefficients of variation of 0.113 and 0.112. No seed produced a degenerate policy, no evaluation episode terminated through frustration-driven dropout, and the rolling-window training reward never collapsed in any run.

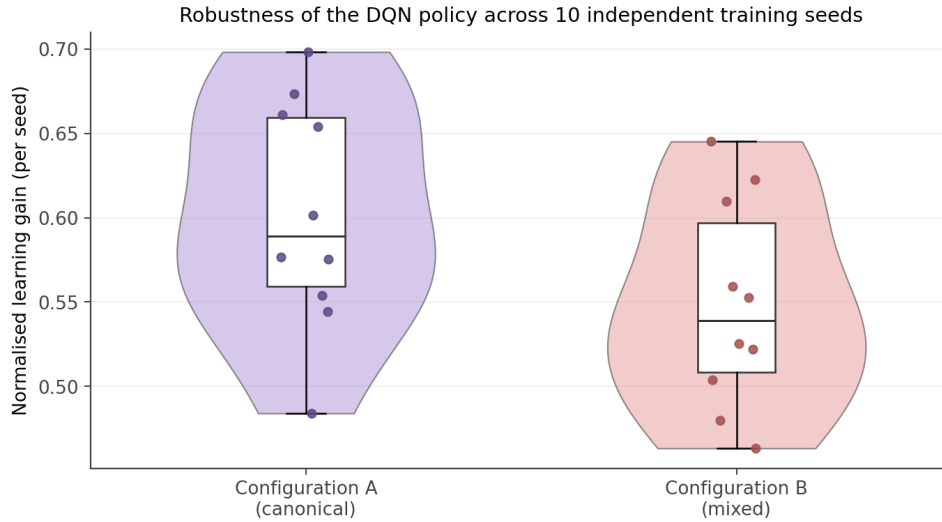


Figure 5.12: Per-seed normalised learning gain for both configurations. Violin shapes show the empirical distribution; the inner box reports the median and interquartile range.

Generalisation behaviour separates the two configurations. The canonical-trained policy reaches a same-population learning gain of 0.602 but its transfer learning gain on the mixed population drops to 0.553, a positive gap of 0.049 in every one of the ten seeds. The mixed-trained policy reaches 0.548 in-population, has a zero gap by construction, and transfers upward to the canonical learner at 0.607. The canonical-only setting therefore overfits the simulator personality, whereas mixed-population training trades a small amount of in-distribution gain for distributionally robust deployment behaviour [119]. The mixed-trained policy of Configuration B is the recommended thesis system.

5.3.4 Comparison with Prior Studies

Cross-study numerical comparison across reinforcement-learning intelligent tutoring systems is rarely meaningful. Studies differ in their simulators, reward functions, state and action spaces, learner and emotion models and evaluation protocols, so equally numbered results do not measure the same quantity. The comparison reported here is therefore qualitative and architectural. Each prior system is positioned on the seven design dimensions on which the present the-

sis takes a stance, namely the use of DQN, an educational or ITS focus, synthetic learners for training, knowledge modelling, emotion modelling, RL-driven adaptation and end-to-end perception-decision integration. Table 5.6 reports the coverage matrix.

Table 5.6: Component coverage of prior reinforcement-learning intelligent tutoring systems on the seven design dimensions of the present thesis. The first column reports bibliography reference numbers; full bibliographic information is available in the References section. ✓ denotes that the component is present and central, ✗ that it is absent, and ~ that it is partial or implicit.

Study	DQN	Edu/ ITS	Synth. stud.	Know. mod.	Emo. mod.	RL adapt.	FER-RL framework	Cov.
[118]	✓	✓	✗	✓	✗	✓	✗	4/7
[143]	✗	✓	✗	✓	✗	✓	✗	3/7
[115]	✗	✓	✓	✓	✗	✓	✗	4/7
[144]	✗	✓	✓	✓	✗	✓	✗	4/7
[145]	✗	✓	✓	✓	✗	✓	✗	4/7
[42]	✗	✓	✗	✓	✓	✓	✗	4/7
[36]	✗	✓	✗	~	✓	✓	✗	4/7
[146]	✗	✓	✗	✓	✓	✗	✗	3/7
[147]	✗	✓	✓	✓	✓	✗	✗	4/7
[14]	✓	✓	✗	✗	✗	✓	✗	3/7
[148]	✓	✓	~	✓	✗	✓	✗	4.5/7
[149]	✓	✓	✗	✓	~	✓	✗	4.5/7
[114]	✓	✓	✗	✓	~	✓	✗	4.5/7
[43]	✗	✓	✗	✗	✓	✓	~	4/7
Proposed FER-RL system (this thesis)	✓	✓	✓	✓	✓	✓	✓	7/7

The matrix is to be read qualitatively. Three patterns emerge. First, prior systems cluster into two largely disjoint groups: deep value-based controllers without an affective channel [118, 143, 115, 144, 145, 14, 148] and affect-aware models that integrate emotion information either without a deep value-based controller or without a closed-loop RL adaptation [42, 36, 146, 147, 43]. The thesis sits at the intersection of the two groups. Second, the most recent emotion-aware DQN systems [114, 149] extend value-based control with an affective input, but the affective signal is treated as a labelled input rather than as the output of a perception layer evaluated on a public benchmark; the studies indicated by ~ in the emotion column include emotion

in the state representation without validating a perception pipeline on a standard dataset. The thesis evaluates the FER perception layer on DAiSEE in Section 5.2 and integrates the resulting affect vector with the decision layer through a documented interface, which is the principal differentiator on the integration column. Third, no numerical advantage is claimed against the entries in Table 5.6 because the simulators and reward functions differ across studies; the contribution is the joint coverage of all seven dimensions rather than a numerical ranking on any single dimension.

5.3.5 Discussion and Educational Interpretation

Three findings drive the discussion.

The DQN policy outperformed both baselines. The learned policy improved on the rule-based expert by an absolute learning gain of 0.088 on Configuration B ($d = 2.01$) and on the random policy by an absolute learning gain of 0.134 ($d = 3.05$); the same ordering holds on every outcome metric and in both populations. The interpretation is that value-based control integrates cognitive and affective signals into a multi-step return more effectively than a hand-crafted heuristic or unstructured sampling. The result is consistent with prior reports that affect-aware controllers outperform rule-based tutors when the reward couples knowledge and affect [114, 43].

Mixed-population training improved robustness and transferability. Configuration A reaches the higher in- population learning gain (0.602 vs 0.548) but transfers downward to mixed dynamics in every seed (mean gap +0.049). Configuration B has a zero in-distribution gap by construction and transfers upward to the canonical learner (0.607). The policy that experienced heterogeneous learners during training generalises more credibly to the heterogeneous populations that classrooms produce, in line with the heterogeneous-learner adaptation literature [150, 21] and the real-world RL framing of generalisation [119].

Policy behaviour reflected affect-aware adaptation rather than static instruction. The affect-conditioned action distribution of the DQN policy (Figure 5.13) selects scaffold on 90% of confused states, harder problem on 99% of bored states, encouragement on 74% of frustrated states and a scaffold-plus-no-action mixture under engagement. Two actions, scaffold (31.7%) and harder problem (45.6%), dominate the population-level distribution; the remaining six actions appear as state-conditional refinements rather than as collapsed or uniform behaviour. The action heatmap aligns with the zone-of-proximal-development principle used by DQN-based cognitive-load tutors [114] and with the affect-aware ITS prescription that scaffolding follows confusion and motivational support follows frustration [123, 122]. The Control-Value framing of academic emotions [7] interprets the pattern as control restoration under negative affect and value maintenance under positive affect.

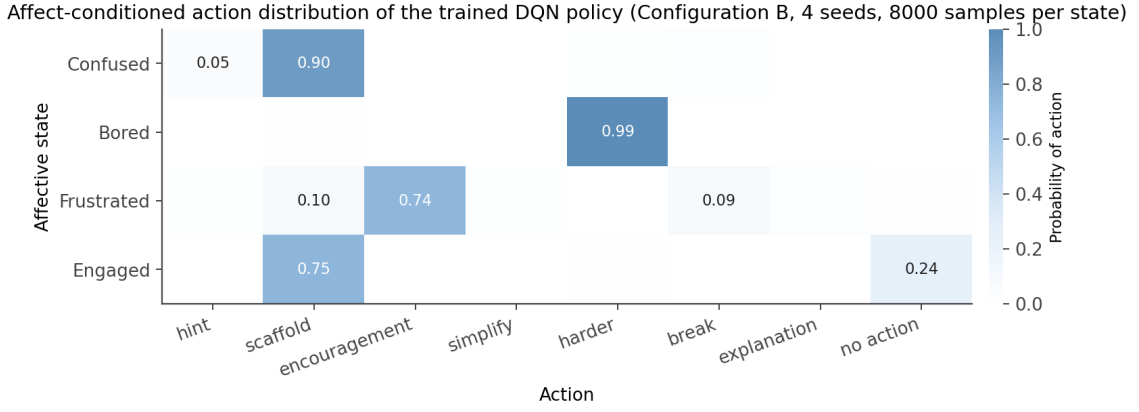


Figure 5.13: Affect-conditioned action distribution of the trained DQN policy on Configuration B. Each row sums to one before omission; entries below 0.05 are hidden.

5.3.6 FER-RL System Integration

Figure 5.14 illustrates the integrated pipeline as a single control loop. The FER module of Part I is the perception layer: at each interaction window it consumes the face-aligned frames and emits a four-channel affect vector $\mathbf{a}_t \in \{0, 1\}^4$ over the labels boredom, engagement, confusion and frustration. The RL module of Part II is the decision layer: at each step it consumes the six-dimensional learner state $\mathbf{s}_t$ and emits a pedagogical action from the action set $\mathcal{A}$. The continuous channels of $\mathbf{a}_t$ enter $\mathbf{s}_t$ directly; the `FERAdapter` interface projects the same vector onto the discrete emotion identifier used for masking. The same interface is used in simulator mode and in real-FER mode, with only the source of $\mathbf{a}_t$ changing.

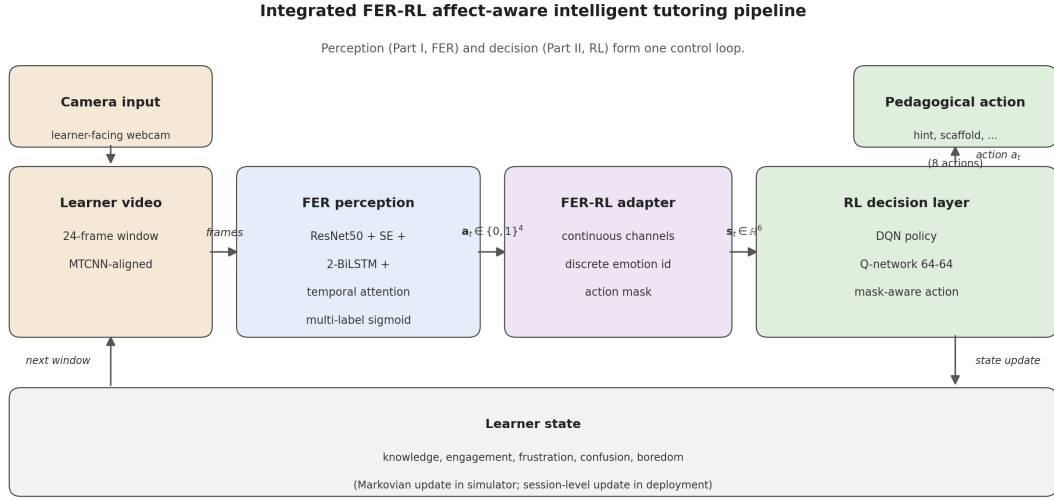


Figure 5.14: Integrated FER-RL pipeline. The perception layer converts a face-aligned window into a four-channel affect vector; the adapter assembles the RL state; the DQN policy emits a pedagogical action that updates the learner state at the next interaction step.

The operational dependency is asymmetric. The decision layer cannot act without affect information, so a failure of the FER module collapses the system to a state-independent policy. The perception layer in turn has no mechanism for changing the learner’s affect; without the decision layer it remains a diagnostic instrument. The two modules are therefore halves of one control loop and the quantitative findings of Part I and Part II jointly characterise the system. The reliability profile of the perception layer reported in Section 5.2.4 (engagement F1 0.975; minority-label F1 between 0.224 and 0.358) defines the operating condition under which the decision layer must function in deployment, and the action mask of the simulator guarantees that catastrophic actions remain inadmissible regardless of perception noise. Educator oversight of the mask, as advocated by [151], and the privacy and integration considerations summarised by [41], complete the deployment envelope of the integrated architecture.

5.3.7 Adaptation Module Summary

- The DQN policy outperforms the rule-based expert and the random baseline on every outcome metric (learning gain, mean reward, success rate) and in both student populations, with Cohen’s d between 1.81 and 3.30 and Holm-corrected p -values below 10^{-2} .
- Convergence is reproducible across the ten seeds (coefficient of variation 0.112 on Configuration B), and no evaluation episode terminated through frustration-driven dropout.
- Mixed-population training transfers upward to the canonical learner and shows no overfitting signal, whereas canonical-only training overfits in every seed (mean gap +0.049);

the mixed-trained policy of Configuration B is the recommended deployment configuration.

- Affect-conditioned action selection is concentrated and pedagogically interpretable, with scaffold and harder problem as the dominant actions and the remaining six actions appearing as state-conditional refinements.

5.4 Chapter Summary

This chapter has reported the evaluation of the two modules that together constitute the affect-aware intelligent tutoring system defended in this thesis. The findings of Part I (perception, Section 5.2) and Part II (decision, Section 5.3) are summarised below.

- **Perception layer.** The proposed ResNet50 with SE block, two-layer BiLSTM and temporal attention reaches Macro F1 0.453, Weighted F1 0.798, Micro F1 0.724 and Samples F1 0.746 on the DAiSEE test split under the standardised pipeline, with per-label F1 of 0.975 on engagement, 0.358 on boredom, 0.255 on frustration and 0.224 on confusion.
- **Decision layer.** The DQN policy is the highest-scoring policy on every outcome metric and in both student populations. The mixed-population policy of Configuration B reaches learning gain 0.548, mean reward 2.031 and success rate 0.295.
- **Effect magnitude.** The advantage over the rule and random baselines is large by Cohen’s convention (d between 1.81 and 3.30) and statistically significant at $p_{\text{Holm}} < 10^{-2}$ on every pairwise test.
- **Robustness and generalisation.** The policy converges reproducibly across ten seeds, avoids frustration-driven dropout and selects actions that vary with the affective state. Mixed-population training transfers upward to the canonical learner and is the recommended deployment configuration.
- **Integrated architecture.** The FER perception layer and the DQN decision layer form one closed control loop: the FER module supplies the affect vector $\mathbf{a}_t$, the adapter assembles the RL state, the DQN policy emits a pedagogical action and the action updates the learner state at the next interaction step. Together they realise an affect-aware adaptive tutoring system in which perception and decision act as complementary halves of a single architecture.

Chapter 6

Conclusion

This thesis developed a single closed-loop adaptive tutoring system in which facial expression recognition (FER) and reinforcement learning (RL) function as inseparable layers of one control architecture rather than as adjacent models. A commodity webcam supplies the perception layer; a multi-label academic FER component trained on the DAiSEE dataset projects the learner’s affective state onto a continuous signal; and a Deep Q-Network (DQN) decision layer consumes that signal directly to select pedagogical actions within a theory-grounded Markov decision process over an eight-action space. The two layers are coupled through the FERAdapter, a unifying abstraction that lets the same policy operate over either simulated or real affect without retraining. The central result is direct: an affect-aware policy raised normalised learning gain by 0.088 over a hand-crafted, affect-keyed expert heuristic on a heterogeneous population of synthetic learners (Cohen’s $d = 2.01$, Holm-corrected $p < 10^{-2}$), and held that advantage across every outcome metric in both populations, with effect sizes from 1.81 to 3.30. Intelligent tutoring systems have modelled what learners know with increasing sophistication while ignoring how they feel; these simulation results suggest that incorporating affect into instructional decisions can improve modelled learning outcomes within the synthetic tutoring environment used in this thesis. Whether these improvements transfer to real learners remains an open question for future work.

The perception layer earns that claim by first repairing the ground beneath it. Published DAiSEE results are mutually incomparable, because preprocessing, frame counts, and balancing differ silently between papers. Retraining six competing architectures from scratch under one fixed pipeline turned that literature into a controlled comparison, and within it the proposed ResNet-50 with Squeeze-and-Excitation recalibration, a two-layer BiLSTM, and temporal attention pooling won, reaching the highest Macro F1 at 0.453 and a mean per-label accuracy of 0.782. Framing academic affect as four independent binary labels rather than one dominant class let the model represent the engaged-yet-confused learner that a single-label head cannot, and the per-label F1 scores record both the achievement and its honest boundary: 0.975 on en-

gement, 0.358 on boredom, 0.255 on frustration, and 0.224 on confusion. The decision layer earns its claim through restraint. Every transition rule in the environment traces to published theory, and the DQN is the only component allowed to learn, so the policy's learning gain of 0.548 against 0.460 for the expert heuristic reflects learned control, not a simulator bent in its favour. The generalisation result carries the most weight: a policy trained on diverse learners transferred upward to the canonical learner at 0.607, while a policy trained on the canonical learner alone fell to 0.553 on the mixed population in every one of ten seeds. Robustness in affect-aware tutoring comes from exposure to learner variability, not from architectural depth, and the system that demonstrates it remains lightweight enough to deploy on commodity hardware.

This thesis shows that benchmark-validated affect perception and theory-grounded decision-making can be joined through a single interface into one reproducible architecture, with each half measured honestly and the connection between them made explicit. For intelligent tutoring, it shows that the affective signal a good teacher reads by instinct can be instrumented, learned, and acted on without surrendering rigour. For affective computing, it shows that emotion recognition pays off not as a dashboard but as a control signal inside a closed loop. For personalised education at scale, it shows that tutors worthy of real learners are built by training on the full variability of the people they will serve. The learner's face has always been the richest signal in the room; this work demonstrates in simulation, in numbers and in architecture, that a machine can be built to read it, reason from it, and earn a place beside the teacher rather than the test.

Chapter 7

Limitations and Future Work

7.1 Limitations

Two boundaries define the scope within which these claims hold. The decision-layer results were obtained inside a synthetic-learner simulation rather than a real classroom, so they establish what the policy achieves against modelled learners, not yet against human ones. The perception-layer results were measured on the DAiSEE benchmark rather than on live webcam streams, so they characterise the model under curated recording conditions rather than the variability of real deployment. The two layers, although joined by a single adapter interface, were never run together as one loop on real affect, so the end-to-end system is validated component by component rather than as a whole. These limitations do not undermine the results; they locate them. Each was also a deliberate methodological choice: a controlled benchmark isolates the perception model from confounded preprocessing, and a theory-grounded simulator isolates the policy from the ethical and logistical cost of experimenting on students before the architecture is ready to meet them. The directions in Section 7.2 address each of these boundaries in turn.

7.2 Future Work

The first direction for future work is classroom deployment of the system. The current work is evaluated in a simulated environment, so testing it with real students is a natural next step. This is not only a practical step but a scientific one, because the policy was trained on simulator-derived affect with clean Gaussian noise and perfectly synchronous timing, whereas real facial emotion detection produces different noise and delays that the policy has never encountered. In real use, the system will face more noise in emotion detection, different student behaviours, and privacy constraints. To address this, the model can run on low-cost devices with on-device processing so that raw video does not need to be stored or shared. In addition, simple safety

rules can be kept under teacher control so that educators can adjust system behaviour during use.

The second direction is to improve the perception part of the system using simple additional signals. Instead of relying only on facial expression, the system can also use low-cost modalities such as gaze direction, head movement, or speech tone (prosody). These signals are easy to collect and can improve robustness when facial emotion detection is uncertain. This matters most for the states the face hides: frustration and confusion reach an F1 of only 0.255 and 0.224, and these are exactly the cases that visual-only recognition misses. This extension does not change the main model, but only adds extra input signals to make predictions more stable.

The third direction is to make the system more flexible in classroom settings using a simple multi-agent setup. Instead of one learning agent only, a second agent can represent the teacher or classroom strategy. The two agents can work together: one focuses on student engagement, and the other controls teaching decisions. This makes the system easier to adapt to real classroom situations where teaching is not fixed and can change depending on student response.

Finally, future work can extend the system from short sessions to longer learning periods. This matters because the current policy reads a fixed-length state vector and cannot reason over long interaction histories, so slow-drifting states such as boredom are effectively invisible to it. In this case, the system would track student progress over multiple lessons instead of one session only. This would make the model closer to real education settings, where learning happens over time rather than in a single interaction.

Appendix

Appendix A

DQN Policy Replay Samples

These three frames are consecutive steps from Episode 2 of a trained DQN policy replay, captured with the replay viewer tool. Each frame substitutes the current state into the thesis reward function $r = w_k \Delta K + w_e E - w_f F - w_b B - w_c C$ to illustrate the policy’s real-time responsiveness across three decisions.

Across these steps, the policy transitions from **scaffold** to **harder problem** as the learner’s knowledge level increases, and then to **explanation** when confusion rises. The sequence reflects the behaviour of the trained DQN policy as it reacts to changes in the observed student state.

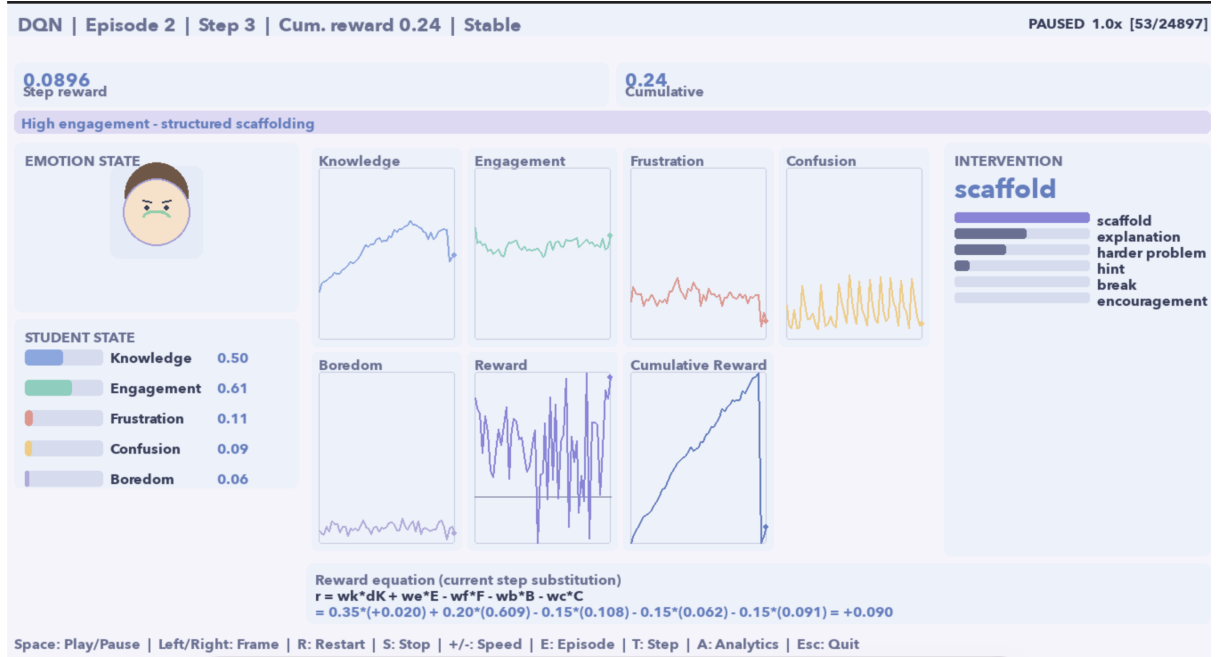


Figure A.1: Episode 2, Step 3, cumulative reward 0.24, with student state Knowledge 0.50, Engagement 0.61, Frustration 0.11, Confusion 0.09, Boredom 0.06. The policy selected **scaffold**, a flow-consistent choice that structures a high-engagement learner without raising difficulty, yielding a step reward of 0.0896.



Figure A.2: Episode 2, Step 4, cumulative reward 0.33, with student state Knowledge 0.54, Engagement 0.58, Frustration 0.13, Confusion 0.17, Boredom 0.11. The policy selected **harder problem** as the learner's knowledge level increased to approximately 0.54, resulting in a rise in confusion to 0.17 following the increase in task difficulty, with a step reward of 0.0845.

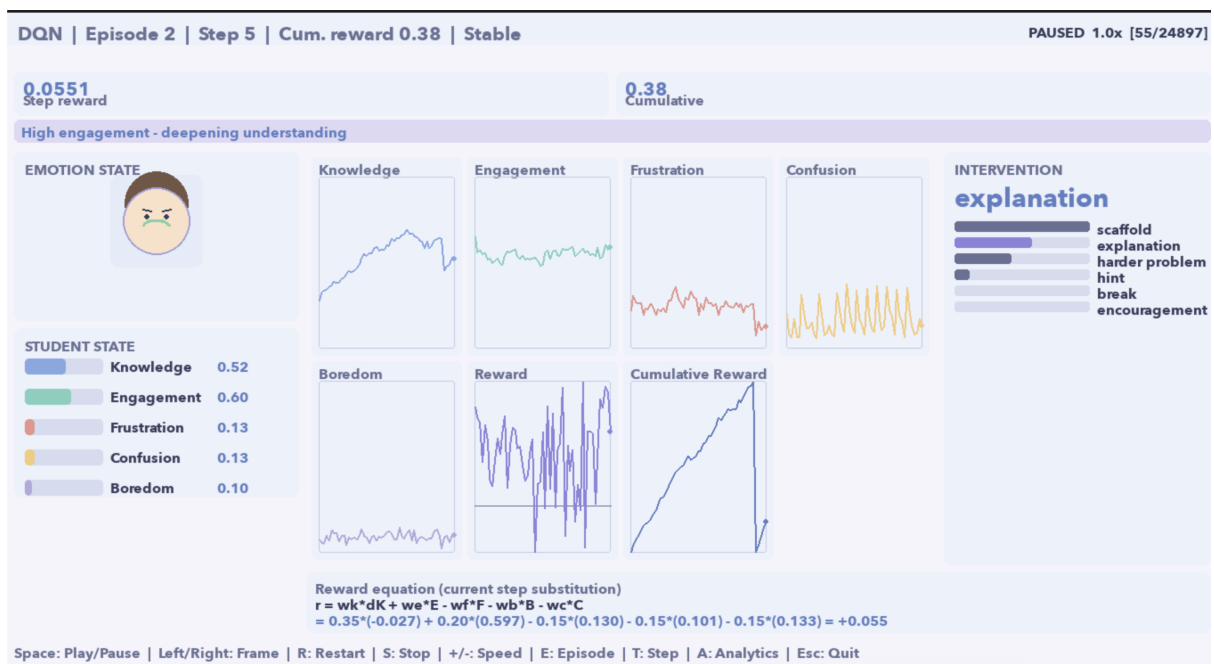


Figure A.3: Episode 2, Step 5, cumulative reward 0.38, with student state Knowledge 0.52, Engagement 0.60, Frustration 0.13, Confusion 0.13, Boredom 0.10. In response to the prior confusion rise, the policy selected **explanation**, which lowered confusion to 0.13 while engagement stayed high, yielding a step reward of 0.0551.

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